

Renewal and changes of law for critical honeycomb walks

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Abstract

For critical honeycomb self-avoiding walk, we prove spatial exponent $3/4$ for the infinite irreducible-bridge law and for bridges of every sufficiently large compatible even length. We establish the corresponding thermal laws at every large discount scale, including all positive length and spatial moments. For unrestricted uniform walks, the endpoint lower law holds on a common set of lengths of natural density one. We also determine the near-critical exponential correlation scale and small-force free-energy exponent, and prove spatial local lower bounds that permit conditioning on a prescribed terminal vertex.

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1 Introduction

A self-avoiding walk of prescribed length and a self-avoiding walk whose length L is penalized by $e^{-L/N}$ need not have the same distribution in space. The first law fixes one coefficient of a generating function; the second sums all its coefficients, with N setting the discount scale. A third law is obtained by joining independent irreducible bridges forever. Understanding the relations among these laws requires control of their normalizing masses, as well as estimates for the individual paths.

We study these changes of law on the regular honeycomb lattice. Realize its vertices as the centers of the triangles in an equilateral triangular tiling with side length one, and put

$$\rho = (2 + \sqrt{2})^{-1/2}.$$

A port is the midpoint of a tiling edge on a domain boundary. A strict bridge starts at a prescribed port on one horizontal tiling line, ends at a port on a higher parallel line, and has all its visited vertices strictly between the two lines. A port path with L visited vertices has critical weight ρ^L . The ports carry no weight, so adjoining bridges concatenate with exactly the product of their weights. The terminal height and horizontal position are free unless specified.

An intermediate horizontal line crossed exactly once determines a unique cut of a bridge. Cutting at all such lines decomposes the bridge into irreducible bridges. Their critical weights sum to one and therefore define a probability law p . Independent p -distributed pieces can be concatenated without further avoidance conditions: successive pieces occupy disjoint open slabs. This construction provides the common probability space used throughout the paper.

For an ordinary finite walk $\gamma = (\gamma_0, \dots, \gamma_n)$, write

$$A(\gamma) = |\gamma_n - \gamma_0|, \quad D(\gamma) = \text{diam}\{\gamma_0, \dots, \gamma_n\}.$$

Uniform full-plane walks start at one prescribed vertex and have n edges. Uniform half-plane walks start at the inward vertex of a prescribed boundary port and have a prescribed number of visited vertices in the strict half-plane, with their last vertex unrestricted. Thermal walks have variable length and weight $\rho^L e^{-L/N}$, normalized over all such walks. Using edge length instead of vertex length in this last law multiplies all weights by one common factor and gives the same probability measure.

Here are the principal spatial conclusions, with the law included in each statement. The notation $X_t = t^{a+o(1)}$ in probability means that for every $\varepsilon > 0$ the event $t^{a-\varepsilon} \leq X_t \leq t^{a+\varepsilon}$ has probability tending to one.

- Under the infinite irreducible-bridge law, the distance of the n th vertex from the initial port is $n^{3/4+o(1)}$ almost surely, and the number of visited vertices within distance R is $R^{4/3+o(1)}$ almost surely (Theorem 2.7). Both uniform and fugacity-weighted finite half-plane laws converge on each fixed initial path to this law (Theorem 2.8 and Lemma 6.1).
- Uniform strict bridges with n visited vertices have terminal distance and diameter $n^{3/4+o(1)}$ in probability at every sufficiently large even n . The parity restriction is part of the port convention. Critical bridges conditioned instead on height h have length $h^{4/3+o(1)}$ in probability (Theorems 3.3 and 3.4).
- Uniform endpoint-free half-plane and full-plane walks have endpoint distance and diameter $n^{3/4+o(1)}$ in probability along a common set of lengths of natural density one. Their upper spatial bound holds at every sufficiently large length, with arbitrarily high polynomial accuracy (Theorems 8.1 and 8.2).
- For thermal half-plane and full-plane walks, $L = N^{1+o(1)}$ and $A, D = N^{3/4+o(1)}$ in probability. Their positive moments have the corresponding exponents: for each fixed $p > 0$, $\mathbb{E}L^p = N^{p+o(1)}$ and $\mathbb{E}A^p, \mathbb{E}D^p = N^{3p/4+o(1)}$ (Theorems 6.4 and 8.1).

The full-plane endpoint lower bound at fixed length retains the density-one qualification. The thermal statement concerns every large discount scale N . These conclusions use different normalization arguments, described below. Nienhuis's analysis of the two-dimensional $O(n)$ model predicts the spatial exponent $\nu = 3/4$ for self-avoiding walk in the $n \rightarrow 0$ polymer limit [15, Eqs. (1)–(2)]; the results here retain the sampling-law and length qualifications stated above.

Bridge theory and the additional estimates

Kesten's normalization for irreducible bridges underlies the infinite-bridge construction of Madras and Slade [11, 14]. Lawler, Schramm and Werner identify this measure as the fixed-length half-space limit and describe its fugacity counterpart [13, Appendix]. Dyhr, Gilbert, Kennedy, Lawler and Passon give explicit finite-prefix descriptions of both limits and identify the strip law by conditioning on a renewal height [5, Propositions 2.2–2.3, 2.9 and 2.10]. These works use vertex conventions on the integer lattice; the strip construction of Dyhr et al. also includes a final bond from the conditioning height to the target height. We adapt the renewal arguments to the present honeycomb port convention and prove the required normalizations here. The infinite law, a fixed-height bridge law and a law conditioned on one exact length are different uses of the construction.

Duminil-Copin and Smirnov proved the value of ρ using a parafermionic observable on the honeycomb lattice [4]. Beaton et al. use a honeycomb mid-edge renewal construction and prove that the critical bridge mass tends to zero as the strip height grows [1, Theorem 10 and Appendix]. Krachun and Panagiotis obtain a positive-power decay bound for this mass and show that a uniform n -step honeycomb walk remains within distance $O(n/\log n)$ of its root with probability tending to one [12].

The sharper finite estimates used below require additional strip and marked-polygon arguments. We state the companion results as estimates of positive walk weights. In Section 2, the finite bridge estimates of Theorem 2.1 and the renewal identities yield joint bounds for the height, length and diameter of one irreducible. Section 5 uses further finite strip and polygon estimates to control avoidance and absolute path masses. The analytic finite estimates are proved in the cited companions; the present paper proves the renewal and geometric transfers needed when the sampling law changes.

The key exponents for one irreducible are

$$\alpha = \frac{9}{16}, \quad \beta = \frac{3}{4}.$$

Write H, L, D for the height, vertex length and diameter of one irreducible. On the main route the finite estimates give

$$1 - \mathbb{E}_p e^{-uL} \asymp u^\alpha, \quad 1 - \mathbb{E}_p e^{-sH} \asymp s^\beta, \quad \mathbb{P}_p\{D > r\} \leq Cr^{-\beta}$$

as $u, s \downarrow 0$ and $r \rightarrow \infty$. Here $\asymp$ means upper and lower bounds with positive constants independent of the parameter. The ratio $\alpha/\beta = 3/4$ explains the spatial scale after many pieces have been joined. Turning this observation into a theorem at one exact length requires more than a sum estimate.

How the normalizations are controlled

Our main exact-length route uses these bounded-factor finite estimates. If $S_L(k) = L_1 + \cdots + L_k$ for independent irreducibles, the local denominator is

$$u_n = \mathbb{P}_p\{S_L(k) = n \text{ for some } k\} \geq cn^{\alpha-1} \quad \text{for all sufficiently large even } n.$$

Section 3 proves this by exponential tilting and Fourier inversion on the compatible length lattice. Fourier concentration for the remaining pieces then pays for sequences with atypical total height or diameter. The identification of the uniform half-plane prefix limit in Section 2 is a separate ratio argument.

A second, self-contained route begins with a logarithmic window of bridge lengths. Its length information is weaker:

$$cu^\alpha(\log(1/u))^{-P} \leq 1 - \mathbb{E}_p e^{-uL} \leq Cu^\alpha.$$

A Fourier difference estimate first gives a long interval on which the renewal mass has the required lower bound. Fresh independent increments then move the remaining length into that interval. The logarithmic window permits enough progress at each stage that the total landing cost is only $n^{-o(1)}$, giving $u_n \geq n^{\alpha-1-o(1)}$. Section 4 proves this alternative under its own hypotheses, with a fixed logarithmic exponent below one and a stopping interval whose width remains large.

Section 5 next establishes the geometric estimates used in the thermal transfer of Section 6. A half-plane walk consists of a list of complete irreducibles followed by a remainder with no further renewal. In a prescribed-prefix probability, the total weight of that remainder cancels exactly. For an unrestricted plane walk we obtain a denominator by putting a nonempty bridge between two exterior remainders. Recovering the first and last single-crossing lines makes this construction injective. To bound small endpoint displacement, we cut a path at its lowest vertex and use the two-end estimate of Corollary 5.11, retaining one independent height-smoothing variable. The two remainder weights then appear in both numerator and denominator and cancel.

The censored-arm estimates in Section 7 give a separate geometric route from the logarithmic finite inputs to absolute spatial bounds. Section 8 then treats moments and uniform walks, where there is no comparable exact cancellation at each length. In the half-plane, disjoint long-prefix events compare masses at nearby lengths. In the full plane, we insert a bridge at a minimum and use censored survival to control the inverse multiplicity. Both arguments average logarithms of the nearby-length masses: the interior terms telescope and leave a boundary error. This is why the resulting lower laws hold on a density-one set.

Next, let $G_t(b)$ be the total weight $(\rho e^{-t})^{\ell(\gamma)}$ of ordinary walks from a fixed vertex o to b , with edge length ℓ . Section 9 proves that the convergence threshold in s of $\sum_b G_t(b) e^{se \cdot (b-o)}$ is $t^{3/4+o(1)}$, uniformly for unit directions e . The fixed-length free energy

$$f_e(s) = \lim_{j \rightarrow \infty} \frac{1}{j} \log \sum_{\ell(\gamma)=j} \rho^j e^{se \cdot (\gamma_j - o)}$$

exists and is $s^{4/3+o(1)}$ as $s \downarrow 0$, uniformly in e . Its two one-sided derivatives are $s^{1/3+o(1)}$, in agreement with Pincus's force-extension scaling relation $1/\nu - 1$ at $\nu = 3/4$ [25, Eq. (I.4)]. Here the response follows from the threshold bounds and convexity; the distinction from fixed-force ballistic results is discussed in Section 9.

Prescribing a terminal vertex requires a further estimate: if $|b_N - o| = N^{3/4+o(1)}$, the thermal law conditioned to end at b_N has length $N^{1+o(1)}$ and diameter $N^{3/4+o(1)}$ in probability, with the corresponding powers for all positive moments (Theorem 9.5). Proposition 9.4 obtains a uniform two-dimensional lattice lower bound from subsequential local limits. A lower partition bound supported in a corridor then permits this conditioning. The separate spatial-rate statement takes distance to infinity at fixed discount and then takes the discount to zero; it does not identify these two orders of limits.

The final three sections develop consequences of other finite inputs. Section 10 retains a length reward in a logarithmically enlarged box and a completed-prefix expectation in a fixed-proportion ball. Section 11 derives the irreducible length tail from signed cylinder and marked-polygon estimates. Section 12 gives a geometric route to the infinite and thermal laws, using future-renewal averaging for the infinite path and recoverable tall-bridge insertion for the thermal path. These routes state their own finite inputs and retain the precision those inputs supply.

2 One probability space for the different laws

The elementary renewal identities must precede the changes of measure. They concern complete irreducible pieces. In particular, no independence between the length, height and displacement of one piece is assumed.

2.1 Ports, strips and the finite input

Fix one port on a horizontal row boundary. A strict bridge of height $h \in \mathbb{N}$ has its other port on the boundary h rows above and all its vertices in the open strip between these boundaries. One row has Euclidean height $d_0 = \sqrt{3}/2$. Write $\mathcal{B}_h$ for this set, with the terminal port unrestricted, and set

$$B_h = \sum_{\gamma \in \mathcal{B}_h} \rho^{L(\gamma)}, \quad M_h = \sum_{\gamma \in \mathcal{B}_h} L(\gamma) \rho^{L(\gamma)}, \quad B_0 = 1.$$

The term at height zero denotes an empty concatenation, not a nonempty walk. For every port path its diameter includes both ports as well as its visited vertices. Heights and transverse displacements use the actual staggered honeycomb port lattice. All paths and all translations below retain their starting port; there is never an extra sum over its position.

Here is the positive finite input for our main route. Its proof belongs to the uniform marked-polygon calculation and its exterior sewing argument.

Theorem 2.1 (Finite bridge estimates). *There are constants $c, C > 0$ such that, for every sufficiently large integer h ,*

$$ch^{-1/4} \leq B_h \leq Ch^{-1/4}, \quad M_h \leq Ch^{13/12}, \quad B_h\{L \geq ch^{4/3}\} \geq ch^{-1/4}.$$

For all $h \geq 1$ and $x \geq 1$, one also has

$$B_h\{D > x\} \leq Cx^{-1/4}.$$

The same estimates hold in the symmetry-related pure directions.

These are the bounded-factor estimates of [17, Theorem 1.1]. All finitely many smaller-height masses and first moments are finite as well: append a fixed straight bridge to inject their paths into a sufficiently tall strip, with bounded weight and additive length changes. We enlarge constants to include these heights. The first moment and the positive mass of long bridges play different roles: the former bounds a truncated irreducible reward, whereas the latter supplies a lower length deficit. Exponent-only versions lead to the weaker, explicitly stated variants later in the paper.

2.2 Factorization and marked rewards

An intermediate row boundary crossed exactly once is a renewal boundary. Cutting a strict bridge at every renewal boundary gives an ordered list of irreducible bridges, each of positive integral height. Conversely, a list of irreducibles concatenates to a strict bridge: its pieces occupy disjoint open strips. Let $\mathcal{I}$ be the set of irreducibles rooted at our fixed port and initially put $p(\gamma) = \rho^{L(\gamma)}$.

Proposition 2.2 (Renewal factorization). *The masses $p(\gamma)$ sum to one. If H, L, X, D denote respectively the height, visited-vertex length, signed transverse displacement and Euclidean diameter of an irreducible, then, for $s > 0$ and $u \geq 0$,*

$$b(s, u) := \sum_{h \geq 0} \sum_{\gamma \in \mathcal{B}_h} \rho^{L(\gamma)} e^{-sh - uL(\gamma)} = \frac{1}{1 - \mathbb{E}_p e^{-sH - uL}}.$$

In particular $b(s, 0) \asymp s^{-3/4}$ as $s \downarrow 0$. Under independent successive samples from p , B_h is the probability that some renewal occurs at height h .

Proof. Unique factorization and multiplication of vertex weights give the geometric series with nonnegative terms. The finite strip estimate makes $b(s, 0)$ finite for $s > 0$, so $\sum p(\gamma)e^{-sH(\gamma)} < 1$. Monotone convergence as $s \downarrow 0$ gives $\sum p \leq 1$. On the other hand, summing $B_h \asymp (1+h)^{-1/4}$ gives $b(s, 0) \asymp s^{-3/4} \rightarrow \infty$, forcing $\sum p = 1$. Positive heights make the events that the k th renewal has height h disjoint as k varies. Summing their probabilities yields the last assertion. $\square$

For a nonnegative function f of an irreducible, define the height-indexed reward

$$p_f(j) = \mathbb{E}_p[f; H = j], \quad B_f(h) = \sum_{\gamma \in \mathcal{B}_h} \rho^{L(\gamma)} \sum_{I \text{ a piece of } \gamma} f(I).$$

Lemma 2.3 (Additive rewards). *With convolution on nonnegative integral heights,*

$$B_f = B * p_f * B, \quad \sum_h e^{-sh} B_f(h) = b(s, 0)^2 \mathbb{E}_p[f e^{-sH}].$$

These identities hold as identities of nonnegative sums, even when a sum is infinite. For $f = L$, $B_f(h) = M_h$.

Proof. Mark one piece in a bridge and then cut immediately before and after it. The pieces before the mark form an arbitrary bridge, the mark is one irreducible, and the pieces after it form another arbitrary bridge. This is a bijection, with multiplicative weights. Tonelli's theorem gives both formulas. Since length is additive across ports, summing the lengths of all pieces gives the full visited-vertex length. $\square$

2.3 Joint increment estimates

The next statement concerns the joint law of the four coordinates, rather than four independent variables. Its exponent-only consequences are sufficient for the prescribed-endpoint applications.

Proposition 2.4. *Put $\beta = 3/4$ and $\alpha = 9/16$. Under p ,*

$$\begin{aligned} 1 - \mathbb{E}_p e^{-sH} &\asymp s^\beta, & p(H > x) + p(D > x) &\leq Cx^{-\beta}, \\ \mathbb{E}_p[L; H \leq x] &\leq Cx^{7/12}, & p(L > n) &\leq Cn^{-\alpha}, & 1 - \mathbb{E}_p e^{-uL} &\asymp u^\alpha. \end{aligned}$$

The constants are uniform for sufficiently small positive s, u and sufficiently large x, n .

Proof. The first assertion follows directly from Proposition 2.2. Since $1 - e^{-H/x} \geq 1 - e^{-1}$ on $H > x$, it gives the height tail.

We first prove the diameter tail, which is not a consequence of the height tail alone. Set $s = A/x$ with a fixed large A . Give all finite bridges their normalized weight proportional to $\rho^L e^{-sH}$; its normalizer is $b(s, 0)$. The mass of bridges of diameter greater than x is at most

$$Cx^{-1/4} \sum_{h \geq 1} e^{-sh} \leq Cx^{-1/4} s^{-1}.$$

Relative to $b(s, 0) \geq cs^{-3/4}$ this is at most $CA^{-1/4}$. Choose A so that this is at most $1/2$. Put $q = \mathbb{E}_p[e^{-sH}; D > x]$ and $d = 1 - \mathbb{E}_p e^{-sH}$. The relative mass of bridge lists containing at least one such piece is exactly

$$1 - \frac{d}{d+q} = \frac{q}{d+q}.$$

A piece of diameter greater than x forces the whole bridge to have that diameter. Thus $q \leq d \leq C_A x^{-3/4}$. On $H \leq x$, $e^{-sH} \geq e^{-A}$; on $H > x$ use the height tail. This proves $p(D > x) \leq Cx^{-3/4}$.

For the reward bound, retain in Lemma 2.3 only the marked pieces of height at most x and outer bridges of heights at most x . Their total height is at most $3x$, whence

$$\left(\sum_{h \leq x} B_h \right)^2 \mathbb{E}_p[L; H \leq x] \leq \sum_{h \leq 3x} M_h \leq Cx^{25/12}.$$

Since the squared factor on the left is at least $cx^{3/2}$, the reward is at most $Cx^{7/12}$. Taking $x = n^{3/4}$ and splitting according to $H > x$ gives

$$p(L > n) \leq p(H > x) + n^{-1} \mathbb{E}_p[L; H \leq x] \leq Cn^{-9/16}.$$

Integration of this tail yields $\mathbb{E}_p(1 - e^{-uL}) \leq Cu^{9/16}$.

For the reverse bound take $s = R^{-1}$ and $u = R^{-4/3}$. On every height $R \leq h \leq 2R$, Theorem 2.1 retains critical mass at least $cR^{-1/4}$ with $L \geq cR^{4/3}$. Consequently

$$b(s, 0) - b(s, u) \geq cR^{3/4} \geq c'b(s, 0).$$

Writing $d_s = 1 - \mathbb{E}_p e^{-sH}$ and $e_{s,u} = \mathbb{E}_p[e^{-sH}(1 - e^{-uL})]$, the renewal identity makes the relative difference equal to $e_{s,u}/(d_s + e_{s,u})$. The displayed lower bound therefore implies $e_{s,u} \geq cd_s \geq cR^{-3/4}$. Finally $\mathbb{E}_p(1 - e^{-uL}) \geq e_{s,u}$ and $R^{-3/4} = u^{9/16}$. $\square$

Remark 2.5. *If the first-length upper bound and the mass of long bridges have only exponent precision, the same proof gives*

$$\mathbb{E}_p[L; H \leq x] \leq x^{7/12+o(1)}, \quad p(L > n) \leq n^{-9/16+o(1)}, \quad 1 - \mathbb{E}_p e^{-uL} = u^{9/16+o(1)}.$$

The bounded-factor height and diameter tails are unchanged. These are the joint-law inputs used in the prescribed-endpoint arguments of Section 9; no endpoint conditioning is implicit in them.

2.4 Infinite concatenation and spatial growth

We record once the elementary conversion from a Laplace deficit to the growth of independent sums.

Lemma 2.6. *Let $Y_1, Y_2, \dots$ be independent copies of a nonnegative random variable Y , and let $0 < a < 1$. If $\mathbb{P}(Y > t) \leq t^{-a+o(1)}$, then almost surely $\sum_{j \leq k} Y_j \leq k^{1/a+o(1)}$. If also $1 - \mathbb{E}e^{-uY} = u^{a+o(1)}$, then almost surely $\sum_{j \leq k} Y_j = k^{1/a+o(1)}$.*

Proof. Fix $\epsilon > 0$ and let $k = 2^j$. Put $t = k^{1/a+\epsilon}$. The probability of a summand above t is at most $kt^{-a+o(1)}$. Moreover $\mathbb{E}[Y \mathbf{1}_{Y \leq t}] \leq t^{1-a+o(1)}$, by integrating the tail, so Markov's inequality bounds the probability that the truncated sum exceeds t by the same expression. This is a summable geometric sequence in j . For the lower bound put $t = k^{1/a-\epsilon}$ and $u = t^{-1}$. Exponential Markov gives

$$\mathbb{P}\{\sum_{i \leq k} Y_i \leq t\} \leq e(\mathbb{E}e^{-Y/t})^k \leq \exp\{1 - kt^{-a+o(1)}\},$$

also summable. Borel–Cantelli, monotonicity between consecutive dyadic k , and a countable sequence of $\epsilon \downarrow 0$ prove the claims. $\square$

Theorem 2.7 (Infinite spatial law). *Concatenate independent irreducibles of law p , and enumerate the resulting vertices as $\gamma_1, \gamma_2, \dots$. Almost surely,*

$$|\gamma_n - \gamma_1| = n^{3/4+o(1)}, \quad \#\{n : |\gamma_n - \gamma_1| \leq R\} = R^{4/3+o(1)}.$$

Proof. Lemma 2.6 and Proposition 2.4 give

$$\sum_{i \leq k} L_i = k^{16/9+o(1)}, \quad \sum_{i \leq k} H_i = k^{4/3+o(1)}, \quad \sum_{i \leq k} D_i \leq k^{4/3+o(1)}$$

on one event of probability one. If vertex n lies in piece $k+1$, it is above the last completed renewal boundary and within the total diameter of the first $k+1$ pieces. The first of the three bounds implies $k = n^{9/16+o(1)}$: both consecutive partial sums have the same exponent, so an exceptionally long current piece creates no missing interpolation interval. The height lower bound and diameter upper bound give the first conclusion. For each fixed $\epsilon > 0$, all sufficiently large n then obey $n^{3/4-\epsilon} \leq |\gamma_n - \gamma_1| \leq n^{3/4+\epsilon}$. Inverting these two inequalities gives the asserted count of vertices in a ball. $\square$

2.5 Uniform half-plane paths have the same prefix limit

The fugacity cancellation will follow directly from factorization in Section 6. Uniform paths of one exact length require a separate ratio argument. The passage from a two-step counting ratio to the prefix limit adapts the argument of Lawler, Schramm and Werner [13, Appendix] and Dyhr et al. [5, Section 2.3, Proposition 2.9] to honeycomb ports; we prove the required ratio by local hexagon flips. Let a_n count paths of n visited vertices starting inward from our bottom port and otherwise unrestricted in the strict upper half-plane; put $a_0 = 1$.

Theorem 2.8. *One has $a_{n+2}/a_n \rightarrow \rho^{-2}$. The uniform half-plane laws converge on each fixed initial path to the infinite law of Theorem 2.7.*

Proof. First $a_n^{1/n} \rightarrow \rho^{-1}$. The upper bound is the honeycomb connective-constant theorem [4]. For the lower bound, let b_m count strict bridges of length $2m$ over all heights. Concatenation at a prescribed length is injective, so $b_{m+j} \geq b_m b_j$; also $b_1 > 0$. The divergence $\sum_m b_m \rho^{2m} = \sum_h B_h = \infty$ and supermultiplicativity imply the required root bound along even lengths. Length-two bridges fill the bounded remainders after repeating a fixed long block. Extending the terminal port by one vertex gives the odd-length bound.

A path is specified, up to bounded initial choices, by its left/right turn word. Runs have length at most four, since five equal turns close an elementary hexagon. Except for an exponentially negligible fraction at length n , there are at least δn maximal runs of length three, for some fixed $\delta > 0$. Indeed, compositions into parts 1, 2, 4 have growth less than ρ^{-1} because $\rho + \rho^2 + \rho^4 < 1$. Assign a sufficiently small positive weight $v < 1$ to every part of length three. The geometric composition series still has growth at most $\lambda < \rho^{-1}$; the words with at most δn such parts number at most $C(v^{-\delta} \lambda)^n$. Choose δ small enough and use the root bound above.

Such a run follows a four-edge arc of a hexagonal face, with exterior entrance and exit edges. The one omitted face vertex cannot be used internally elsewhere, because two of its neighbors are already saturated. Discard faces incident to either endpoint and the bounded number of runs at the ends of the word. The remaining faces are wholly in the strict half-plane: a honeycomb face intersecting its boundary has at most three vertices strictly above it. Each retained four-edge arc may therefore be replaced by the complementary two-edge arc.

Call a wholly allowed face a slot if it avoids the endpoints, is traversed in one consecutive two- or four-edge arc using exterior entrance and exit edges, and has all its remaining vertices unused. A flip changes the length by two. Color faces with finitely many colors so that faces of one color have disjoint vertex sets. For a fixed color the slot set is unchanged by its flips. Each resulting equivalence class with k slots has $\binom{k}{j}$ paths of lengths $n_0 + 2j$, where j is the number of four-edge arcs.

Put $\theta = \rho^2/(1 + \rho^2)$. For each color and each fixed $\epsilon > 0$, paths of length n for which $k \geq \epsilon n$ but $|j/k - \theta| \geq \epsilon$ have exponentially negligible proportion. To see this, weight every path by ρ^{length} . In each class shift j towards $k\theta$ by $\lfloor \eta n \rfloor$ steps, with a sufficiently small fixed $\eta > 0$, using the same sign for each of the two deviation classes. The binomial successive ratios show that the new weighted class count exceeds the old by e^{cn} . The target lengths are $n \pm 2\lfloor \eta n \rfloor$, whose total critical counts are $e^{o(n)}$ by the root limit; the original denominator is also $e^{o(n)}$. The claim follows. Summing over colors, the fraction of four-edge arcs among all slots therefore tends in probability to θ . Colors with fewer than ϵn slots contribute at most a fixed multiple of ϵn , while the total slot count is at least $\delta n - O(1)$ with probability tending to one.

Double-count the edges representing one flip between lengths n and $n + 2$, weighting an edge by the reciprocal of the larger endpoint slot count. One flip changes that count by at most a fixed constant, since only neighboring faces can change their status. The mean incident weight tends to $1 - \theta$ at length n and to θ at length $n + 2$. The incident sums are at most one, so the negligible exceptional paths do not affect the limits. Hence $a_n(1 - \theta + o(1)) = a_{n+2}(\theta + o(1))$, proving the ratio limit.

Fix an ordered list of k irreducibles, of total length m . Paths which start with this list and then remain strictly above its top boundary have exactly a_{n-m} possible continuations. Here m is even. The probability of this event tends to ρ^m , the product of the specified irreducible masses. For fixed k , the events over all such lists are disjoint and their limiting probabilities sum to one. Truncation to a finite set of lists therefore proves convergence on every fixed initial path. $\square$

3 Conditioning on one exact length

The sum of all bridge masses at lengths near n does not bound the mass at length n . Local renewal estimates belong to classical infinite-mean renewal theory; Garsia and Lamperti and Caravenna and Doney treat regularly varying tails [6, 3]. Our finite inputs give comparable powers, without a regular-variation limit. We therefore prove the local lower bound directly by exponential tilting and Fourier inversion, then apply it to the joint height-length-diameter law. The probability statement holds for every exponent $0 < a < 1$; the restriction $a > 1/2$ enters only when we bound the contribution from small piece counts in the bridge application.

3.1 A local renewal lower bound with bounded-factor tails

Lemma 3.1 (Annular mass and concentration). *Let X be a positive integer-valued random variable of span one, and let $0 < a < 1$. Suppose there are constants $c, C > 0$ such that*

$$\mathbb{P}(X > x) \leq Cx^{-a}, \quad cu^a \leq 1 - \mathbb{E}e^{-uX} \leq Cu^a$$

for all sufficiently large x and all sufficiently small $u > 0$, respectively. Then, for some fixed $M > 1$ and $c_1 > 0$, every sufficiently large x satisfies

$$\mathbb{P}(x/M < X \leq x) \geq c_1 x^{-a}.$$

If $S_k = X_1 + \dots + X_k$ is a sum of independent copies, then, for all $k \geq 1$ and $t \geq 1$,

$$\sup_j \mathbb{P}(S_k = j) \leq C_1 k^{-1/a}, \quad \mathbb{P}(S_k > t) \leq C_1 k t^{-a}.$$

Proof. Evaluate the Laplace deficit at A/x . Its contribution from $X > x$ is at most Cx^{-a} , while its contribution from $X \leq x/M$ is at most

$$\frac{A}{x} \mathbb{E}[X; X \leq x/M] \leq CAM^{a-1}x^{-a}.$$

Choose A large enough that the lower deficit $cA^a x^{-a}$ dominates the first term, and then M large enough to dominate the second. The remaining contribution is at most $\mathbb{P}(x/M < X \leq x)$, proving the claim.

Let $\phi(\theta) = \mathbb{E}e^{i\theta X}$. Use two independent copies in $1 - |\phi(\theta)|^2 = \mathbb{E}[1 - \cos(\theta(X - X'))]$. Keep X' in a fixed bounded set of positive probability and X in the annulus $[\epsilon/(M|\theta|), \epsilon/|\theta|]$, with a fixed small $\epsilon > 0$. For sufficiently small $|\theta|$, the phase is bounded above by a small constant and below in magnitude by a fixed positive constant depending only on ϵ, M . The annular bound therefore gives $|\phi(\theta)| \leq e^{-c|\theta|^a}$. Away from zero, span one gives a uniform bound strictly below one. Fourier inversion yields

$$\sup_j \mathbb{P}(S_k = j) \leq (2\pi)^{-1} \int_{-\pi}^{\pi} |\phi(\theta)|^k d\theta \leq Ck^{-1/a}.$$

Finally, either some $X_i > t$ or $\sum_i X_i \mathbf{1}_{X_i \leq t} > t$. The union bound and Markov's inequality, using $\mathbb{E}[X; X \leq t] \leq Ct^{1-a}$, give the sum-tail estimate. $\square$

Theorem 3.2 (Local renewal mass). *Under the hypotheses of Lemma 3.1, for every sufficiently large integer m ,*

$$V_m := \sum_{k \geq 1} \mathbb{P}(S_k = m) \geq c_2 m^{a-1}.$$

The constants depend only on the law and the constants in the hypotheses. No regular-variation limit of its tail is required.

Proof. For $u > 0$ tilt the law by

$$p_u(j) = \frac{e^{-uj} \mathbb{P}(X = j)}{Z(u)}, \quad Z(u) = \mathbb{E}e^{-uX}.$$

Write μ_u, v_u for its mean and variance. Tail integration gives the upper bounds for its first three moments. The annular mass at scale u^{-1} gives the lower bounds for the first two moments; for the variance, apply the identity $2v_u = \mathbb{E}_u(X - X')^2$, keeping one variable bounded and the other in that annulus. Thus

$$\mu_u \asymp u^{a-1}, \quad v_u \asymp u^{a-2}, \quad \mathbb{E}_u |X - \mu_u|^3 \leq Cu^{a-3}.$$

Let ϕ_u be the tilted characteristic function. The same two-copy argument, now using the annulus of size u^{-1} for $|\theta| \leq u$ and of size $|\theta|^{-1}$ for $u \leq |\theta| \leq \theta_0$, gives uniform bounds

$$|\phi_u(\theta)| \leq \begin{cases} e^{-cv_u \theta^2}, & |\theta| \leq u, \\ e^{-c|\theta|^a}, & u \leq |\theta| \leq \theta_0, \\ 1 - c, & \theta_0 \leq |\theta| \leq \pi. \end{cases} \quad (1)$$

In the first annulus the tilt is bounded below by a positive constant, and the phase is comparable to $|\theta|/u$. In the second it is again bounded below because $u/|\theta| \leq 1$. The final line follows from span one and the total-variation convergence $p_u \rightarrow p$.

Fix a large constant A and put $u = A/m$. Consider integers k with

$$|m - k\mu_u| \leq \frac{1}{10}\sqrt{kv_u}.$$

There are at least $c_A m^a$ such integers for all large m . Indeed their center is $m/\mu_u \asymp_A m^a$ and their interval width is comparable to $\sqrt{(m/\mu_u)v_u}/\mu_u \asymp_A m^a$. On this interval $ku^a \asymp A$, and $\sqrt{kv_u} \asymp_A m$.

We verify a uniform local lower bound under the tilted law. Write $\sigma = \sqrt{kv_u}$. On $|\theta| \leq \omega/\sigma$, Taylor expansion of the centered characteristic function gives its Gaussian approximation with error at most $C_\omega A^{-1/2}$, since

$$\frac{k\mathbb{E}_u|X - \mu_u|^3}{(kv_u)^{3/2}} \leq \frac{C}{\sqrt{ku^a}} \leq \frac{C}{\sqrt{A}}.$$

After the change of variables $t = \sigma\theta$, Fourier inversion over this interval therefore differs by at most $C_\omega A^{-1/2}/\sigma$ from

$$\frac{1}{2\pi\sigma} \int_{-\omega}^{\omega} e^{-t^2/2} e^{-it(m-k\mu_u)/\sigma} dt.$$

The latter has real part at least c/σ , uniformly for the allowed k , once ω is large enough. The portion $\omega/\sigma < |\theta| \leq u$ is at most $Ce^{-c\omega^2}/\sigma$ by the first line of (1). The portion $u < |\theta| \leq \theta_0$ is at most $Cue^{-c'A}$, by substituting $|\theta| = ut$ into the second line; relative to $1/\sigma$ this is at most $C\sqrt{A}e^{-c'A}$. The final portion is exponentially small in k . Choose first ω and then A large enough. We obtain

$$\mathbb{P}_u(S_k = m) \geq c_A/m.$$

Removing the tilt multiplies this by $e^{um}Z(u)^k$. Since $um = A$ and $k(1 - Z(u)) = O(A)$, this factor is bounded below by a positive constant depending only on A . Summing over the $c_A m^a$ choices of k proves the theorem. $\square$

3.2 Uniform bridges at every compatible length

Theorem 3.3. *Choose a strict bridge uniformly from all bridges with n visited vertices, rooted at a fixed bottom port, with terminal height and port free. For every $\xi > 0$, along every sufficiently large even integer n ,*

$$\mathbb{P}_n\{n^{3/4-\xi} \leq A \leq D \leq n^{3/4+\xi}\} \longrightarrow 1.$$

Here A is the Euclidean distance between the ports and D the diameter of the visited trace together with its ports. Replacing either by its vertex-endpoint convention changes it by a bounded amount.

Proof. Port bridge lengths are even: their first and last triangles have opposite types. The irreducible law admits two- and four-vertex zigzags in one row, so $X = L/2$ has atoms at 1 and 2 and has span one. Proposition 2.4 verifies the hypotheses of Theorem 3.2 with $a = \alpha = 9/16$. Thus, writing $m = n/2$, the total critical bridge mass at that length is

$$V_m = \sum_{k \geq 1} \mathbb{P}_p\{\sum_{i \leq k} L_i = 2m\} \geq cm^{\alpha-1}.$$

All bridges of this length have the same critical weight, so normalization by V_m gives the uniform law.

We first restrict the number k of pieces. For $k \geq 2$ split the list into two comparable halves. If the total X -sum is m , one half is at least $m/2$. Lemma 3.1 bounds that probability by $Ckm^{-\alpha}$; the other independent half pays at most $Ck^{-1/\alpha}$ for the required residual length. Therefore, for any fixed $\delta > 0$,

$$\sum_{2 \leq k \leq m^{\alpha-\delta}} \mathbb{P}(S_k = m) \leq Cm^{-\alpha} \sum_{k \leq m^{\alpha-\delta}} k^{1-1/\alpha} \leq Cm^{\alpha-1-\delta(2-1/\alpha)}.$$

The exponent $2 - 1/\alpha = 2/9$ is positive. The $k = 1$ term is at most $Cm^{-\alpha} = o(m^{\alpha-1})$. At the other end, exponential Markov at discount $1/m$ gives

$$\mathbb{P}(S_k = m) \leq e \exp(-ckm^{-\alpha}),$$

so the sum over $k > m^{\alpha+\delta}$ is negligible. Here $k \leq m$ because $X \geq 1$.

For the remaining k , total height below $t = m^{3/4-\xi}$ costs at most

$$\mathbb{P}\{\sum H_i \leq t\} \leq e \exp(-ckt^{-\beta}) \leq e \exp(-cm^{\beta\xi-\delta}),$$

which is negligible if $\delta < \beta\xi$. This step does not impose a length condition, so it is a valid upper bound for the exceptional length mass as well.

For the diameter upper bound put $r = m^{3/4+\xi}$. If $\sum_{i \leq k} D_i > r$, one of the two halves has diameter sum exceeding $r/2$. The diameter tail and the same truncation argument as in Lemma 3.1 bound this by $Ckr^{-\beta}$. Pay for the exact length using the independent other half. Summing over $k \leq m^{\alpha+\delta}$ gives

$$Cr^{-\beta} \sum_{k \leq m^{\alpha+\delta}} k^{1-1/\alpha} \leq Cm^{\alpha-1-\beta\xi+\delta(2-1/\alpha)} = o(V_m)$$

for sufficiently small δ . Independence was used between disjoint sets of pieces, never between the length and diameter of a piece. Finally $d_0 \sum H_i \leq A \leq D \leq \sum D_i + O(1)$. Apply the argument with slightly smaller tolerances to absorb fixed geometric constants. $\square$

3.3 Critical bridges at a fixed height

This conditioning requires a different denominator, B_h , but only a simpler part of the preceding local argument.

Theorem 3.4. *Under critical weights normalized on strict bridges of exact height h with free terminal port,*

$$L = h^{4/3+o(1)}, \quad D = h^{1+o(1)}$$

in probability as every integer $h \rightarrow \infty$. The same holds after restricting to any event whose critical mass is at least a fixed positive fraction of B_h .

Proof. The mean length is $M_h/B_h \leq Ch^{4/3}$, proving the upper length tail at every positive exponent slack. The height variable has Laplace deficit comparable to s^β and tail $O(x^{-\beta})$. Its support is an unbounded subset of the integers. The modulus-one characters therefore form a finite set on the Fourier torus; the same near-origin estimate translates to each of them. Lemma 3.1's Fourier proof gives $\sup_j \mathbb{P}(\sum_{i \leq k} H_i = j) \leq Ck^{-1/\beta}$. Splitting into halves as above shows

$$\sum_{k \leq h^{\beta-\delta}} \mathbb{P}\{\sum_{i \leq k} H_i = h\} \leq Ch^{-\beta} \left(1 + \sum_{2 \leq k \leq h^{\beta-\delta}} k^{1-1/\beta}\right) = o(h^{\beta-1}) = o(B_h).$$

For $k > h^{\beta-\delta}$, length below $h^{4/3-\epsilon}$ costs at most $\exp\{1 - ch^{\alpha\epsilon-\delta}\}$ by the length deficit. There are at most h possible counts, so choosing $\delta < \alpha\epsilon$ proves the lower length tail. The diameter is at least $d_0 h$; its upper tail at $h^{1+\epsilon}$ has relative mass at most $Ch^{-\epsilon/4}$ by Theorem 2.1. A restriction retaining a fixed fraction of the denominator multiplies all these vanishing upper bounds by only a constant. $\square$

4 Conditioning on length from logarithmic finite estimates

We now obtain exact-length bridge laws from weaker finite length information: a logarithmic window and a first moment restricted to bridges whose width is at most a fixed multiple of their height. Fourier estimates first produce a whole interval of renewal masses with a common lower bound. Fresh independent blocks then move the remaining target length into that interval at subpolynomial cost. This route also gives the neighboring-length ratio and local convergence of the finite bridge laws. The later geometric estimates and unrestricted changes of law use the finite inputs stated here, but do not enter this conditioning argument.

Throughout this section the critical weight is ρ^L , where L counts visited centers and boundary ports carry no weight. The strict bridge starts at one fixed port and its terminal port is free. Height H is a positive integer; one band has Euclidean height $\sqrt{3}/2$. Its transverse displacement is denoted by X , with $2X$ integral on the row-staggered lattice. Write D for its diameter. The irreducible law p and its independent concatenation are those of Proposition 2.2. Dependence between H, X, L, D within one irreducible is allowed. Put

$$a = \frac{3}{4}, \quad \vartheta = 1 - a = \frac{1}{4}, \quad d_* = \frac{4}{3}, \quad \alpha = \frac{a}{d_*} = \frac{9}{16}.$$

We use ϑ for the boundary exponent to keep it distinct from the height-tail exponent a .

4.1 The finite estimates and a logarithmic renewal denominator

The finite input is supplied by the calibrated cylinder analysis [18, Theorems 1.1–1.2 and Proposition 11.2]. We state its precise form here, so that the conditioning argument can be read without revisiting that representation.

Proposition 4.1 (Finite calibrated input). *Let B_h be the critical mass of strict bridges of height h from one fixed port, and let $B_0 = 1$. Then $B_h \asymp (1 + h)^{-1/4}$. For sufficiently large fixed T , bridges confined to transverse width Th have mass at least $c_T h^{-1/4}$ and first-length mass at most $C_T h^{13/12}$. The arch-plus-bridge mass in a height- h strip with diameter larger than $r \geq Ch$ is at most $Ch^{-1/4} e^{-cr/h}$. Since a self-avoiding path of diameter r has at most Cr^2 centers, this estimate also controls every fixed length moment in the tail. In particular the mass at $D > h \log^2 h$, including any fixed length power, is smaller than every inverse power of h .*

The half-plane arch mass at prescribed horizontal gap g satisfies $A_g \asymp g^{-5/4}$; its lower bound remains valid with diameter at most Cg for a sufficiently large fixed C . The normalized irreducible law satisfies

$$1 - \mathbb{E}_p e^{-tH} \asymp t^{3/4} \quad (0 < t \leq 1), \quad p(D > r) \leq Cr^{-3/4} \quad (r \geq 1).$$

For h large, put $R_0 = h(\log h)^{1/64}$ and

$$l_- = h^{4/3}(\log h)^{-1/8}, \quad l_+ = h^{4/3}(\log h)^{1/2}.$$

For a fixed finite C ,

$$\sum_{j \leq 2R_0} B_j \{l_- \leq L \leq 2l_+\} \geq h^{3/4} (\log h)^{-C}. \quad (2)$$

The estimate sums over heights and terminal ports; it does not prescribe either one of them.

Lemma 4.2 (Length deficit). *There is a finite P such that, for all sufficiently small $t > 0$,*

$$ct^\alpha (\log(1/t))^{-P} \leq q_L(t) := 1 - \mathbb{E}_p e^{-tL} \leq Ct^\alpha.$$

Proof. Take $t = h^{-4/3}$. On the length window in (2), $1 - e^{-tL} \geq c(\log h)^{-1/8}$. Summing this damping loss over the $O(h(\log h)^{1/64})$ allowed heights shows that some height $1 \leq h_* \leq 2R_0$ has loss at least $h^{-1/4} (\log h)^{-C'}$. For a bridge with irreducible lengths L_i , $1 - e^{-\sum_i L_i} \leq \sum_i (1 - e^{-L_i})$. The marked reward identity of Lemma 2.3 therefore bounds this loss above by

$$\sum_{j \leq h_*} \mathbb{E}_p [1 - e^{-tL}; H = j] \sum_{b+d=h_*-j} B_b B_d \leq Ch_*^{1/2} q_L(t).$$

Here $B_j \asymp (1+j)^{-1/4}$ gives the convolution bound. Since $h_* \leq 2h(\log h)^{1/64}$, we obtain

$$q_L(h^{-4/3}) \geq ch^{-3/4} (\log h)^{-P}$$

for some finite P , proving the lower estimate.

For the upper estimate fix a sufficiently large confinement constant T in Proposition 4.1. At each height $d \leq h$, its confined bridges have mass at least $c_T d^{-1/4}$ and first-length mass at most $C_T d^{13/12}$. Choosing a large fixed C' and restricting to $L \leq C' h^{4/3}$ preserves at least half the mass: the discarded fraction is at most a constant times $d^{4/3} / (C' h^{4/3})$. On the retained paths $e^{-tL} \geq e^{-C'}$. Summing over $d \leq h$ gives damped bridge mass at least $ch^{3/4}$. By renewal the total damped bridge mass over all heights is $1/q_L(t)$, so $q_L(h^{-4/3}) \leq Ch^{-3/4}$. $\square$

The upper deficit also gives the length tail and truncated mean:

$$p(L > x) \leq \frac{q_L(1/x)}{1 - e^{-1}} \leq Cx^{-\alpha}, \quad \mathbb{E}_p[L; L \leq x] \leq Cx^{1-\alpha} \quad (x \geq 1).$$

Lemma 2.6 therefore gives both bounds for length and height, and the upper bound for diameter. Writing $S_Y(k) = \sum_{i=1}^k Y_i$ for each coordinate, we obtain almost surely

$$S_H(k) = k^{4/3+o(1)}, \quad S_L(k) = k^{16/9+o(1)}, \quad S_D(k) \leq k^{4/3+o(1)}.$$

Later slabs cannot re-enter below a completed height. The interpolation between consecutive complete pieces in Theorem 2.7 therefore proves the same infinite-path conclusions from these weaker inputs: mass $r^{4/3+o(1)}$ in a radius- r ball and distance $n^{3/4+o(1)}$ at center time n .

Theorem 4.3 (Exact compatible lengths from a logarithmic window). *Let $u_n = p(\exists k : S_L(k) = n)$. For every sufficiently large even n ,*

$$u_n \geq n^{-7/16-o(1)}, \quad \frac{u_{n+2}}{u_n} \longrightarrow 1.$$

Conditioning the independent sequence on hitting n gives the uniform strict bridge of length n , with height and terminal port free. Under this law the height, endpoint distance, and maximum radius are $n^{3/4+o(1)}$ in probability. The finite bridge laws converge locally to the independent-irreducible law.

Proof. A block law from the finite window. The height deficit gives

$$p\{S_H(k) \leq 2R_0\} \leq e \exp(-ckR_0^{-a}).$$

Consequently, the contribution of counts $k > h^a \log^2 h$ to (2) is negligible. Dividing the remaining window mass by this number of counts yields a block whose probability of lying in the length window is bounded below by a negative power of $\log h$. Since $l_-/l_+ = (\log h)^{-5/8}$, fix any $b \in (5/8, 1)$, say $b = 3/4$. For each sufficiently large r , choose h so that l_+ is comparable to r and $2l_+ \leq r/2$. Then there is a deterministic integer $k(r)$ such that

$$p\{r(\log r)^{-b} \leq S_L(k(r)) \leq r/2\} \geq (\log r)^{-C}. \quad (3)$$

Fix one such choice for every r . We will use these blocks to advance through a large remaining gap without passing its target.

A whole interval of local renewal mass. Port lengths are even, and two- and four-center one-row irreducibles show that $L/2$ has span one. Let ϕ be its characteristic function. The length deficit with logarithmic loss implies, near zero,

$$1 - |\phi(\theta)|^2 \geq c|\theta|^\alpha (\log(1/|\theta|))^{-P'}.$$

To verify this estimate, set $T = 1/|\theta|$ and $z = T/(\log T)^K$. The deficit bounds give $\mathbb{E}_p \min(1, L/z) \geq cz^{-\alpha}(\log T)^{-P}$. Lengths above T contribute at most $CT^{-\alpha}$; those below $z/(\log T)^{K'}$ contribute at most $Cz^{-\alpha}(\log T)^{-K'(1-\alpha)}$, by the truncated mean bound. Choose K and then K' large enough that these contributions leave a fixed fraction of the lower bound. On the remaining range the phase $\theta(L-2)/2$ has magnitude between a negative power of $\log T$ and a fixed constant less than one. In $1 - |\phi(\theta)|^2 = \mathbb{E}_p[1 - \cos(\theta(L-L')/2)]$, keep this range and the atom $L' = 2$. The cosine bound proves the claim. Away from zero, the atoms at $L/2 = 1, 2$ give $|\phi| < 1$.

Fourier inversion now gives, uniformly in the target and for every $k \geq 1$,

$$\sup_j p(S_L(k) = j) \leq C(1+k)^{-1/\alpha} \log^C(2+k), \quad (4)$$

$$\sup_{j \text{ even}} |p(S_L(k) = j+2) - p(S_L(k) = j)| \leq C(1+k)^{-2/\alpha} \log^C(2+k). \quad (5)$$

Indeed split the Fourier integral at $k^{-1/\alpha}(\log k)^M$ for large k . The inner interval has that length; for the difference its integrand has the additional factor $|e^{i\theta} - 1| \leq |\theta|$. For sufficiently large fixed M , the remaining integral near zero is smaller than any prescribed power of k^{-1} , and the part away from zero is exponentially small. Increasing the constants handles bounded k .

For a small fixed $c_0 > 0$, put

$$f_n(j) = \sum_{c_0 n^\alpha \leq k \leq 2c_0 n^\alpha} p(S_L(k) = j).$$

The upper deficit bound gives

$$p\{S_L(k) > n/3\} \leq \frac{1 - (1 - q_L(1/n))^k}{1 - e^{-1/3}} \leq Ckn^{-\alpha} \leq \frac{1}{2}$$

when c_0 is sufficiently small. Thus $\sum_{0 \leq j \leq n/3} f_n(j) \geq cn^\alpha$, and some even $j_0 \leq n/3$ satisfies $f_n(j_0) \geq cn^{\alpha-1}$. By (5),

$$|f_n(j+2) - f_n(j)| \leq Cn^{\alpha-2}(\log n)^{C_1}.$$

This preserves half the lower bound on an even interval $[m, m+w] = [j_0, j_0+w] \subset [0, n/2]$, where $w \geq c'n(\log n)^{-C_1}$. Positive increments make the events $S_L(k) = j$ disjoint in k on one independent sequence. Hence $u_j \geq f_n(j) \geq cn^{\alpha-1}$ throughout this interval.

Landing before the final renewal hit. Aim for a partial sum in $[n - m - w, n - m]$. At a current sum $x < n - m - w$, put $r = n - m - x$ and expose the next $k(r)$ fresh pieces. On the event in (3), the new remaining gap is between $r/2$ and $r - r(\log r)^{-b}$. In particular the target $n - m$ is not passed. Stop as soon as the gap is at most w , so the partial sum lies in the required landing interval.

Before stopping every gap lies in $[w, n]$, and its logarithm is comparable to $\log n$. At most $J = O((\log n)^b(1 + \log \log n))$ successful stages suffice, since each reduces the gap by a fraction at least $(\log n)^{-b}$ and $n/w \leq C(\log n)^{C_1}$. The choice $k(r)$ depends only on the exposed past, so every block end is a stopping time and the unused pieces remain independent with law p . At each unfinished stage the conditional success probability is at least $(\log n)^{-C}$. Declaring all later stages successful once landing has occurred gives

$$p\{\text{all required stages succeed}\} \geq \exp\{-C(\log n)^b(1 + \log \log n)^2\} = n^{-o(1)}.$$

The fixed inequality $b < 1$ is essential here. At the landing time the remaining length to n belongs to $[m, m + w]$. The fresh continuation therefore hits n with probability at least $cn^{\alpha-1}$, and

$$u_n \geq n^{\alpha-1-o(1)}.$$

Conditioned count and spatial bounds. Every length- n bridge has critical weight ρ^n , so conditioning the sequence on hitting n gives exactly the uniform strict-bridge law. We first show that its number of pieces lies between $n^{\alpha-\epsilon}$ and $n^{\alpha+\epsilon}$ with high probability, for each fixed $0 < \epsilon < \min(\alpha, 1 - \alpha)$; larger tolerances follow at once.

The tail and truncated mean give $p\{S_L(j) > t\} \leq Cjt^{-\alpha}$ by the union bound and Markov's inequality. For $k \geq 2$, split the pieces into two comparable groups. If $S_L(k) = n$, one group has length at least $n/2$, at cost $Ckn^{-\alpha}$; the independent other group pays at most $Ck^{-1/\alpha} \log^C(2+k)$ for the residual length, by (4). Consequently

$$\sum_{2 \leq k \leq n^{\alpha-\epsilon}} p(S_L(k) = n) \leq Cn^{-\alpha} \sum_{k \leq n^{\alpha-\epsilon}} k^{1-1/\alpha} \log^C(2+k) = n^{\alpha-1-\epsilon(2-1/\alpha)+o(1)} = o(u_n).$$

Here $2 - 1/\alpha = 2/9 > 0$. The one-piece term is at most $Cn^{-\alpha} = o(u_n)$ because $2\alpha - 1 = 1/8 > 0$. For the upper cutoff, the lower deficit gives

$$p(S_L(k) = n) \leq e \exp\{-ckn^{-\alpha}(\log n)^{-P}\}.$$

Summing over $k > n^{\alpha+\epsilon}$ is negligible; there are at most n possible counts since lengths are positive.

It suffices to take $0 < \xi < 3/4$. Choose $\epsilon > 0$ sufficiently small. At the retained counts, the height deficit gives

$$p\{S_H(k) < n^{3/4-\xi}\} \leq \exp\{1 - cn^{a\xi-\epsilon}\},$$

which remains negligible after summing counts and dividing by u_n if $\epsilon < a\xi$. For the upper spatial bound put $r = n^{3/4+\xi}$. The diameter tail similarly gives $p\{S_D(j) > r\} \leq Cjr^{-a}$. If $S_D(k) > r$, one of the two groups has diameter sum greater than $r/2$. Test this event in that group and the exact residual length in the other. By independence between groups and (4),

$$\begin{aligned} \sum_{n^{\alpha-\epsilon} \leq k \leq n^{\alpha+\epsilon}} p\{S_D(k) > r, S_L(k) = n\} &\leq Cr^{-a} \sum_{k \leq n^{\alpha+\epsilon}} k^{1-1/\alpha} \log^C(2+k) \\ &= n^{\alpha-1-a\xi+\epsilon(2-1/\alpha)+o(1)} = o(u_n) \end{aligned}$$

for sufficiently small ϵ . Neither split assumes independence between the coordinates of one piece. The total height bounds endpoint distance below, and $S_D(k)$ bounds the whole trace above. Absorbing fixed geometric constants by slightly smaller tolerances proves the spatial claims.

The neighboring-length ratio and local limit. In $u_{n+2} - u_n$, counts outside $[n^{\alpha-\epsilon}, n^{\alpha+\epsilon}]$ contribute $o(u_n)$ by the same absolute bounds for both target lengths. On the retained counts, (5) bounds the sum of successive differences by

$$C \sum_{k \geq n^{\alpha-\epsilon}} k^{-2/\alpha} \log^C(2+k) = n^{(\alpha-\epsilon)(1-2/\alpha)+o(1)} = o(u_n)$$

for small ϵ . Thus $u_{n+2}/u_n \rightarrow 1$ on even lengths. For a fixed tuple of k irreducibles of total length l , its initial-tuple probability under the length- n bridge law is $\rho^l u_{n-l}/u_n$ for $n > l$, which tends to ρ^l . For fixed k these limiting probabilities sum to one. A sufficiently large k determines any prescribed finite initial path, or the path inside a fixed ball, since all later pieces lie above the completed height. Finite-set truncation of the tuples proves the asserted local convergence. $\square$

Endpoint conventions and the half-plane limit. The half-plane paths considered here start at the inward center adjacent to a fixed wall port and otherwise stay strictly above that wall; their terminal center is free. They are precisely the paths in Theorem 2.8. Its slot-flip ratio and the disjoint-prefix factorization therefore prove that their uniform laws converge locally to our independent-irreducible law. The proof applies in center length; a bounded change to edge length does not change its conclusion.

The exact-length conclusion also transfers to uniform n -edge vertex bridges from a fixed vertex, with endpoints the height extrema (ties allowed, or take a strict initial minimum). Extend each to a strict midport bridge by these same bounded endpoint extensions, of length $m = n+1, n+2$ or $n+3$, with bounded multiplicity and start-port choices. At even m the exact-length estimates above give mass of bridges violating either spatial cutoff with fixed slack at most $m^{\alpha-1-c(\xi)}$, $c(\xi) > 0$, $\alpha = 9/16$ (both the diameter and small-piece-count bounds have strict power savings). Constant endpoint adjustments preserve this with slightly reduced slack. Conversely prepend to a strict bridge one down center below its start port, or for prescribed up initial type also one lower adjacent up center before this down center, as vertex root; append one outward up center through the top port or not. After translation this realizes $n = m + \varepsilon_- + \varepsilon_+$ with $\varepsilon_- \in \{0, 1\}$ determined by initial type and $\varepsilon_+ \in \{0, 1\}$ freely chosen so every large n uses even m ; multiplicity and weight losses are bounded. (If both extrema must be strict, append one or two outward centers instead.) The total vertex-bridge mass is thus $\geq n^{\alpha-1-o(1)}$, proving the same typical exponent $3/4$ there.

5 Geometric estimates before a change of law

This section has two related purposes. We first derive irreducible estimates and the laws conditioned on bridge height or on two boundary ports, using the finite spatial and polygon estimates below together with renewal factorization. This route does not use the first-length estimate or the mass of long bridges from Theorem 2.1; it therefore also explains what remains available without those stronger inputs.

We then estimate the absolute critical mass of unrestricted paths, before any probability normalization. Splitting such a path at a turn produces two bridges that must avoid each other. We estimate this avoidance under the independent irreducible law while retaining separate, unused increments for matching a height or testing a length. The resulting turning-chain estimate gives the total-count and rapid-travel bounds used in the thermal and prescribed-endpoint laws. Its two-bridge form also allows arbitrary terminal pieces to be attached without consuming the height variable needed to compare their final endpoints.

Throughout this section a port path has weight ρ^L , where L is its number of visited vertices. A port itself has no weight. Heights are integers measured in row spacings. Write $B_h(x)$ for the mass of strict bridges from a fixed bottom port to displacement (h, x) , and $B_h = \sum_x B_h(x)$. The horizontal coordinate uses the fixed port lattice; it may be staggered between successive rows. For a bridge or an irreducible, H is its height, X its signed horizontal displacement, and D its Euclidean diameter. For a nonnegative function f , write $p(f) = \int f dp$; on events p denotes probability. The law p and the factorization into independent irreducibles are those of Proposition 2.2. We use $q(t) = p(1 - e^{-L/t})$.

5.1 The finite geometric inputs

Here are the finite estimates used in this section. They are the slab, arch, and polygon theorems of *Radial transfer estimates and polygon length laws for honeycomb walks*. We state their full probabilistic content so that the subsequent changes of law can be read independently of their analytic proofs [16, Theorems 1.1, 12.1, 12.2 and Proposition 3.1].

1. For $h \geq 1$ and every terminal port,

$$ch^{-1/4} \leq B_h \leq Ch^{-1/4}, \quad B_h(x) \leq Ch^{-5/4}, \quad B_h[D > Mh] \leq C(Mh)^{-1/4} \quad (M \geq 1). \quad (6)$$

Let $g : [0, 1] \rightarrow \mathbb{R}$ be continuous and piecewise linear, with $g(0) = 0$, and fix $\epsilon > 0$. The raw bridge mass restricted to

$$\{(x, y) : 0 \leq y \leq h, |x - hg(y/h)| \leq \epsilon h\}$$

is at least $c(g, \epsilon)h^{-1/4}$ for all sufficiently large admissible h , in the cut convention of [16, Theorem 12.2]. The lower port is fixed and the upper port is free. For a compact family of such graphs in the uniform topology, both the constant and the lower threshold can be chosen uniformly. In particular this is a fixed-proportion corridor estimate, without a power loss.

2. The mass a_b of upper-half-plane arches between ports separated by b along their boundary satisfies $a_b \asymp b^{-5/4}$. The same lower bound holds after restricting the diameter to Cb , for one fixed C . Also $\sum_{b \geq 1} a_b < \infty$.
3. Let $\mathcal{P}$ be simple unoriented polygons modulo translations, with weight $w(P) = \rho^{|P|}$. As $r \rightarrow \infty$,

$$\sum_{D(P) \geq r} w(P) = r^{-2+o(1)}, \quad \sum_{D(P) \geq r} |P|w(P) = r^{-2/3+o(1)}, \quad \sum_{D(P) \leq r} |P|^2 w(P) \leq r^{2/3+o(1)}. \quad (7)$$

4. Put $e = (1, 0)$ and $a = (1/2, \sqrt{3}/2)$, and write $P_{ij} = ie + ja$ and $R_{ij} = P_{ij} + (e + a)/3$ for the two cell types. Use the same definitions after a lattice symmetry. On the cylinder obtained by quotienting by Ne , count polygons only modulo axial translation by a , and let $S(P)$ be the span of their cell-row indices j . For every fixed $\epsilon > 0$, $q \geq 0$, and $M > 0$,

$$\sum_{P: S(P) \geq N^{1+\epsilon}} (1 + |P|)^q w(P) = O(N^{-M}) \quad (8)$$

for both contractible and winding polygons. In particular a marked polygon carries its actual mark multiplicity, which can be absorbed by the length factor.

The companion locators are Theorem 12.2 for the slab and corridor estimates, Theorem 12.1 for boundary arches, Theorem 1.1 for the polygon tails and second moment, and Proposition 3.1 for thin-cylinder span. Every use below concerns these positive masses; no conditional mean or prescribed-terminal lower bound is implicit in (6).

5.2 Tails and concentration of an irreducible

The slab mass determines the height scale of one increment. Its pointwise bound and corridor tightness also control horizontal movement. Both coordinates will be needed when two separately sampled bridges are joined.

Lemma 5.1. *There are constants $c, C > 0$ such that, for $r \geq 1$,*

$$p(D > r) \leq Cr^{-3/4}, \quad (9)$$

$$p(cr < H < Cr) \geq cr^{-3/4}, \quad (10)$$

$$p(cr < |X| < Cr, H < Cr) \geq cr^{-3/4}. \quad (11)$$

Consequently, for $Y = H, D$, or $|X|$, $p(Y \wedge r) \leq Cr^{1/4}$. Also $1 - p(e^{-H/r}) \asymp r^{-3/4}$.

Proof. Put $\vartheta_r = 1 - p(e^{-H/r})$. The renewal identity gives

$$\vartheta_r^{-1} = \sum_{h \geq 0} e^{-h/r} B_h \asymp r^{3/4}.$$

After normalization, this bridge sum consists of a geometric number of increments with one-increment law proportional to $e^{-H/r}p$. Its number of increments has mean of order $r^{3/4}$. The inequality $1 - e^{-H/r} \geq (1 - e^{-1})\mathbf{1}_{\{H \geq r\}}$ first gives the height tail.

The diameter of the whole bridge divided by r is tight in this normalized sum, by (6), summation over heights, and the exponentially small contribution of heights much larger than r . An increment of diameter greater than Mr forces the same diameter event for the whole bridge. The elementary formula for a geometrically stopped list therefore gives $p(e^{-H/r}; D > Mr) \leq C_M \vartheta_r$ for one sufficiently large fixed M . On $H \leq r$ the damping is bounded below; the already proved height tail handles $H > r$. Rescaling by the fixed M proves (9), and integration of that tail proves the truncated first-moment bounds.

The lower bound for ϑ_r cannot be supplied by very small or very large heights. Indeed their contributions to $p(1 - e^{-H/r})$ are at most $C\delta^{1/4}r^{-3/4}$ for $H \leq \delta r$ and $CM^{-3/4}r^{-3/4}$ for $H > Mr$. Choosing δ small and M large proves (10).

In the normalized bridge sum, heights in $[r, 2r]$ have probability bounded below. At any such height, the pointwise bound in (6) shows that terminal positions in an interval of width δr have probability at most $C\delta$. Thus the total horizontal displacement has magnitude at least cr with probability bounded below. The expected sum of $\min(|X_i|, \delta r)$ over the geometric list is at most $C\delta^{1/4}r$; increments of diameter greater than Mr have arbitrarily small probability when M is large. Hence, with probability bounded below, the list contains an increment with $c'r < |X_i| < C'r$ and $H_i < C'r$. Its expected number is an upper bound for that probability. The geometric mean is of order $r^{3/4}$ and the tilted probability of this event is at most a constant times its p probability. This proves (11). $\square$

Lemma 5.2 (Spatial atoms). *For $k \geq 1$, sums of k independent irreducibles satisfy*

$$\sup_y p^{\otimes k} \{ \sum_{i=1}^k (H_i, X_i) = y \} \leq Ck^{-8/3}, \quad \sup_h p^{\otimes k} \{ \sum_{i=1}^k H_i = h \} \leq Ck^{-4/3}.$$

Let Γ be the lattice generated by the possible (H, X) increments, and let Γ_1 be the subgroup generated by their differences. On the dual torus of Γ , the characteristic function φ has modulus one only at the finitely many characters annihilating Γ_1 . In a neighborhood of each such character,

$$|\varphi(\theta)| \leq 1 - c \operatorname{dist}(\theta, \Gamma_1^\perp)^{3/4}.$$

Proof. Use independent increments Y, Y' in $1 - |\varphi(\theta)|^2 = p \otimes p[1 - \cos(\theta \cdot (Y - Y'))]$. Keep Y' in a fixed bounded set of positive probability. At scale $r = \epsilon/|\theta|$, Lemma 5.1 supplies mass $cr^{-3/4}$ with the coordinate whose coefficient has larger absolute value between fixed positive multiples of r in magnitude. For the height shell, discard $D > Mr$ with fixed M sufficiently large: the diameter tail removes at most $CM^{-3/4}r^{-3/4}$, leaving a fixed fraction of that shell mass. Both coordinates are now bounded by a fixed multiple of r ; the horizontal shell already has this property. Reflection replaces X by $-X$ without changing H . One of these two signs avoids cancellation between the two coordinates in the scalar product. Choose the fixed $\epsilon > 0$ small enough to avoid wrapping the resulting phase around 2π . The integrand is then bounded below by a positive constant on a fixed fraction of this mass. This gives the claimed loss near zero. The same argument using only H proves the one-dimensional loss.

The support differences span two dimensions: otherwise reflection would first force an affine relation involving only H , and (10) rules that out. A finite subset of the differences therefore generates a rank-two sublattice. Thus Γ_1 has finite index in Γ . Equality in the triangle inequality for $p(e^{i\theta \cdot Y})$ occurs exactly when that character is constant on the support, which gives precisely the stated finite set. At each of these characters its modulus is a translate of the modulus near zero. Away from their neighborhoods compactness gives a bound strictly below one. Fourier inversion, with $\int_{\mathbb{R}^d} e^{-ck|\theta|^{3/4}} d\theta = O(k^{-4d/3})$, proves both atom estimates. No span-one assumption in two dimensions is being made. $\square$

5.3 Closing two bridges into a polygon

Let D_h^* be the mass of ordered disjoint pairs of bridges from two fixed adjacent bottom ports to arbitrary ordered distinct top ports at height h . Let D_h be its restriction to adjacent top ports, summed over their common translate. These are pair masses, rather than probability laws.

Lemma 5.3. *For each fixed C , as $r \rightarrow \infty$,*

$$\sum_{r \leq h \leq 2r} D_h^*[D(\gamma_1 \cup \gamma_2) \leq Cr] \leq r^{-1/4+o(1)}, \quad (12)$$

$$\sum_{r \leq h \leq 2r} B_h[L; D \leq Cr] \leq r^{25/12+o(1)}. \quad (13)$$

In particular

$$p(L; D \leq r) \leq r^{7/12+o(1)}, \quad q(t) \leq t^{-9/16+o(1)}. \quad (14)$$

Proof. For a polygon of diameter in $[cr, C'r]$, let K count horizontal row cuts crossed exactly twice. Mark s of these cuts, allowing repeated marks, and order their distinct levels. Outside the extreme cuts there are two arches. Between successive cuts there are two bridges. Anchor the first cut and its left crossing port, sum its arch gap using $\sum_b a_b < \infty$, and drop all further disjointness restrictions. Each internal gap contributes at most $\sum_{t \leq C''r} B_t^2 \leq C\sqrt{r}$. Repeated marks and the choices of the extreme cuts contribute a fixed polynomial in r whose degree does not depend on s . Consequently

$$\sum_{D(P) \leq C'r} w(P)K(P)^s \leq C_s r^{C_0+s/2}.$$

Apply Hölder with arbitrarily large fixed s and the first two bounds in (7). For the length-weighted measure use also $|P| \leq Cr^2$ in the high-moment bound. We obtain

$$\sum_{cr \leq D(P) \leq C'r} w(P)K(P) \leq r^{-3/2+o(1)}, \quad \sum_{cr \leq D(P) \leq C'r} |P|w(P)K(P)^2 \leq r^{1/3+o(1)}.$$

Close a pair counted by D_h^* with a fixed bounded lower cap and an independent upper arch. The latter has mass at least $cr^{-5/4}$ and diameter at most $C'r$; the two exterior half-planes keep the closures disjoint from the bridges. The lower cut is at bounded distance from the bottom of the resulting polygon. An output polygon has at most CK preimages, since choosing its upper cut recovers both bridges and both caps. This proves (12).

For (13), place a second bridge in a disjoint box to the right. Its starting port has order r choices and its confined mass is at least $cr^{-1/4}$. Close with exterior arches above and below, each of mass at least $cr^{-5/4}$. The added mass per input is at least $cr^{-7/4}$. Both separating cuts recover the construction with multiplicity at most CK^2 , and the input length is bounded by the output polygon length. The preceding weighted bound gives $r^{7/4+1/3+o(1)} = r^{25/12+o(1)}$.

Finally surround an irreducible of diameter at most r by confined bridges whose heights each range over $[r, 2r]$. Each side has mass at least $cr^{3/4}$. In the output bridge the sum of the lengths of all possible marked irreducibles is at most its full length. Applying (13) on a bounded number of height bins therefore gives $p(L; D \leq r) \leq r^{25/12-3/2+o(1)}$. In $q(t) \leq t^{-1}p(L; D \leq r) + p(D > r)$ take $r = t^{3/4}$. $\square$

5.4 Avoidance of adjacent initial pieces

Two independent increment lists begun at adjacent ports need not stay disjoint. Define E_k to be the event that their first k increments can be extended to two disjoint bridges ending at one common height, while remaining their initial k renewal pieces. Equivalently the extensions are strict bridges above the terminal renewal cuts. This event concerns the finite prefixes; those extensions need not be the independently sampled continuations.

Lemma 5.4. *For each $\delta > 0$ there is C_δ such that $p^{\otimes \mathbb{N}} \otimes p^{\otimes \mathbb{N}}(E_k) \leq C_\delta(1+k)^{-1+\delta}$.*

Proof. We prove this bound by induction, for $0 < \delta < 1$. Set $r = k^{4/3}$ and expose $k_0 = \lfloor k/2 \rfloor$ increments on each side. Call their endpoint coordinates (X_i, t_i) . If the sum of their diameters exceeds Mr , the sum of the diameters truncated at Mr also does. On E_k every earlier paired prefix satisfies E_{a-1} . Markov's inequality and independence of the next increment give an upper bound

$$\frac{1}{Mr} \sum_{i=1}^2 \sum_{a \leq k_0} \Pr(E_{a-1}) p(D \wedge Mr) \leq C_\delta M^{-3/4} A_\delta k^{-1+\delta},$$

where A_δ is the induction constant. Choose M large enough that this is at most one quarter of the proposed bound.

Label the lower endpoint path 1, so $t_1 \leq t_2$, and reflect horizontally if necessary so that it starts at the left port. For small fixed $d > 0$, let m be the leftmost coordinate of the second history between heights t_1 and $t_1 + dr$, with $m = +\infty$ if it has no point there. We discard $|m - X_1| \leq ur$, where $u > 0$ will be small. Condition on both histories through $\lfloor k/4 \rfloor$ increments and then on the rest of the second history. For each possible $t_1 = O(Mr)$, the prohibited interval for X_1 has width $2ur + O(1)$. The two-dimensional atom bound on the still independent part of the first history shows that this conditional probability is at most $C_M(u + r^{-1})$. The induction bound for $E_{\lfloor k/4 \rfloor}$ pays for the earlier histories.

Next discard the case $m < X_1 - ur$. Were a disjoint common-height extension possible, the left crosscut would separate the second path from the far left of the slab. Thus its continuation above t_1 would have to pass to the left of a point of the second history in this band; at height t_1 such a point is already excluded by the single crossing there. In the next $\lfloor sk \rfloor$ increments, however, the first path, with probability as close to one as desired, gains height greater than dr while its total diameter sum is less than ur . Indeed the latter exceptional probability is at most $Csu^{-3/4}$ by truncation and the diameter tail. The probability of no height increment exceeding dr is at most $\exp(-csd^{-3/4})$, by the height shell lower bound. Choose s small after u , and d small after s . These fresh increments fit among the first k and are independent of the exposed histories. They contradict the required leftward passage. Conditional on those histories their failure probability is uniformly small. The induction bound at $\lfloor k/4 \rfloor$, with u, s, d chosen in that order, makes the total of these two discarded cases, including reflected cases, at most another quarter of $A_\delta k^{-1+\delta}$.

For each remaining pair of histories the strip just above t_1 of height dr , to the left of $X_1 + ur$, is free of the second history. Extend the first history through this strip to the left of the box containing both histories, then upward. Extend the second upward in a separate corridor. If $t_2 < t_1 + dr$, its tip lies to the right of the cleared strip; if $t_2 \geq t_1 + dr$, the first path has reached its separate corridor before the second extension begins. These are piecewise linear guides with fixed positive separation. The corridor input supplies joint mass at least $cr^{-1/2}$ to every common terminal level in $[C_M r, 2C_M r]$. The diameter stays at most $C'_M r$.

In an output pair the original histories are recovered by cutting after exactly k_0 irreducibles on both paths. Hence there is no multiplicity factor for those cuts. Summing the order r terminal levels and using (12), the mass of the remaining histories is at most $r^{-3/4+o(1)} = k^{-1+o(1)}$. For large k this fits in the remaining half of the induction bound. Increasing A_δ handles the finitely many smaller k . $\square$

5.5 Matching the two ends of a marked pair

The previous lemma controls initial avoidance. The next estimate also pays for matching the far endpoints. Its mark may be any nonnegative function of one whole irreducible; this uniformity is what permits both length and diameter tests later.

Proposition 5.5. *For every nonnegative function f of an irreducible, with $p(f) < \infty$, and every $\epsilon > 0$,*

$$\sum_{R \leq h \leq 2R} D_h \left[\sum_{i \text{ in the first path}} f(I_i) \right] \leq C_\epsilon R^{-5/4+\epsilon} p(f). \quad (15)$$

Either path can carry the mark. If the two bottom ports instead have any prescribed separation, while the top ports remain adjacent, the right side becomes $C_\epsilon R^{-1/2+\epsilon} p(f)$. In that second estimate, requiring the marked path to contain at most $R^{3/4-\delta}$ increments improves the bound by $R^{-\delta+\epsilon}$, with the exponent slacks combined in the constant.

Proof. Fix a small $\eta > 0$. Lists with more than $R^{3/4+\eta}$ increments have superpolynomially small mass: after removing the marked slot, exponential Markov with $e^{-H/R}$ gives $\exp(-cR^\eta)$. There are at most $2R$ increments on either list, so summing all counts and possible marks still bounds this part by $p(f)R^{-M}$ for any fixed M . We henceforth impose the count cutoff.

Bin the numbers of increments before and after the marked slot on the first path, each increased by one, at dyadic scales a_-, a_+ . There are $O(a_- a_+)$ choices of these two numbers. Fix them, and put $K = \max(a_-, a_+)$. Near each end of the second path choose a grid of more than $2/\eta$ height thresholds in a fixed small fraction of R . The counts to their first reaching renewals are positive

and at most $2R$. At some consecutive pair the counts differ by a factor at most R^η . Bin the lower count at scale c_e , where $e = -, +$ specifies the end. Sum over these finitely many threshold choices and logarithmically many bins at the end.

At each end expose $m_e = \lfloor R^{-2\eta} \min(a_e, c_e) \rfloor$ increments on both paths. If necessary reduce this by a fixed factor to account for the added one in a_e ; zero histories cost one. For large R these are disjoint coordinate blocks and do not include the mark. On the second path they lie before the selected lower height thresholds. On the first path that extra restriction is unnecessary: the actual full disjoint bridges supply the extensions required by the prefix event. At the upper end read the lists backwards in relative coordinates from adjacent ports. Their still unknown common translation is immaterial to this test. Lemma 5.4 supplies the independent cost

$$R^{O(\eta)} \prod_{e=-,+} \min(a_e, c_e)^{-1}. \quad (16)$$

The choices of counts and of the marked slot cost $O(a_- a_+)$. To cancel that cost against the two history factors, we will obtain a matching gain $\min(1, c_e/K)$ at each end. We first bound the central part of the second path, then match its displacement using an independent batch of increments on the first path.

Fix the histories. Sum over the remaining count on the second path and let $F(y)$ be its central bridge kernel, where y is its two-dimensional displacement. Its height is comparable to R . From either end, within $O(c_e R^\eta)$ central increments it must gain at least $c_\eta R$ height, by the gap between the two selected thresholds. Retain these requirements and drop the other restrictions. The pointwise slab estimate gives $F(y) \leq CR^{-5/4}$.

We need the following improvement on a small displacement ball U_b , valid for $1 \ll b \ll c_\eta R$ and either subset S of the two ends:

$$\sum_{y \in U_b} F(y) \leq CR^{-5/4} b^2 \prod_{e \in S} (C_\eta c_e R^\eta b^{-3/4}). \quad (17)$$

To prove it, append at each indicated end an independently chosen bridge of height in $[b, 2b]$ and diameter at most Cb . Its mass, with terminal position and height summed, is at least $cb^{3/4}$. The enlarged displacement lies in a ball of radius $C'b$, hence has at most Cb^2 possible values, each with mass $CR^{-5/4}$. In any output, a possible removed attachment ends within $O(b)$ height of the new endpoint. Moreover that cut precedes the first renewal at or beyond height $c_\eta R/2$ from the new endpoint by at most $O(c_e R^\eta)$ renewal steps. Thus there are at most that many possible cuts at each indicated end. Dividing by the attachment masses proves (17). Positivity is used both in adjoining these bridges and in dropping all further restrictions.

The first path still has order K unexposed increments outside the mark and the histories. If $K \leq R^{5\eta}$, the pointwise bound on F suffices, since all additional factors below cost only $R^{O(\eta)}$. Otherwise choose $b_0 = \lfloor KR^{-4\eta} \rfloor$ slots in each of $\lceil 30/\eta \rceil$ disjoint batches. Put $b = R^\eta b_0^{4/3}$. The count cutoff implies $b \leq R^{1-3\eta}$, up to fixed factors. By the diameter tail and truncation, a batch has displacement norm greater than b with probability at most $CR^{-3\eta/4}$. The batches are independent, so the chance that all of them fail is $O(R^{-10})$.

On that exception use the pointwise bound on F . Otherwise take a union over the fixed finite set of possible successful batch labels. For each label, fix everything outside that batch and use its two-dimensional atom bound $Cb_0^{-8/3}$ together with (17). At the end e use the improvement when $c_e < K$. Since $b_0^{-8/3} b^2 = R^{2\eta}$ and $c_e R^\eta b^{-3/4} \leq R^{O(\eta)} c_e / K$, the resulting matching cost is

$$C_\eta R^{-5/4+O(\eta)} \prod_{e=-,+} \min(1, c_e/K). \quad (18)$$

The estimate is uniform in the value of the marked increment. The mark can therefore be integrated separately at cost $p(f)$.

Multiply (18) by (16) and the $O(a_-a_+)$ count and mark choices. Each end contributes at most one because

$$a_e \min(1, c_e/K) \leq \min(a_e, c_e).$$

Summing the threshold and dyadic labels costs only logarithmic powers. Taking η sufficiently small in terms of ϵ proves (15).

For prescribed separated bottom ports, retain exactly the same central kernel and matching proof. The displacement to be matched is merely shifted by that prescribed separation. The bottom avoidance test is now absent, so its factor in (16) is lost. The remaining count factor is bounded by

$$\frac{a_-a_+ \prod_e \min(1, c_e/K)}{\min(a_+, c_+)} \leq a_- \leq R^{3/4+\eta}.$$

This proves the $R^{3/4+\epsilon}$ loss. If the marked path has at most $R^{3/4-\delta}$ increments, then $a_- \leq CR^{3/4-\delta}$, which gives the stated improvement. This is the only change in the argument; no bottom avoidance probability is retained implicitly. $\square$

The next consequence recovers the length-deficit exponent from the spatial and polygon inputs alone. Proposition 2.4 gave the stronger bounded-factor estimate using the finite first-length and long-bridge estimates.

Corollary 5.6. *The length deficit satisfies $q(t) = t^{-9/16+o(1)}$.*

Proof. The upper bound is (14). For a lower bound fix $\eta > 0$. On polygon diameters in $[r, r^{1+\eta}]$, the length-weighted mass is $r^{-2/3+o(1)}$. Lengths less than $r^{4/3-\eta}$ contribute at most $r^{-2/3-\eta+o(1)}$ by the unweighted tail. Lengths greater than $r^{4/3+2\eta}$ contribute at most

$$r^{-4/3-2\eta} \sum_{D(P) \leq r^{1+\eta}} |P|^2 w(P) \leq r^{-2/3-4\eta/3+o(1)}.$$

Thus the unweighted mass with both diameter and length in these windows is at least $r^{-2-2\eta+o(1)}$.

Choose a symmetry-related row normal retaining a fixed fraction with vertical span comparable to diameter. A bounded hexagonal-face modification at its lowest and highest vertices turns the polygon into two disjoint bridges with adjacent ports at each end. Here is the local operation. In coordinates $P_{ij} = iu + jv$, $R_{ij} = P_{ij} + (u+v)/3$, with $u = (1, 0)$ and $v = (1/2, \sqrt{3}/2)$, the neighbors of P_{ij} are $R_{ij}, R_{i-1,j}, R_{i,j-1}$. A lowest polygon vertex is a P and uses both ascending bonds. At the leftmost lowest such vertex, take the adjacent face below and to its right. Its overlap with the polygon is a path of one or two edges and includes every occupied vertex of that face. Symmetric difference replaces it by the complementary face path, creating a bounded exterior lower cap. The reflected operation creates the upper cap. Remove the two caps. Simplicity forces the remainder to be exactly two through bridges; otherwise one cap would bound a separate cycle. The faces are disjoint for large r , and length, diameter, and weight change by bounded amounts. Anchoring the lower left port, the cuts and caps recover the input with bounded multiplicity. This suffices for the mass lower bound, whether or not a tie convention makes the map injective.

The output heights lie between cr and $Cr^{1+\eta}$. Set $t = r^{4/3-2\eta}$. Across both output lists, $\sum_i (1 - e^{-L_i/t}) \geq 1 - e^{-\sum_i L_i/t}$ is bounded below by a positive constant. Apply Proposition 5.5 with $f = 1 - e^{-L/t}$ to both paths and sum the height bins. It gives $r^{-2-2\eta+o(1)} \leq r^{-5/4+o(1)} q(t)$, hence $q(t) \geq r^{-3/4-2\eta+o(1)}$. Letting η decrease proves the lower exponent. $\square$

5.6 Height conditioning and fixed boundary ports

These two applications keep their distinct normalizing masses. A bridge with terminal port summed has mass $h^{-1/4}$; an arch with both boundary ports prescribed has mass $r^{-5/4}$. The missing bottom avoidance test in Proposition 5.5 is exactly what changes the accounting between them.

Proposition 5.7. *Under the critical law on strict bridges of height h , with terminal port unrestricted, $L = h^{4/3+o(1)}$ and $D = h^{1+o(1)}$ in probability. The same holds after any restriction retaining a fixed positive fraction of B_h .*

Proof. For counts in $[k, 2k)$, split each list into two groups of comparable size. If their total height is h , one group gains at least $h/2$, at cost $C \min(1, kh^{-3/4})$; the other matches the remaining height at cost $C(1+k)^{-4/3}$. There are $O(k)$ count choices. Thus counts at most $h^{3/4-\delta}$ have total mass at most $h^{-1/4-2\delta/3+o(1)}$, negligible compared with B_h . Bounded counts follow directly from the height tail. Counts exceeding $h^{3/4+\delta}$ have superpolynomially small mass by exponential Markov with $e^{-H/h}$.

On the remaining counts, length at most $t = h^{4/3-\epsilon}$ costs at most $e \exp(-kq(t))$. Choose $\delta < 9\epsilon/32$; the deficit estimate makes this superpolynomially small. For length greater than $t = h^{4/3+\epsilon}$, mark $1 - e^{-L_i/t}$, whose sum is bounded below on that event. The other increments match height at cost $Ck^{-4/3}$. Summing marks and count choices gives $Ck^{2/3}q(t)$ in a dyadic bin. For sufficiently small δ this is $o(h^{-1/4})$. Finally $D \geq ch$ deterministically, and (6) gives $B_h[D > h^{1+\epsilon}] = o(B_h)$. Division by any retained fixed fraction of B_h preserves every conclusion. $\square$

Proposition 5.8. *For the critical upper-half-plane law between two prescribed boundary ports at separation r , one has $L = r^{4/3+o(1)}$ and $D = r^{1+o(1)}$ in probability. Any restriction retaining a fixed positive fraction of the arch mass has the same exponents.*

Proof. The denominator is $a_r \asymp r^{-5/4}$. First discard arches of height at most $r^{1-\epsilon}$. Glue two independent such arches, one reflected into the lower half-plane, to make a marked polygon. This squares their total mass, up to bounded weight factors. Choose a symmetry-related period of order $r^{1-\epsilon}$ whose vertical component exceeds the polygon's span. Its projection is simple, and anchoring the first marked edge recovers the lift. The transverse span is of order at least r , because of the original boundary gap and the nonparallel period. The possible marks cost at most a length factor. The cylinder estimate (8), with arbitrarily large M , makes the squared mass and hence the mass itself smaller than every power of r .

For every remaining arch perform only the upper extremal-face operation described in the proof of Corollary 5.6. Removing the new upper cap leaves two disjoint bridges from the original prescribed bottom ports to adjacent top ports. Weights and lengths change boundedly and the inverse multiplicity is bounded. At height $h \in [R, 2R]$, Proposition 5.5 with the bottom test omitted gives $R^{-1/2+o(1)}p(f)$ for a mark on either path. Take $f = \min(1, H/R)$: its sum on either path is bounded below, while $p(f) \leq CR^{-3/4}$. The total mass of the bin is therefore $R^{-5/4+o(1)}$. Summing $R > r^{1+\epsilon}$ is negligible relative to a_r .

In the remaining height range $[r^{1-\epsilon}, r^{1+\epsilon}]$, requiring one path to have count at most $R^{3/4-\delta}$ adds the factor $R^{-\delta+o(1)}$. Choose ϵ small after δ ; these pairs are negligible. For larger counts, a total length at most $r^{4/3-\delta'}$ is superpolynomially unlikely under the free increment law, by $\exp(-kq(r^{4/3-\delta'}))$. One may drop all pair constraints here; each count is at most $2R$, so their choices cost only a polynomial. Taking ϵ, δ small enough in terms of δ' proves the lower length bound.

For the upper length bound mark $f = 1 - e^{-L/t}$ with $t = r^{4/3+\delta'}$. Summing the two paths in the typical height bins gives $R^{-1/2+o(1)}q(t)$, which is $o(r^{-5/4})$ when ϵ is sufficiently small. For diameter greater than $r^{1+\delta'}$, use instead $f = \min(1, D/r^{1+\delta'})$. The sum of increment diameters dominates

each path diameter, and Lemma 5.1 bounds its expectation by $Cr^{-3(1+\delta')/4}$. The same calculation applies. The lower diameter bound follows from the prescribed boundary gap. Bounded offsets in the cap construction are absorbed by decreasing exponent slacks. A retained positive fraction of the denominator changes none of the estimates. $\square$

5.7 A joint estimate for turning chains

We now estimate path pieces before choosing any probability normalization. A *turn* joins two consecutive pieces that both run away from their joining vertex toward the same side in height. We first convert these pieces into strict port bridges, then integrate their irreducible coordinates while retaining the necessary avoidance tests.

Lemma 5.9 (Converting endpoints to ports). *Suppose a simple path segment stays between the heights of its endpoints. If its span is sufficiently large, bounded endpoint changes convert it into a strict port bridge. At a turn between two such segments, the changes can be made jointly so that their ports are adjacent and their initial parts remain disjoint through a fixed positive fraction of the smaller span. Changes at opposite ends of a large-span segment occur in disjoint height zones.*

The original ordered segments, with one initial anchor, can be recovered from the resulting ordered bridges and at most constantly many local choices per segment. Lengths and endpoint heights change by bounded amounts per end. Bounded-span segments can instead be represented by positive-height bridges and a class of one-row paths with bounded total critical mass and an exponential length tail.

Proof. Use the P, R coordinates in the proof of Corollary 5.6. At a top endpoint an R extends by its upward bond to the next cut; a top P must have been reached along its unique descending bond and can be trimmed to the port immediately below. No other top P is internal to the segment, since it has only one descending bond. Reflect at bottom endpoints.

At a common top turn the pivot is an R_{ij} using both descending bonds. Use the face immediately above and to its right, with lower path $R_{ij}, P_{i+1,j}, R_{i+1,j}$. If the other uppermost $R_{i+1,j}$ is occupied, it too uses both descending bonds. The union's overlap with this face is therefore a path of one or two edges containing all its occupied face vertices. Symmetric difference with the face, followed by removal of the new upper cap, creates disjoint ends at adjacent ports. Reflect at a bottom turn. Only boundedly many bonds change. For large spans the far-end modification cannot affect disjointness within, say, one quarter of the smaller original span from this turn.

If individual trimming makes the endpoint levels meet or cross, the segment was confined to one row. Its induced graph has degree at most two, so there are at most constantly many paths of each length from its endpoint. Their critical weights have an exponential tail because $\rho < 1$. Otherwise the result is a positive-height bridge. Record the case, each bounded local change, and the offsets between consecutive ports. There are constantly many such records per piece, and reversing them recovers the original path. The next anchor is determined by the preceding piece and its record; no free translation factor is added. $\square$

Consider a chain of m bridge nodes, each with at most two tested ends. A tested turn requires the corresponding prefixes, read in the same upward direction, to be disjoint for a fixed positive fraction of the smaller height. Fix their local records. At node i , restrict height to $[cR_i, CR_i]$ and irreducible count to $[k_i, 2k_i)$, with fixed $c, C > 0$. All scales and counts are at most C_0h . Put

$$K_i = R_i^{3/4}, \quad \kappa_i = \min(k_i, K_i), \quad s_{ij} = \min(\kappa_i, \kappa_j).$$

The scales R_i need not be comparable between different nodes.

Proposition 5.10 (Independent coordinates at a turn). *For every sufficiently small fixed $\eta > 0$, the joint critical mass, including all actual count choices in the bins, is at most*

$$h^{C_1\eta(m+1)} \prod_i \frac{\kappa_i^2}{K_i} \prod_{ij \text{ tested}} s_{ij}^{-1} \quad (19)$$

for h sufficiently large depending on η . Constant factors per node are included in this bound.

At each sufficiently large-count node, the proof leaves an independent group of at least ck_i increments available for a concentration test. An interval of at most a_i allowed total heights contributes

$$C \min(1, a_i \kappa_i^{-4/3}), \quad (20)$$

provided the interval is measurable outside that group, with any fixed external data allowed. A proved concentration bound for another additive statistic, such as length, may be used on this group instead. A different independent group supplies

$$C e^{-ck_i q(l)} \quad (21)$$

when the node's total length is at most l .

Several height factors may be used for a specified vector of heights and then summed over an explicitly bounded admissible set, or in a successive integration order with the stated measurability property. The estimate does not condition on the reserved group's own horizontal displacement. Bounded-count nodes have the trivial bounded-factor version of the height estimate. One-row paths have $K_i = \kappa_i = 1$ and bounded mass, with no atom or independent length test asserted.

Proof. Fix every actual count. At $e = ij$ put $d_e = \min(R_i, R_j)$ and $s_e = s_{ij}$. If d_e is bounded in terms of η , drop this test at a bounded cost. Otherwise place more than $3/\eta$ consecutive thresholds within a small fixed fraction of d_e from each incident end, with gaps at least $c_\eta d_e$. The counts to their first reaching renewals lie in $[1, C_0 h]$. At some consecutive pair their ratio is at most h^η , since otherwise their product would exceed the count cap. Bin the lower crossing counts dyadically at $c_{i,e}$ and $c_{j,e}$, and put $c_e = \min(c_{i,e}, c_{j,e})$.

Once these labels are fixed, the first $\ell_e = \lfloor h^{-2\eta} c_e \rfloor$ slots at both ends are deterministic coordinate sets. On the actual event their heights are below the lower thresholds, and the actual disjoint paths extend them to a common cut within the tested range. Opposite-end histories are disjoint coordinate blocks: each uses a vanishing fraction of the node's count. Lemma 5.4 supplies the independent cost

$$C_\eta h^{3\eta} / c_e. \quad (22)$$

This includes $\ell_e = 0$. If $c_e \geq h^{-4\eta} s_e$, it already costs at most $C_\eta h^{7\eta} / s_e$.

If $c_e < h^{-4\eta} s_e$, designate an incident node attaining this smaller bin. On it take the deterministic group of slots

$$\ell_e + 1, \dots, \lceil 2h^\eta c_e \rceil$$

read from that end. It contains the later threshold renewal, so its height sum is at least $c_\eta d_e$. It has at most $C c_e h^\eta \leq C h^{-3\eta} s_e \leq C h^{-3\eta} \kappa_i$ slots. Call it a rapid group. Such groups are disjoint from the histories and from any opposite-end rapid group. The height tail and its truncated first moment imply

$$\Pr(H_1 + \dots + H_n \geq u) \leq C \min(1, n u^{-3/4}). \quad (23)$$

The rapid group therefore costs at most $C_\eta c_e h^\eta / d_e^{3/4} \leq C_\eta h^\eta c_e / s_e$. Multiplying by (22) pays s_e^{-1} . We now discard the original crossing-index restrictions: only necessary events on the deterministic independent coordinate blocks are retained.

The histories removed at one node have total height less than a small fixed multiple of R_i . Their complement must gain at least $c_0 R_i$. It consists of at most two rapid groups and the remaining slots. Partition those remaining slots into eight groups of nearly equal size. For sufficiently large fixed counts, each has order k_i slots. Some one of these at most ten groups gains height at least $c'_0 R_i$. Take a finite union over a label selecting that height-carrying group. *After* fixing the label, reserve three different free groups for count damping, concentration, and the length test. None is the height-carrying group.

Write $b_i = \min(1, k_i/K_i)$. If a free group carries the height, (23) charges it by Cb_i , independently of the rapid groups. If the carrier is the rapid group for e on node i , charge its larger height directly at scale R_i . The cost $C_\eta c_e h^\eta / K_i$ supplies both required factors, because

$$c_e/K_i \leq \min(1, k_i/K_i) c_e/s_e. \quad (24)$$

This follows from $s_e \leq \kappa_i$ in either regime for k_i/K_i . The other rapid group, if any, remains independent. Also $d_e^{3/4} = \min(K_i, K_j) \geq s_e$; thus this calculation allows unrelated height scales.

The reserved damping group has height at most CR_i . Exponential Markov and $1 - p(e^{-H/R_i}) \geq c/K_i$ give Ce^{-ck_i/K_i} . All previously charged events use other slots. Conditional on coordinates outside the concentration group, its product law is unchanged. Lemma 5.2 bounds its height atom by $Ck_i^{-4/3} \leq C\kappa_i^{-4/3}$, proving (20). The same statement holds with any proved concentration function for an additive statistic of these slots. On the separate length group,

$$\Pr\{\sum L_j \leq l\} \leq e(1 - q(l))^{ck_i} \leq Ce^{-ck_i q(l)}.$$

These bounds multiply because their coordinate sets are disjoint. For several height constraints, condition outside all their groups, bound the joint atom by the product, and sum the allowed height vectors. Successive interval constraints follow by successive integration with the same uniform bounds.

For bounded counts the incident turn factors are bounded. Drop them and apply (23) to the whole list. The height factor is then bounded below by a positive constant, so its insertion changes only the constant; no large-count length test is needed. There are $O(k_i)$ actual counts at each node, and

$$k_i \min(1, k_i/K_i) e^{-ck_i/K_i} \leq C\kappa_i^2/K_i.$$

For $k_i \leq K_i$ this is immediate; otherwise use $xe^{-cx} \leq C$. Thresholds, crossing-count bins, height-carrier labels, and local records number at most $(C_\eta \log^C h)^{C(m+1)}$. These choices and the power losses above are at most $h^{C_1 \eta(m+1)}$ for large h . This proves the proposition. $\square$

Corollary 5.11 (Two ends with a reserved height variable). *Suppose two bridges start at adjacent prescribed ports and stay disjoint through a fixed positive fraction of the smaller height. Their heights and counts lie in the independent bins above, all bounded by CN^2 . For every $\epsilon > 0$ their critical mass is at most*

$$C_\epsilon N^\epsilon F, \quad F = \frac{\kappa_1^2 \kappa_2^2}{K_1 K_2 \min(\kappa_1, \kappa_2)}. \quad (25)$$

An interval of $C(1+r)$ allowed heights on node i , measurable outside its reserved group, adds $\min(1, (1+r)\kappa_i^{-4/3})$. In particular this applies to the difference of the final endpoint heights after any two terminal pieces have been fixed up to translation. For one nonempty bridge the factor is κ_i^2/K_i , with the same optional concentration test.

Proof. Take $h = C'N^2$ in Proposition 5.10 and choose η sufficiently small in terms of ϵ . Once terminal pieces and local records are fixed, their final-height constraint reads $H_i \in [H_j + b - r - C_0, H_j + b + r + C_0]$. The interval depends on the other bridge and fixed external data, hence is measurable outside the selected group. No horizontal displacement of that group has been fixed. The one-node assertion is the proposition with no turn test. $\square$

5.8 Absolute critical mass of unrestricted paths

The preceding estimate pays for every turn in a chain. Many turns therefore make an ordered extremum decomposition inexpensive, even though we impose no length damping. This is the point at which the bridge-unfolding method of Hammersley and Welsh is useful [7, 14].

Proposition 5.12. *Let a_n be the total critical mass of plane walks with n visited vertices, rooted at one specified vertex, or counted up to translations with both possible initial vertex types. For every $\delta > 0$,*

$$a_n \leq C_\delta e^{n^\delta}.$$

The same bound holds for the mass with length at most n . In the extremum decomposition below, a bounded number of specified pieces may be omitted and counted separately; all other pieces still cost at most $C_\delta e^{n^\delta}$, including their count, position, and local records.

Proof. Split a path at a global minimum of its height and read both parts outward from that minimum. In either part, end the first piece at the last maximum of the remaining path, the next at its last minimum, and continue. Each piece stays between its endpoint heights. The spans decrease strictly after a possible initial equality, since the selected extreme cannot occur again in the remaining path. Every honeycomb edge has nonzero vertical projection in these coordinates.

In a dyadic span bin $[R, 2R)$ there are at most CR pieces in a chain. Convert their endpoints by Lemma 5.9. Test turns only between consecutive pieces in the same bin. An omitted piece splits that bin into runs; no test crosses an omission. At a run of m pieces put $K = R^{3/4}$ and $x_i = \kappa_i/K \leq 1$, absorbing fixed comparability constants into the per-node constants. The product in (19), before its exponent loss, is

$$Kx_1x_m \prod_{i < m} \max(x_i, x_{i+1}). \quad (26)$$

There are only $O(\log R)$ possible count bins per piece.

Fix a small $\zeta > 0$. If fewer than $m/4$ nodes have $x_i \geq R^{-\zeta}$, at least $m/2 - O(1)$ neighboring pairs have both labels smaller than this threshold. They supply the factor $R^{-\zeta(m/2 - O(1))}$. If instead $b \geq m/4$ nodes have large labels, use the reserved height atom at those nodes. Their atom bounds are at most $CR^{-1+4\zeta/3}$. Their original spans are ordered and lie in a range of CR levels. Modified heights differ from them by specified bounded offsets. Hence, conditional on the other coordinates, the number of allowed vectors for these b heights is at most

$$C^b \binom{CR+b}{b}.$$

The product atom bound from Proposition 5.10 therefore costs at most $C^m R^{4\zeta b/3}/b!$, since $b \leq CR$. This is a joint atom calculation followed by a count of ordered vectors, not a repeated conditioning on already fixed bridge heights.

Choose η sufficiently small after ζ . The factors $R^{C_1\eta(m+1)}$, the count bins, and the local records are absorbed by the preceding decay whenever $m > R^{C_2\zeta}$, for one sufficiently large absolute C_2 . For

example, in the second case $b! \geq (b/e)^b$ beats $R^{(4\zeta/3+C\eta)m}$; in the first case choose $C\eta < \zeta/8$. These long-run contributions are summable. For $m \leq R^{C_2\zeta}$, even bounding every displayed factor and choice separately costs at most $\exp(R^{C_3\zeta})$, with a slightly larger absolute C_3 . Thus that bound covers the whole bin.

The bins occur in decreasing order. Multiplying over dyadic scales up to Cn gives $\exp(Cn^{C_3\zeta})$, because the powers sum geometrically. Choose ζ small enough for the requested δ , and absorb bounded bins and both halves into the constant. If a fixed number of pieces is omitted, the number of runs per bin increases by only a fixed amount. Choosing their bins and positions costs a polynomial in n , and the same argument covers the remainder. Root choice or allowing both vertex types changes a constant only. $\square$

5.9 Paths that travel too far with too few steps

The total-count bound is deliberately weak. To suppress rapid travel we cut a tall bridge into pieces on one intermediate height scale. A single crossing permits exact height matching. A repeated crossing is replaced by a short reverse piece, whose two turns pay for the loss of that exact matching. This is why the height atom and the short-length test had to remain independent in Proposition 5.10.

Proposition 5.13 (Absolute fast-path estimate). *For each sufficiently small $\epsilon > 0$ there are $c_\epsilon > 0$ and n_ϵ such that, for $n \geq n_\epsilon$,*

$$\sum_{L(\gamma)=n, D(\gamma) \geq n^{3/4+\epsilon}} \rho^{L(\gamma)} \leq e^{-n^{c_\epsilon}}. \quad (27)$$

More precisely, uniformly for $d \geq n^{3/4+\epsilon}$,

$$\sum_{L(\gamma)=n, D(\gamma) \geq d} \rho^{L(\gamma)} \leq \exp(-c_\epsilon d/n^{3/4}). \quad (28)$$

Walks are rooted at one fixed vertex, or counted in the translation convention of Proposition 5.12. These are absolute masses before division by any finite or thermal partition.

Proof. First consider a port bridge of length at most $n + O(1)$ and height at least $d \geq cn^{3/4+\epsilon}$. Decreasing ϵ slightly absorbs the fixed c . Put $h = \lfloor n^{3/4-\gamma} \rfloor$, where $\gamma = \epsilon/10$. Starting upward, track the running maximum until the first drop of h from it, if such a drop occurs. Place a turn at the last vertex attaining that maximum before the drop. From there track the running minimum until the first rise of h , and repeat with directions alternating. Consecutive pieces stay between their endpoint levels, have spans at least h , and have no move against their direction of more than $h + O(1)$. The first and last pieces point upward since the whole path is a bridge. Choosing the last extremum makes the same statements valid in the portion preceding each trigger.

Within every such run place transverse cuts at deterministic spacings of order $10h$, leaving margins of at least $5h$ from the ends and no residual gap greater than a fixed multiple of h . A run too short for a cut is kept as one piece. At a cut crossed exactly once, split at that port. At a repeatedly crossed cut of an upward run, let M be the largest height before its last crossing, attained at a selected pivot before that crossing. Let m be the smallest height from that pivot to the end of the run. Then

$$M > q > m, \quad M - m \leq h + O(1),$$

where q is the cut level. The minimum occurs before the last crossing; after that crossing all heights are above q . Split at the two pivots. The intervening descending segment is a *repair piece*. Before

the upper pivot the path never exceeds it, and after the lower pivot it never goes below that lower level. Thus the neighboring upward pieces are again slab pieces. Reflect this construction in a downward run. The cut spacing and drawdown bound put different repairs in disjoint ordered zones.

Figure 1 shows the local reason for the reverse piece: cutting only at the last crossing would leave the preceding piece above its terminal level, so it would not be a strict bridge.

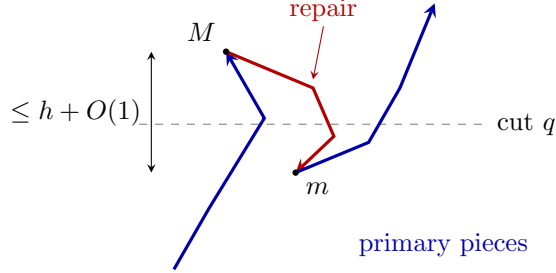


Figure 1: A repeated crossing of a transverse cut in an upward run. The segment from the selected maximum M to the following minimum m is read downward; the neighboring pieces are read upward. The drawing is schematic and suppresses individual lattice edges. The drawdown bound controls the repair's height, while the separated cut levels keep distinct repairs in ordered disjoint zones.

Call the pieces in the original direction primary pieces. Their spans are between fixed positive multiples of h ; repair spans are at most Ch . After the local port changes, test every turn with sufficiently large smaller span. If P is the number of primary pieces, then

$$cd/h \leq P \leq Cn/h. \quad (29)$$

There are at most $P - 1$ repairs, and the total modified length is at most $n + CP$.

Apply Proposition 5.10 with $K = h^{3/4}$. Write a primary label as $\kappa_i = Kx_i$. A repair of height scale $a \leq Ch$ has $K_{\text{rep}} = a^{3/4}$; put $t = K_{\text{rep}}/K$ and $y = \kappa_{\text{rep}}/K \leq t$. Fixed comparability constants can be absorbed into all bounds below, so the normalized labels are at most one. For a single-crossing cut without a repair use the formal values $a = K_{\text{rep}} = \kappa_{\text{rep}} = 1$. This virtual node has unit mass and no avoidance test; its algebraic node and seam factors are also one, since every actual $\kappa_i \geq 1$.

At either kind of cut, smooth the height of the preceding primary piece. Its endpoint lies within $O(a)$ of the cut, where the cut level is specified relative to the run's starting height by the discrete cut pattern. We claim that this supplies the factor

$$C \min(1, (t/x_i)^{4/3}). \quad (30)$$

Bounded-count nodes use the bounded-factor height estimate in Proposition 5.10. At the other nodes, use this constraint only when $a\kappa_i^{-4/3} < 1$, and fix all coordinates outside the selected reserved groups. Their height sums have a joint atom bound equal to the product of their individual bounds $C\kappa_i^{-4/3}$ from Proposition 5.10. The allowed height vectors can be counted in traversal order: once the earlier selected sums in the run are given, the current endpoint constraint allows at most Ca values for the next sum. The run's starting height cancels from this relative constraint. Thus the number of allowed vectors is at most the product of these Ca factors. Multiplying by the joint atom bound, and dropping the constraints at the unselected nodes, proves the product of (30), since $a\kappa_i^{-4/3} = (t/x_i)^{4/3}$. Each primary receives at most one factor. This count already includes its possible endpoint heights. The turn tests use only their separate history and rapid groups in adjacent-port coordinates; none fixes the displacement of a selected reserved group.

For completeness, the product calculation is as follows. A direct turn between primaries contributes $\max(x_i, x_{i+1})$ after the primary factors have been collected. A cut with a repair contributes

$$Q = \frac{\max(x_i, y) \max(x_{i+1}, y)}{t} \min(1, (t/x_i)^{4/3}) \leq \max(x_{i+1}, y).$$

If $x_i \leq t$, use $y \leq t$; if $x_i > t$, the remaining factor is $(t/x_i)^{1/3} \leq 1$. This proves the inequality in both cases, including the virtual-node convention. The full product is bounded by Kx_1x_P times these factors and constant bases. Every connecting factor is bounded, and is $O(h^{-s})$ if all its incident labels are less than h^{-s} . The patterns of runs and cuts, repair scales, count bins, and local records cost at most $h^{C\eta(P+1)}$.

Take $s' = \epsilon/2$ and $s = 9\epsilon/64$. For small ϵ these choices satisfy $9s'/16 > s$ and

$$\frac{n}{h^{4/3-s'}} = o(n^{3/4+\epsilon}/h).$$

Set $l = h^{4/3-s'}$. Since the total length is at most $n + CP$, only $o(P)$ actual nodes have length greater than l . Designate a superset of these nodes; all possible designations cost at most 2^{2P} . On every other actual node with normalized label at least h^{-s} , the separate short-length group supplies

$$Ce^{-ck_i q(l)} \leq \exp(-h^{c_s}),$$

for some $c_s > 0$: indeed $k_i \geq Kh^{-s}$ and $Kh^{-s}q(l) = h^{9s'/16-s+o(1)}$. Virtual and bounded one-row nodes have small labels.

If a positive fraction of labels are large, this gives stronger than exponential decay in $P \log h$. Otherwise a positive fraction of the connections have all incident labels small. Each large or designated long node can spoil at most two such connections, so those connections supply h^{-s} each. Choose η sufficiently small after s . After summing all the records and designations, the contribution with P primary pieces is at most $\exp(-cP \log h)$. Equation (29) gives the tall-bridge bound

$$\exp(-cd \log h/h). \tag{31}$$

All constants are uniform for d in the stated range.

Finally take an unrestricted path of diameter at least d . One of the finitely many symmetry-related row normals has vertical span at least cd . In its extremum decomposition from Proposition 5.12, an initial piece on one of the two halves reaches the global maximum from the global minimum. Omit that piece and its neighboring turn tests. Its individual port conversion has length at most $n + O(1)$ and height at least $c'd$; its mass is bounded by (31). The remaining pieces, including their choices and records, cost at most $C_\delta e^{n^\delta}$ for every fixed $\delta > 0$. Choose δ smaller than the power gain in $d/h \geq cn^{\epsilon+\gamma}$. That cost is absorbed by (31). This proves (27). It also proves the uniform bound (28), since $h \leq n^{3/4}$ and the absorbed exponent is a fixed fraction of $d \log h/h$. $\square$

6 From renewal pieces to the unrestricted thermal law

We now prove the full-plane thermal assertion. The proof separates three tasks: identify the half-plane remainder, construct a recoverable family giving the plane denominator, and control paths whose endpoints have small vertical separation. The geometric estimate used in the last task is stated explicitly before the proof.

6.1 Conventions and prior geometric estimates

Let $N \geq 2$, $z = \rho e^{-1/N}$, and write $w_z(\gamma) = z^{L(\gamma)}$ for vertex length. We count oriented plane walks modulo lattice translations and allow both initial vertex types. Reflection exchanges those types, so the normalized law agrees, after the appropriate reflection and translation, with the thermal law from any fixed vertex. In particular these identifications preserve length, diameter, endpoint distance and absolute endpoint-height difference. If Z_N is this total vertex-weighted mass and Z_N^{edge} is the edge-weighted sum from one fixed vertex, then

$$Z_N = 2zZ_N^{\text{edge}}.$$

This bounded common factor has no effect on the probability laws or moment exponents.

Fix a horizontal row normal. Height is measured in row units; Euclidean vertical distance is d_0 times height, where $d_0 = \sqrt{3}/2$. Let T, L, Δ be the height, vertex length and diameter of an irreducible bridge. The critical probability law is p , and

$$q_N = 1 - \mathbb{E}_p e^{-L/N}.$$

We use the following earlier conclusions of the geometric and irreducible analysis.

1. Let B_h be the total *critical* mass $\sum \rho^{L(\gamma)}$ of strict bridges from one bottom port at exact height h , with terminal port summed. Then

$$B_h \leq C(1+h)^{-1/4}, \quad B_0 = 1, \quad q_N = N^{-9/16+o(1)}. \quad (32)$$

The error in the last exponent is an ordinary deterministic asymptotic. It is sufficient here; no bounded-factor length estimate is needed for this thermal transfer.

2. Write a_m for total critical vertex-weighted mass of plane walks of vertex length m , in the translation convention just specified. For every $\delta > 0$,

$$a_m \leq C_\delta \exp(m^\delta). \quad (33)$$

For every sufficiently small fixed $\eta > 0$, there are $c_\eta > 0$ and m_η such that

$$\sum_{\substack{L(\gamma)=m \\ D(\gamma) > m^{3/4+\eta}}} \rho^m \leq \exp(-m^{c_\eta}) \quad (m \geq m_\eta). \quad (34)$$

This is an *absolute critical mass* bound, before division by any probability denominator.

3. We use the two-bridge specialization of the preceding turning-chain estimate. Its exact content for this section is as follows. Consider two bridges with adjacent prescribed bottom ports which stay disjoint for a fixed positive fraction of the smaller height. Restrict each height to an interval $[cH_i, CH_i]$ and each irreducible count to a dyadic interval $[k_i, 2k_i)$, for $i = 1, 2$, with fixed $c, C > 0$. The two scales H_1, H_2 vary independently; they need not be comparable to each other. All heights and counts are at most a fixed multiple of N^2 . Put

$$K_i = H_i^{3/4}, \quad \kappa_i = \min(k_i, K_i), \quad F = \frac{\kappa_1^2 \kappa_2^2}{K_1 K_2 \min(\kappa_1, \kappa_2)}.$$

For every fixed $\varepsilon > 0$, the critical mass in these bins, including the choices of the counts and the bounded endpoint modifications, is at most $C_\varepsilon N^\varepsilon F$. At each bridge with sufficiently

large count, the proof reserves a group of irreducibles for a height-concentration test. At bounded counts the displayed height factor below is bounded below by a positive constant, so its inclusion only changes the overall constant. Suppose an interval of at most $C(1+r)$ allowed total heights on bridge i is measurable with respect to the other bridge, the local records, any externally attached remainders fixed up to translation, and the increments of bridge i outside this group. Imposing that interval changes the joint mass bound to

$$C_\varepsilon N^\varepsilon F \min\{1, (1+r)\kappa_i^{-4/3}\}. \quad (35)$$

The reserved group is unused by the avoidance and large-height estimates, and its horizontal displacement remains unconditioned and is summed over. The estimate is therefore a joint bound; it does not assert independence after conditioning a bridge on its total height. With a single nonempty bridge the corresponding factor is κ_i^2/K_i and the same optional height factor is available. Bounded one-row pieces cost a constant and need no smoothing. There is no additional translation sum at a joining port.

The third input is stronger than a product of unconditioned bridge masses. Its proof uses adjacent-prefix avoidance, height tails and a reserved height-concentration estimate. The bridge-mass and one-piece estimates in (32) follow from Theorem 2.1 and Proposition 2.4, respectively. In this section we use it only for two ends created at one minimum. The turning-chain estimate of Proposition 5.10, its two-end specialization in Corollary 5.11, and Propositions 5.12 and 5.13 provide this rule and the absolute count and fast-path bounds. Those arguments concern unnormalized masses and do not use the thermal conclusion. The later uniform-law transfer uses these estimates as well.

6.2 The terminal remainder and an exact cancellation

A *half-plane end* begins at a prescribed bottom port, visits vertices strictly above its line and ends at an arbitrary vertex. Its last piece after the highest once-crossed positive row boundary is a half-plane end with no positive renewal boundary. If there is no such boundary, the whole end is the last piece. Let U_N denote the sum of z -weights of these no-renewal ends, and let H_N denote the sum over all half-plane ends. Both are finite by (33); also $U_N \geq z$, from the end consisting of its first vertex.

Lemma 6.1 (Half-plane factorization). *With the conventions above,*

$$H_N = \frac{U_N}{q_N}.$$

If a particular ordered tuple of irreducibles has total vertex length ℓ , the probability under the normalized half-plane z -law that it is the initial tuple equals z^ℓ .

Proof. Factor at all once-crossed positive row boundaries. The factors before the last remainder are strict irreducible bridges, and this factorization is unique. Conversely, adjoining any finite list of irreducibles and a no-renewal end gives a half-plane end. The interiors of successive pieces lie in disjoint height slabs, and the remainder lies above the last joining line. Thus weights multiply and no avoidance correction is needed. A list of j unspecified irreducibles has z -mass $(1 - q_N)^j$, whence

$$H_N = U_N \sum_{j \geq 0} (1 - q_N)^j = U_N / q_N.$$

For a prescribed tuple, all possible later pieces together form an arbitrary half-plane end. Its numerator is $z^\ell H_N$, and division by H_N gives the claimed probability. $\square$

This is the same factorization that defines the infinite renewal law, with the honeycomb port convention supplying the adaptation of the finite-prefix argument in [5, Propositions 2.2–2.3]. For a fixed tuple, $z^\ell \rightarrow \rho^\ell$. Moreover the number of complete irreducibles has the geometric law $q_N(1 - q_N)^j$, so it tends to infinity in probability. Hence the fugacity prefix law converges to independent critical irreducibles. This cancellation does not involve a local estimate at any prescribed total length. Theorem 8.1 uses this cancellation to prove the half-plane thermal length, spatial and positive-moment laws.

6.3 A recoverable lower bound for the plane partition function

Lemma 6.2. *For all sufficiently large N ,*

$$Z_N \geq (q_N^{-1} - 1)U_N^2 \geq \frac{U_N^2}{2q_N}.$$

Proof. Take a nonempty bridge from a fixed lower port to an arbitrary upper port. At the lower port attach the reverse of a no-renewal end in the lower exterior half-plane. At the upper port attach a no-renewal end in the upper exterior half-plane. The three pieces have disjoint vertex sets, their weights multiply, and the result is an oriented plane walk.

The construction is injective in the translation convention. In the resulting path, the lowest and highest horizontal row boundaries crossed exactly once are the two attachment lines. There are no single-crossing lines farther outward because both attached ends have no renewals. Recover those two lines from the output, cut at their crossing ports, and translate the lower port to the fixed anchor. This recovers the central bridge and both ends. Internal renewals of the central bridge do not affect this recovery.

The total z -mass of all nonempty bridges is

$$\sum_{j \geq 1} (1 - q_N)^j = q_N^{-1} - 1.$$

Multiplying by the two end masses proves the first bound. Since $q_N \rightarrow 0$, the second follows. $\square$

The factor U_N may be large and is not estimated separately. Its appearance twice in this lower bound is what allows the next numerator estimate to be useful.

6.4 Small endpoint displacement has small relative mass

Let $V(\gamma)$ be the absolute difference of the endpoint heights in row units. Since $d_0 V \leq A \leq D$, a lower bound for V gives an endpoint-distance lower bound.

Lemma 6.3 (Two-end estimate). *For every $\varepsilon > 0$, all sufficiently large N and every $1 \leq r \leq N^2$,*

$$\sum_{\substack{\gamma \text{ plane} \\ L(\gamma) \leq N^2, V(\gamma) \leq r}} w_z(\gamma) \leq C_\varepsilon N^\varepsilon (1 + r)^{3/4} U_N^2.$$

Proof. If either endpoint attains the minimum height, use that endpoint as the minimum, reversing orientation if necessary. Otherwise cut at the first internal vertex attaining the minimum. This gives either one half-plane end or two disjoint half-plane ends. The passage between vertex endpoints and ports uses bounded local modifications, with bounded inverse multiplicity and bounded weight factors, uniformly for $\rho/2 \leq z < \rho$.

Here is the local modification, illustrated in Figure 2. Write

$$u = (1, 0), \quad v = (1/2, \sqrt{3}/2), \quad P_{ij} = iu + jv, \quad R_{ij} = P_{ij} + (u + v)/3.$$

The neighbors of P_{ij} are $R_{ij}, R_{i-1,j}, R_{i,j-1}$. An internal lowest vertex must be of type P , because a vertex of type R has only one ascending bond. Translate the minimum to P_{00} . Keep the branch toward $R_{-1,0}$ and move the start of the branch toward R_{00} to P_{10} . If P_{10} is absent, append its free bond to R_{00} . If it is present, it cannot be an endpoint, by our initial choice of case. Minimality then forces it to use both ascending bonds, so this branch already reaches it through R_{00} ; trim that bounded initial segment. Attach the two incoming half-edges from below. The resulting ends start at adjacent ports and remain disjoint. The modification changes only boundedly many vertices, preserves their far endpoints and is undone from a bounded local record.

If a minimum is an endpoint of type P , its incoming bottom port gives the one-end representation directly. If it has type R and the path is nontrivial, its first bond is the unique ascending bond; trim that bond and start at its midpoint. No internal vertex can be a lowest R . If the other endpoint is also a lowest R , trim its last descending bond as well. All remaining vertices then lie strictly above the new starting cut. A singleton walk is handled separately at bounded cost. These operations change the endpoint heights by at most a fixed constant.

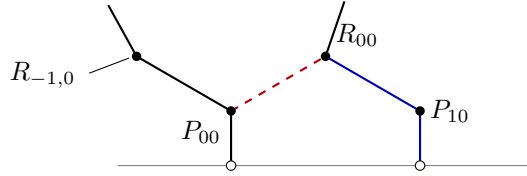


Figure 2: Separating the two branches at an internal minimum. When P_{10} is absent, replace the dashed bond by the blue bond and start the second branch at the right port. When P_{10} is already present, remove the two-bond initial segment through R_{00} instead. The continuations drawn above are schematic.

First suppose there is one end. On the event $V \leq r$, its last renewal is at height at most $C(1+r)$. After removing its no-renewal remainder and dropping the restriction on that remainder's final vertex, the mass is at most

$$CU_N \sum_{h \leq C(1+r)} B_h \leq C'(1+r)^{3/4} U_N \leq C''(1+r)^{3/4} U_N^2,$$

using (32) and $U_N \geq \rho/2$.

Now suppose there are two ends. Remove their no-renewal remainders and fix those remainders up to translation. Their z -weights will be summed at the end. Discard all avoidance constraints involving a remainder and the remaining total-length restriction, but retain the resulting $O(N^2)$ cap on bridge heights and piece counts. Drop the bridge damping factors $e^{-L_i/N} \leq 1$, so their weights are critical and the third input applies. The remaining two bridges have adjacent starts. If both are nonempty, they remain disjoint throughout the common height range. Put their heights and counts into dyadic bins and use the input's notation. If $a = \min(\kappa_1, \kappa_2)$ and $M = \max(\kappa_1, \kappa_2)$, then $K_i \geq \kappa_i$ gives

$$F = \frac{\kappa_1^2 \kappa_2^2}{K_1 K_2 a} \leq M.$$

If M is bounded by the fixed count threshold of Proposition 5.10, the bound without a concentration test is already $O_\varepsilon(N^\varepsilon)$ and suffices. Otherwise choose a bridge i attaining M , with a fixed tie rule;

its count is large enough to supply the reserved group. In the integration for (35), fix the other bridge, the local records and the increments of bridge i outside its reserved group; the two remainder shapes are already fixed. If h_j is the other bridge's height, the endpoint-height condition restricts the total height of bridge i to

$$[h_j + b - r - C_0, h_j + b + r + C_0],$$

where C_0 bounds the endpoint-height changes and b is determined by the two remainder endpoint heights and the fixed local records. Subtracting the fixed height of the other increments gives an interval of the same size for the reserved group's height. Its horizontal displacement is still summed over, so this is precisely the permitted concentration test. Thus (35) bounds the mass in this pair of bins by

$$C_\varepsilon N^\varepsilon M \min\{1, (1+r)M^{-4/3}\} \leq C_\varepsilon N^\varepsilon (1+r)^{3/4}.$$

The last inequality follows by considering $M \leq (1+r)^{3/4}$ and its complement. If only one bridge is nonempty, its factor $\kappa^2/K \leq \kappa$ gives the same estimate: bounded κ needs no concentration test, and larger κ permits the calculation above. If neither is nonempty, the mass is at most one before the remainder weights are restored.

Because $L \leq N^2$, there are only $O((\log N)^4)$ pairs of height/count bins. Choose the exponent slack in the geometric input smaller than the desired ε and absorb this factor. Summing the two fixed remainder weights gives U_N^2 . Their translations were determined by the bridge tips; no free placement factor is introduced. This proves the bound. $\square$

6.5 Length, spatial scale and moments

Theorem 6.4 (Unrestricted thermal law). *Choose an ordinary plane honeycomb walk from a fixed vertex with probability proportional to $(pe^{-1/N})^n$, where n is its edge length. For every $\xi > 0$, its vertex length L , endpoint distance A and diameter D satisfy*

$$\mathbb{P}_N\{N^{1-\xi} \leq L \leq N^{1+\xi}, \quad N^{3/4-\xi} \leq A \leq D \leq N^{3/4+\xi}\} \longrightarrow 1.$$

For every fixed $p > 0$,

$$\mathbb{E}_N L^p = N^{p+o(1)}, \quad \mathbb{E}_N A^p = N^{3p/4+o(1)}, \quad \mathbb{E}_N D^p = N^{3p/4+o(1)}.$$

Proof. We use the equivalent vertex-weighted translation-class law. First record a consequence of (33). For each $\epsilon > 0$, $p \geq 0$ and $B > 0$,

$$\sum_{m > N^{1+\epsilon}} m^p a_m e^{-m/N} = O_{\epsilon,p,B}(N^{-B}). \quad (36)$$

Indeed choose $0 < \delta < \epsilon/(1+\epsilon)$. Uniformly for $m \geq N^{1+\epsilon}$, one has $m^\delta \leq m/(2N)$ for all large N . The summands are then at most $C_\delta m^p e^{-m/(2N)}$, whose sum has faster than polynomial decay. In particular, paths with $L > N^2$ have negligible absolute mass and negligible mass after multiplication by any fixed length power. Since Lemma 6.2 gives $Z_N \geq c/q_N$, this remains true after normalization.

Take $r = N^{3/4-\delta}$ with fixed $0 < \delta < 3/4$. Lemmas 6.2 and 6.3 give

$$\mathbb{P}_N\{V \leq r, L \leq N^2\} \leq C_\varepsilon N^\varepsilon q_N (1+r)^{3/4} = N^{\varepsilon-3\delta/4+o(1)}.$$

Choose $\varepsilon < 3\delta/4$ and add the tail just discarded. It follows that $V \geq N^{3/4-\delta}$ with probability tending to one. After slightly decreasing the exponent tolerance to absorb d_0 , the same lower bound holds for A .

We next obtain the upper spatial estimate from the absolute fast-path bound. Fix $\xi > 0$. Choose small positive ϵ, η such that

$$(1 + \epsilon)(3/4 + \eta) < 3/4 + \xi.$$

For lengths $m \leq N^{1/2}$, the deterministic estimate $D \leq Cm$ already gives the desired upper bound for large N . For $N^{1/2} < m \leq N^{1+\epsilon}$, any violation of $D \leq N^{3/4+\xi}$ implies $D > m^{3/4+\eta}$. Summing (34) over this range gives a faster than polynomially small absolute mass. Lengths above $N^{1+\epsilon}$ are handled by (36). Division by Z_N proves the upper spatial assertion.

The same argument gives the lower length estimate, rather than assuming it from the discount scale. Fix $0 < \xi < 1$, choose $\eta, \delta > 0$ so small that

$$(1 - \xi)(3/4 + \eta) < 3/4 - \delta, \quad \delta < 1/4.$$

On the typical event $A \geq N^{3/4-\delta}$, a walk of length $m \leq N^{1/2}$ is impossible for large N . If instead $N^{1/2} < m < N^{1-\xi}$, that endpoint event forces $D > m^{3/4+\eta}$, up to fixed unit factors absorbed by the strict inequality above. The summed exceptional mass is again negligible by (34). Hence $L \geq N^{1-\xi}$ with high probability. The upper length bound follows from (36). Larger tolerances follow from smaller ones.

For moments, the lower probability estimates give the respective lower exponents. For the upper length moment, restrict to $L \leq N^{1+\epsilon}$ and use (36) on its complement:

$$\mathbb{E}_N L^p \leq N^{p(1+\epsilon)} + o(1).$$

For the spatial upper moments, repeat the fast-path split with the additional factor $D^p \leq C_p L^p$. On the exceptional paths of polynomially large length, the factor is harmless in the sum of (34); on the damped long-length tail use (36) with this p . The remaining paths obey $D \leq N^{3/4+\xi}$, apart from the deterministic short-length class. Thus

$$\mathbb{E}_N A^p \leq \mathbb{E}_N D^p \leq N^{p(3/4+\xi)} + o(1).$$

Let the positive slacks decrease to zero. This completes the proof. $\square$

The proof used the same irreducible law as exact-length conditioning, but a different denominator. It required no estimate of U_N , no local renewal mass at length N , and no inversion of a density-one statement. The part shared with exact conditioning is the renewal model and its increment estimates; the geometric recovery and two-end smoothing are the additional work needed for the unrestricted thermal law.

7 Censored arms and absolute spatial estimates

The density-one uniform law will require a bound on the number of ways to undo an insertion into a walk. Two proposed cuts are valid only if the exterior arms remain disjoint when one is reflected back toward the other. We estimate this constraint using independent irreducible histories stopped just before they reach a prescribed height. Lemma 7.4 gives their survival probability; the kernel calculation in Section 7.3 transfers it to weighted bridge families. These are the inputs to the insertion argument in Section 8.2.

We use the finite calibrated estimates of Section 4, with $a = 3/4$, $\vartheta = 1/4$, $d_* = 4/3$, and $\alpha = 9/16$. The same histories also give an independent proof of the absolute spatial estimates proved in Section 5. We include that proof to retain a route from the calibrated inputs alone. Throughout, $\mu(\omega) = \rho^{|\omega|}$ counts visited centers, ports carry no weight, and the source is fixed. All masses precede normalization.

7.1 Finite inputs and the rapid-travel bound

The absolute estimate supplied by this route is the following. It will control upper tails and positive moments after the probability normalizers have been identified.

Theorem 7.1 (Absolute suppression of rapid travel). *For every fixed sufficiently small $\xi > 0$ there are $c, C > 0$ such that the unnormalized critical mass of all plane self-avoiding walks rooted at a fixed center obeys*

$$\sum_{R \leq \text{diam } \omega \leq 2R, |\omega| \leq R^{d_* - \xi}} \mu(\omega) \leq C \exp(-R^c). \quad (37)$$

Here $|\omega|$ counts centers; changing to edge count changes the weight by a fixed factor. The proof will follow the survival and kernel estimates below. We first record the finite geometric inputs and their concentration consequences. We will only need sufficiently large length/distance parameters; all constants below may change by fixed geometric/unit factors.

Keep horizontal integral levels y . We use $B_m \asymp (1+m)^{-\vartheta}$, the exponential diameter cutoff for strict bridges in strips, $\Pr(D > r) \leq Cr^{-a}$, $1 - \mathbb{E} \exp(-tH) \asymp t^a$ and $1 - \mathbb{E} \exp(-tL) = t^{\alpha+o(1)}$. In particular $\Pr(H \geq r) \asymp (1+r)^{-a}$. For the lower tail comparison, in $\mathbb{E} \min(1, H/T) \gtrsim T^{-a}$ the contribution of $H < \delta T$ is at most $C\delta^{1-a}T^{-a}$ by the upper bound integrated in height; take δ sufficiently small.

Angular growth and finite chord moments. The finite calibrated input also supplies angular growth [18, Section 11, Recoverable half-plane growth]. For two signed lattice normals making angle $\pi/3$, fix a sufficiently small relative transverse tolerance $\gamma > 0$. There are positive constants $r_\gamma, K_\gamma, c_\gamma, C_\gamma, C_{1,\gamma}$ such that, for $r \geq r_\gamma$ and each admissible grid target $S \geq K_\gamma r$, the construction starts with prefixes at scale r from the fixed source port on the first wall and continues them to the exact wall of the second normal at distance S , with terminal port free. All centers stay in the first open half-plane and strictly before the target wall; the mass is at least $c_\gamma S^{-1/4} (S/r)^{-C_{1,\gamma}}$ and the diameter at most $C_\gamma S$. During continuation the transverse distance from the second-normal ray through the selected prefix endpoint is at most γu , where u is the second-normal level measured from the original source. The joining cuts are recovered as the first qualifying singly crossed walls. The finite flux identity applies on the simple polygonal domains used below; its coefficient at an exit is the real part of $\exp(i(3/8) \text{ turn})$, after a fixed common phase rotation if needed.

Finally, the finite visit identity and planar nesting bound give [18, Proposition 11.1]

$$\sum_{\omega \text{ a boundary chord in } Q_R} \rho^{L(\omega)} L(\omega) \leq CR^{25/12}, \quad (38)$$

where Q_R is a convex lattice domain with boundedly many sides and diameter $O(R)$, and both boundary ports are summed. Averaging over order- R translates of a fixed boundary source gives the $CR^{13/12}$ bound used below for paths of diameter at most R . Using a growing diameter cutoff may add logarithmic factors; it does not produce an unrestricted bounded-factor first bridge moment.

Concentration of independent increments.

Lemma 7.2 (Height and planar displacement atoms). *For $k \geq 1$ independent whole irreducibles, with atoms taken on the actual support lattice,*

$$\sup_j \Pr(S_H(k) = j) \leq Ck^{-1/a}, \quad \sup_w \Pr(S_{(H,X)}(k) = w) \leq Ck^{-2/a}.$$

Proof. We first obtain a lower tail in the transverse direction. The angular growth input constructs a positive family of strict bridges of height h whose transverse endpoint displacement has magnitude at least ch . Here are the two stages of that construction.

Start from the bottom port with normals $n = y$ and u making angle $\pi/3$, at scale c_*h . Continue to each exact u -level in $[(2 - 2\delta_*)h, (2 - \delta_*)h]$. Choose the fixed $\delta_* > 0$ small, then c_* and the relative transverse error much smaller than δ_* . The initial part has diameter $O(c_*h)$, and the u -normal ray gains y -height at slope $1/2$. Thus every path remains in $0 < y < h$ and ends at transverse distance at least ch , with remaining height between $\delta_*h/4$ and $2\delta_*h$. Each target has mass at least $ch^{-\vartheta}$.

Keep only the prefix to the first qualifying single-crossing u -cut in that target bin, testing the confinement and terminal-position conditions just stated. The total discarded suffix mass is at most $\sum_{j \leq Ch} B_j \leq Ch^a$. Summing first over the order- h targets therefore leaves a family of distinct prefixes of mass at least a positive constant. From each such prefix, use the angular construction with normals u, n to reach $y = h$ exactly. This extension lies strictly beyond the selected u -cut, has diameter $O(\delta_*h)$, and has mass at least $ch^{-\vartheta}$. Its starting scale is a small fixed fraction of the remaining height, so its diameter constant is independent of δ_* . For small enough δ_* , the full path remains in $y > 0$ and retains transverse displacement at least $c'h$. The joining cut is recoverable. Hence

$$B_h\{|X_{\text{total}}| \geq c'h\} \geq ch^{-\vartheta}.$$

To pass from this bridge family to one irreducible, use $|X_{\text{total}}| \leq \sum_i |X_i|$. On the displayed event, $\sum_i \min(1, |X_i|/h)$ is bounded below by a positive constant. The marked-reward identity and the bridge-mass bound give

$$\begin{aligned} ch^{-\vartheta} &\leq C \sum_{j \leq h} \mathbb{E}_p[\min(1, |X|/h); H = j] \sum_{b+d=h-j} B_b B_d \\ &\leq Ch^{2a-1} \mathbb{E}_p \min(1, |X|/h). \end{aligned}$$

Thus $\mathbb{E}_p \min(1, |X|/h) \geq ch^{-a}$.

The off-axis construction and the height estimate must now be combined uniformly over directions. Put $f_r(z) = \min(1, |z|/r)$. For a unit vector (v_1, v_2) let $W = v_1 H + v_2 X$. Reflection in the starting normal preserves the irreducible law and changes X to $-X$. Since

$$f_r(A + B) + f_r(A - B) \geq f_r(A), \quad f_r(A + B) + f_r(A - B) \geq f_r(B),$$

and at least one $|v_i|$ is at least $1/\sqrt{2}$, we obtain uniformly in (v_1, v_2)

$$\mathbb{E} f_r(W) \geq cr^{-a}, \quad p(|W| > r) \leq Cr^{-a}.$$

For the second bound use $|W| \leq CD$ with the fixed coordinate units. Apply the first bound at scale δr . Choose fixed $\delta > 0$ small enough that $c\delta^{-a} > 2C$. The contribution of $|W| > r$ is then absorbed. The contribution of $|W| < c'r$ is at most $C\delta^{-1}(c')^{1-a}r^{-a}$, by the integrated upper tail. Choosing c' sufficiently small after δ proves

$$p(c'r \leq |W| \leq r) \geq c''r^{-a}$$

with constants independent of the direction.

Let $\phi(t) = \mathbb{E} e^{it \cdot (H, X)}$. For small $t \neq 0$ use the last annulus in direction $t/|t|$ with $r = 1/|t|$, and symmetrize against an independent copy of bounded diameter. The absolute phase difference is between $c'/2$ and 2 for sufficiently small $|t|$. Consequently

$$1 - |\phi(t)|^2 = \mathbb{E}[1 - \cos(t \cdot ((H, X) - (H', X')))] \geq c|t|^a.$$

The random vector $(H, 2X)$ lies in $\mathbb{Z}^2$. Its support differences have rank two: otherwise a nonzero projection would be constant, contrary to the directional annuli. A rank-two subgroup of $\mathbb{Z}^2$ has finite index. Hence on the Fourier torus $|\phi| = 1$ at finitely many points. At each such point all support phases agree, so the modulus nearby is a translate of the modulus at zero; on the compact complement it is uniformly less than one. Fourier inversion thus bounds every atom by a constant times

$$\int_{\mathbb{R}^2} e^{-ck|t|^a} dt = C' k^{-2/a}.$$

The one-dimensional argument for H , allowing its own finite set of peaks, gives $Ck^{-1/a}$. No conditioning on the length of any irreducible has been used. $\square$

Lemma 7.3 (Short bridges at a specified height). *For fixed small $\delta > 0$, some $v_0, \sigma_0 > 0$ give uniformly for $S^{1-v_0}/2 \leq m \leq CS$,*

$$B_m(|\omega| \leq S^{d_*-\delta}) \leq Cm^{-\vartheta} S^{-\sigma_0}. \quad (39)$$

Proof. Here and below $B_m(\cdot)$ denotes restricted bridge mass. First sum over counts $k > m^{a-\theta}$. Exponential damping in length bounds the short-length event by

$$p\{S_L(k) \leq S^{d_*-\delta}\} \leq \exp\{1 - kq_L(S^{-d_*+\delta})\}.$$

For $m \geq S^{1-v_0}/2$, choose $\theta, v_0 > 0$ sufficiently small after δ . The exponent is then at most $-Sc$ for some $c > 0$. There are at most m counts.

For $k \leq m^{a-\theta}$, put $t_0 = m^{1-z}$ with $0 < z \ll \theta$. If all height jumps are at most t_0 , positive exponential weighting bounds the probability of attaining m by

$$e^{-m/t_0}(1 + Ct_0^{-a})^k \leq \exp\{-m^z + Cm^{-\theta+az}\},$$

which is smaller than every inverse power of m . Otherwise select a jump exceeding t_0 and apply the height atom bound to the other increments. Its cost is at most $Ckm^{-a(1-z)}k^{-1/a}$; for $k = 1$, the height tail gives the same bound directly. Summing these terms yields

$$Cm^{-a(1-z)} \sum_{k \leq m^{a-\theta}} k^{1-1/a} \leq Cm^{a-1-\theta(2-1/a)+az}.$$

Since $2 - 1/a > 0$, choose z so that the exponent has a fixed saving below $a - 1 = -\vartheta$. The comparison $m \geq S^{1-v_0}/2$ turns that saving into $S^{-\sigma_0}$, as required. $\square$

7.2 Two censored arms

Lemma 7.4 (Censored two-arm survival). *Start two independent sequences of irreducibles from the same bottom port p with first center v . On each keep its history of completed pieces up to heights **strictly less** than g , not including the first jump reaching or passing g . Fix roles left, right. Impose the following constraints whenever $0 < d_1 < d_2 < g$ are cut heights in the left, right history respectively. The prefixes to them, apart from the shared start through v , must go disjointly (leaving v on the two respective sides). At level d_1 all crossings of the right prefix must be to the right of x_1 , the terminal horizontal coordinate of the left prefix. Write $f(g)$ for the probability of these constraints. Then for each $\eta > 0$*

$$f(g) \leq C_\eta(1 + g)^{-a+\eta}. \quad (40)$$

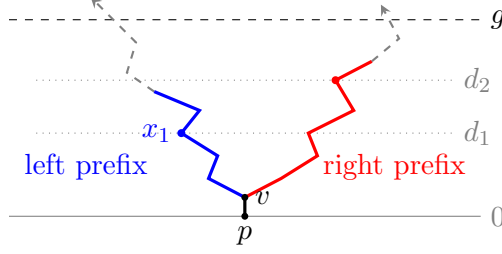


Figure 3: The two histories share their first center v . Only complete irreducibles ending strictly below g are retained; the dashed continuations are the omitted crossing pieces. At a tested left renewal level d_1 , all crossings of the right prefix ending at a higher tested level d_2 must lie right of x_1 . The paths are schematic, rather than lattice embeddings.

Proof. There are two estimates to combine. A test with substantial transverse clearance is rare by a positive flux comparison. After any candidate for the last such test, each later test costs a power through either a large height increment or a small transverse interval. Those later costs use fresh coordinate blocks.

Test scales. Put $s = g$ large. Fix constants $0 < \ell \ll \lambda \ll b \ll \kappa_0 \ll 1$, in particular $b > 2\lambda$, $\lambda \geq \ell$, and $\kappa_0 > b + 2\lambda + a\ell$. These inequalities ensure that the normal heights are separated, all tests lie below g , and each proximity test has the saving used below. Use $r_j = s^{\kappa_0 + j\ell}$, $0 \leq j \leq J = \lfloor (1 - 2\kappa_0)/\ell \rfloor$. Test at piece indices sampled independently, uniform over integers $[A_{i,j}^a, 2A_{i,j}^a]$ on the two uncensored sequences, where $A_{1,j} = r_j$, $A_{2,j} = r_j s^b$ respectively. These index ranges are increasing and separated for large s . Write d_i at a test for its cumulative heights; call them normal if each is in $[A_{i,j} s^{-\lambda}, A_{i,j} s^\lambda]$, implying $1 \ll d_1 \ll d_2 < g$. The chance of falling below such a range at any test is superpolynomially small by height damping. At normal heights we may also require both prefix diameters at most $d_i \log^2 s$, except at superpolynomially small cost (sum the exponential strip diameter tail at the possible exact heights; a prefix at the given index has precisely its irreducible-product mass).

Separated tests and positive flux. Call a test separated if these normality, diameter and two-prefix constraints hold and the horizontal clearance between x_1 and the leftmost right-prefix crossing at d_1 is at least χd_2 , where $\chi = s^{-\kappa_0}$. Unconditionally, its probability is at most

$$C r_j^{-a} s^{\kappa_0 + a\lambda} (1 + \log s)^C. \quad (41)$$

Here are the gluing details. Sort d_2 into dyadic bins $d_2 \asymp H$. Translate both paths together so the line at d_1 becomes $y = 0$, and a lattice corner chosen strictly in the clearance gap becomes the origin; this allows at least $c\chi H$ placements per input. The top line is now $y = \Delta = d_2 - d_1 \asymp H$. Work inside a parallelogram of triangles with this top, bottom at $-CH$ (rounded outward), and slant side walls allowing all paths by width $O(H \log^2 s)$ with ample margin. Slit it along $y = 0$ from its left wall to the origin (split the corresponding ports into two boundary copies). Joining the prefixes at v , removing the common initial half-edge, gives a simple chordal arc from a top wall port to the lower lip of the slit. It has no other boundary contact. To mark the choice of v among deepest vertices, use the hexagonal face centered at the bottom-left corner of v 's triangle. Its only vertices at height $\geq y(v)$ are v , a second minimum-height candidate at its left, and their common higher neighbor. The edge from v to that neighbor is used; if the second candidate is visited it must also use its edge

to the same neighbor (minimum height and not an endpoint). Thus exactly a contiguous one- or two-edge arc of the face is present. Replace by the complementary face arc, giving a unique deepest vertex and hence identifying the face. This lies far from the domain boundaries. The change and joining have bounded weight costs. Inverse multiplicity is bounded since we can undo at that face and recover both original prefixes by translating v back (and the slit placement is determined).

Figure 4 shows the resulting domain and the clipping operation used in the flux comparison.

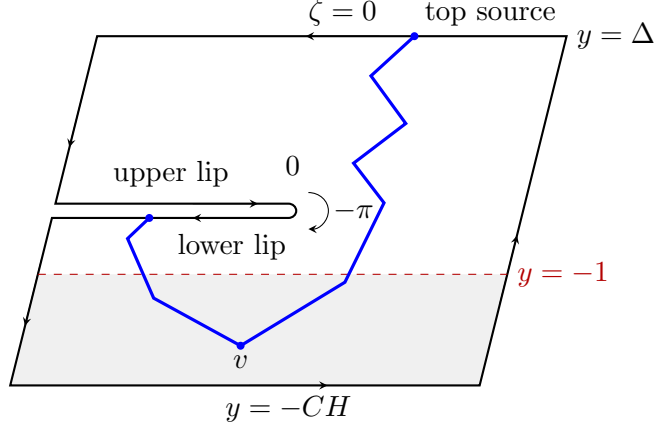


Figure 4: The slit domain for a separated test, before the bounded modification at the deepest face. The joined prefixes form the blue arc from the top source to the lower lip; their original common minimum v lies below the clipping line. Boundary arrows specify the counterclockwise traversal, including its clockwise half-turn at the slit tip. The slit is drawn with positive thickness, and the vertical spacing is not to scale. The lifted-angle calculation in the proof supplies positivity of the flux coefficients.

From each fixed top source, the mass in this slit domain of arcs as produced is $\leq CH^{-\vartheta}$. Indeed they are lost on clipping to $y \geq -1$. The boundary extension identity from that source has uniformly positive real coefficients before and after clipping, unchanged to retained exit ports. Traversing counterclockwise starting west on the top, the lifted tangent travels down the left side, east along the upper lip, turns clockwise by π to west along the lower lip, continues down the left wall then around via the bottom, right wall and top. All lifted boundary tangent angles ζ relative to the start lie in $[0, 2\pi]$. Closing an arc along that boundary gives its turn as $\zeta - \pi$: the two endpoint half-edge corners each add $\pi/2$. Consequently its real flux coefficient is $\cos((3/8)(\zeta - \pi)) \geq \cos(3\pi/8) > 0$, which supplies the required uniform positivity.

More explicitly, the finite extension identity sums with the slit adjacencies forbidden (detour reversals valid since the source on the top is approached from outside any such cycle); slit ends of half-edges are boundary terms. For the simple arc turn calculation one can thicken the slit by an arbitrarily narrow indentation, trimming terminal half-edges at lip ports slightly if needed: other arc edges stay off the slit, including its corners, and the lifts at source and exit are as stated. Thus this also follows by closing in a simple boundary polygon.

Positive lost mass is bounded by the new exits' mass, and those exits are a subset of strict downward bridges of height $\Delta + 1$. This proves the claim. Summing over $O(H \log^2 s)$ top ports and $O(H)$ choices of Δ , and dividing by placements, gives $O(\chi^{-1} H^a \log^2 s)$ cumulative input mass. The probability of each prefix pair at the test has, besides its bridge mass, index factors at most $Cr_j^{-a}(r_j s^b)^{-a}$ by uniform sampling (piece counts uniquely specified). Since $H \leq Cr_j s^{b+\lambda}$, this gives (41).

Fresh blocks after a candidate last separated test. Choose an index p_0 and require only that its test is separated. Condition on all sampled indices and on both sequences through the test at p_0 . We do not condition on its being the last separated test. For $p_0 = -1$, expose no increments. We bound the event that the survival constraints hold and that no later test is separated.

At every $j > p_0$, either a cumulative height exceeds $A_{i,j}s^\lambda$, or the test has normal heights and its left endpoint lies within χd_2 to the left of the minimum right-prefix crossing at level d_1 . The small-height and excessive-diameter cases were already discarded. In the large-height case, some fresh index block q , with $p_0 < q \leq j$, has gained at least $c_J A_{i,j}s^\lambda$. The exposed height at p_0 is negligible compared with this threshold. Assign such a side and block to the test; there are only a bounded number of assignments, since J is fixed. If several tests select one block, keep its requirement for the largest j .

A block on side i has at most $2A_{i,q}^a$ increments. The height-sum tail therefore bounds its assigned event by

$$C_J s^{-a((j-q)\ell+\lambda)} \leq C_J s^{-a\ell(j-q+1)}.$$

Thus this one block pays for every test in the index interval $[q, j]$. For each index outside the union of these intervals we use the transverse proximity condition instead.

To estimate a proximity test, fix the entire relevant right sequence and all earlier left blocks. The last left block contains order r_j^a fresh increments, so its joint endpoint atom is at most $C r_j^{-2}$ by Lemma 7.2. There are at most $C r_j s^\lambda$ allowed left heights. At each one, the right sequence specifies a transverse interval of length at most $s^{-\kappa_0} r_j s^{b+\lambda}$, which is at least one and hence also bounds the number of lattice sites up to a constant. This gives the uniform conditional bound

$$C r_j^{-2} (r_j s^\lambda) (s^{-\kappa_0} r_j s^{b+\lambda}) = C s^{b+2\lambda-\kappa_0} \leq C s^{-a\ell}.$$

The left blocks used here are distinct and have not been assigned large-height events. Their bounds multiply after the right sequence has been fixed. The required large-height estimates for the right blocks then multiply by their independence. Integrating left blocks in reverse order gives the same conclusion for assigned left blocks and proximity tests together. If $\mathcal{C}$ is the union of assigned index intervals, the resulting cost is at most

$$C_J s^{-a\ell|\mathcal{C}|} s^{-a\ell(J-p_0-|\mathcal{C}|)} = C_J s^{-a\ell(J-p_0)}.$$

Overlaps between assigned intervals only improve this upper bound.

Combining the two estimates. Every surviving pair has either a last separated test or none. For a candidate $p_0 \geq 0$, multiply its unconditional bound (41) by the uniform fresh-block bound just proved. For no separated test use that bound with $p_0 = -1$. Summing over the finitely many candidates gives

$$\begin{aligned} f(s) &\leq C_J s^{-a\ell(J+1)} + C_J \sum_{p_0=0}^J r_{p_0}^{-a} s^{\kappa_0+a\lambda} (1 + \log s)^C s^{-a\ell(J-p_0)} + s^{-M} \\ &\leq s^{-a+O(\kappa_0)+o(1)}. \end{aligned}$$

Here M can be arbitrarily large, accounting for the discarded cases. Choose κ_0 and then the smaller parameters so that the exponent loss is less than the prescribed η . Enlarging the constant covers bounded s and proves (40). □

7.3 Summing the cores between censored ends

The preceding lemma is a probability estimate for two independent histories. A bridge of prescribed height does not have independent ends. We now express its positive weight relative to the independent history laws, keeping a kernel for the pieces between them. Uniform bounds on that kernel will justify using the survival estimate in weighted counts.

Endpoint conventions and the retained turn constraints. Decompose a walk into weak bridges with center endpoints, allowing ties in height and cuts at singly crossed wall ports. Extend each piece to exterior wall ports with an outward half-edge and, if necessary, one additional center at each end. At a turn the common vertex is a height extremum, with both used edges pointing toward the interior. The same outward port can therefore be used for both virtual extensions. At a lower turn they both begin $p \rightarrow v$ and then leave on opposite slants, as in Lemma 7.4. Their virtual starts are allowed to coincide. Multiplying the strict-piece weights costs at most C^k for $O(k)$ pieces, accounting for duplicated or added centers. Recording the local extensions and trimming choices also costs C^k and recovers the original anchored walk. Each span changes by a bounded amount.

At a turn use either the vacuous threshold $g = 1$ or a threshold well below both adjacent heights. Then the tested prefixes and their continuations stay before the far endpoint modifications. They are disjoint after v because they come from disjoint arms of the original walk. For the order condition, continue the left prefix to its first hit of the higher tested level d_2 . This is a strip crosscut meeting its renewal level d_1 only once. The right arm leaves v to its right and cannot cross it, so all right-prefix crossings at d_1 lie to the right of the left endpoint. Thus the survival constraints hold. Reflection or reversal handles other turns, with at most a constant choice of roles per turn.

The density relative to independent histories. Write $T(t) = p(H \geq t)$ and $p_j = p(H = j)$. If ξ is a list of complete irreducibles of total height $b < g$, its probability as the history censored before g is

$$P_g(\xi) = \mu(\xi)T(g - b).$$

The extra factor records that the next, omitted jump reaches or passes g . In particular $T(g - b) \geq cg^{-a}$.

Consider a strict bridge of height h . Let ξ_L, ξ_R be the histories read inward from its two ends, with heights $b < g_L$ and $d < g_R$. Put $T_L = T(g_L - b)$ and $T_R = T(g_R - d)$. We choose the thresholds so that $h - b - d \asymp h$. Relative to $P_{g_L} \otimes P_{g_R}$, the remaining mass is bounded by the sum of two kernels:

$$K_h^{\text{same}}(\xi_L, \xi_R) = \frac{p_{h-b-d}}{T_L T_R},$$

$$K_h^{\text{distinct}}(\xi_L, \xi_R) = \mathbb{E}[B_{h-b-d-Y_L-Y_R}; h - b - d - Y_L - Y_R \geq 0 \mid Y_L \geq g_L - b, Y_R \geq g_R - d].$$

Here Y_L, Y_R are independent copies of H . In the first case the same irreducible crosses both thresholds; in the second, the two crossing irreducibles are distinct and the bridge between them has height $h - b - d - Y_L - Y_R$. A total-length upper bound on the original piece may also be imposed on this middle bridge, replacing B_m by its restricted mass. All other restrictions can be dropped for an upper bound.

These formulas follow by irreducible factorization. They keep the survival-tail denominators supplied by the censored laws; those factors cannot be omitted when passing from probabilities to weighted bridges. No final transverse coordinate is prescribed. The terminal port is fixed by translating the reversed suffix to the end of the middle piece, so there is no additional placement factor.

Integration rule for a chain. For a chain of pieces take the product reference measure $\mathbf{P}$ of all its left and right censored-history laws. Retain only the chosen turn indicators $\mathbf{1}_{E_v}$. Each indicator involves a disjoint pair of end histories, aligned at their common turn. Choose their thresholds from coarse data, independently of the fine height variables still to be summed. With $K_i = K_i^{\text{same}} + K_i^{\text{distinct}}$, the weighted chain sum is bounded, up to its recorded C^k endpoint cost, by

$$\int \left[\sum_{\mathbf{h} \in \mathcal{A}} \prod_i K_i(h_i; \xi_{i,L}, \xi_{i,R}) \right] \prod_v \mathbf{1}_{E_v}(\boldsymbol{\xi}) d\mathbf{P}(\boldsymbol{\xi}),$$

where $\mathcal{A}$ denotes the retained height constraints. If the bracket is at most C_* uniformly in the histories, this is at most

$$C_* \prod_v f(g_v).$$

Thus the order is: fix the histories, sum the kernels over fine heights, and only then integrate the independent turn indicators. The physical weighted bridge family itself is not asserted to have independent ends. Both the volume bound below and the insertion-label argument use this rule; the latter needs only its one-ended version.

Proposition 7.5 (Critical mass in a bounded region). *Let V_R denote the mass of all center walks from a fixed center with diameter at most R . We have for every $\nu > 0$*

$$V_R \leq \exp(C_\nu(1 + R)^\nu). \quad (42)$$

Proof. Split at a lowest point (polynomial placement cost), then decompose each direction into weak bridges by repeatedly cutting at the last occurrence of the opposite extremum of the remaining path. On each chain spans are decreasing, strictly except possibly the second equal to the first. Group consecutive spans in dyads $[s, 2s)$. In such a group of size k , $1 \leq k \leq Cs$, only impose shared-turn constraints *within* the group. Strict heights $h \asymp s$ there differ only boundedly from the weak spans. Assume s large (bounded small dyads cost a constant by the ordinary bridge bounds); take common threshold $g = \max(1, \lfloor c_* s/k \rfloor) \asymp s/k$ for small fixed c_* .

Call a piece exceptional in this sum if the threshold jumps coincide or are distinct with middle height $m < g$. Its kernel summed over the height dyad costs at most $Cs^a k^{-2a}$. Indeed the coincident case uses the tail at $h - b - d \gtrsim s$ summed at distinct heights, and the two T factors. In the other case one jump is of order s , giving conditional probability $\leq C(g/s)^a$; sum also $\sum_{m < g} B_m \leq Cg^a$. For ordinary pieces ($m \geq g$) the kernel per fixed h costs $\leq Cs^{-\vartheta}$: for $m \asymp s$ this is direct, for m smaller use $B_m \leq Cg^{-\vartheta}$ and the large jump factor $(g/s)^a$ again. For $t = k - r$ ordinary pieces with r exceptional, we retain height ordering constraints among just the ordinary ones, giving at most $C^k s^t / t!$ ordinary lists (original spans are ordered on the discrete grid, $t \leq Cs$, and bounded perturbations cost boundedly per piece). Exceptional heights sum without ordering. Multiply now by $k - 1$ pair factors (40). Uniformly from the fixed group start the total, absorbing subset choices, is at most

$$C_\eta^k s^{ak} k^{-2ar} (t!)^{-1} (s/k)^{(-a+\eta)(k-1)} \leq s^a (C'_\eta s^\eta / k^\vartheta)^k,$$

with the last bound also covering the sum over types ($1/t! \leq \exp(k)/k^t$, $2a > 1$). Summing k and multiplying over $O(1 + \log R)$ dyads on each chain proves (42), taking η sufficiently small depending on ν (terms with k above a constant times $s^{2\eta/\vartheta}$ already decay). We overcount freely between groups/chains. $\square$

7.4 Subdivision for fast crossings

Proof of Theorem 7.1. An input of (37) contains a weak bridge between a minimum and maximum in some signed normal direction of span $H \asymp R$, since the three lattice directions control diameter; take the sign so it is oriented from low to high. We call this coordinate y again. The exterior pieces cost at most V_{2R}^2 up to constants; it suffices to estimate such weak bridges from their fixed first vertex. Pay $O(R)$ for specifying the last height and normalize the first height by translation.

Take $\delta = \xi/2$, small $0 < \sigma \ll \xi$, integer $S \asymp R^{1-\sigma}$, so $R^{d_*-\xi} = o(S^{d_*-\delta})$. First divide the bridge into swings with drawdown/up threshold $s_0 = S/10$. Track running maxima from the beginning; on first detecting a fall by at least s_0 from one, mark the last corresponding peak before detection and switch to tracking running minima starting at detection, with next switch symmetrically after a rise, marking the trough, and so on. Use the marked extrema (not the detection points) and the two endpoints to cut. Every resulting swing is a weak bridge of span $\geq s_0$, with adverse increments between any earlier and later point of that swing strictly less than s_0 . Indeed, e.g. on an upswing the initial trough-to-rise-detection part lies above the trough, and strictly within s_0 above it before detection. From detection to the peak the running maximum rule gives the bounds; all heights stay above the trough and at most the peak on both parts. The reversed argument works going down, and the global endpoint extrema handle first/last swings (after detecting a fall there must still be a rise by threshold before or at the terminal maximum). Without any switch the span itself is H .

Inside each swing use all absolute grid levels spaced by S (multiples of S in wall coordinates) with interior margin $\geq 2s_0$ from both extrema. Here is the subdivision in upward orientation, symmetrically otherwise. For each such level l , if there is just one crossing, cut there at the port. Otherwise take a highest vertex A_l before the last upward crossing of l , and a lowest B_l after A_l in this swing. Then $y(A_l) > l > y(B_l)$, both are between the first and last crossings (a return occurs after A_l , all vertices after last crossing are above). The path between them is a descending weak bridge of span $< s_0$; mark it as an inserted piece. These crossing zones for successive levels occur in order since a backtrack of size S is impossible. All intermediate pieces are upward weak bridges: e.g. from B_l to A_{l+S} all vertices respect those extremal bounds, with the analogous inequalities at single crossings and swing endpoints. These **main** pieces have spans between cS and CS , by the margins, offsets $< s_0$ and the minimum full swing span if no cut. Thus throughout the original bridge there are $k \geq cH/S$ main pieces and at most k inserts.

Use strict virtual pieces with the bounded endpoint extensions described above. Specify their types/order, signs, and the following boundary data, at cost $R^{O(1)}(C(1 + \log R))^{Ck}$ before any fine coordinates at turns:

- At a turn between swings, label the coarse height interval (grid of size S) and assign boundary scale $D_j = S$.
- At the two turns for an insert, label the grid level and a common dyad $D_j \asymp 1 + |y(A_l) - y(B_l)|$ (same at both). Each height has offset $O(D_j)$ from that level.
- At the single-cut ports label the grid level; take $D_j = 1$ there and at the prescribed endpoint heights.

Coarse coordinates cost boundedly many choices per piece from those before, since increments are $O(S)$. We can include any finite endpoint/port adjustment types as labels. Thus each virtual boundary height w_j still to sum ranges over $O(D_j)$ possibilities (already fixed at external endpoints and single cuts), and for fixed next w_{j+1} has bounded multiplicity per intervening strict-piece height h , their signed difference. We restrict $h \asymp P$, where $P = S$ on main pieces, $P = D_j = D_{j+1}$

on inserts; other fine height ordering restrictions will not be needed. Every strict piece length is $\leq S^{d_*-\delta}$ by the global input bound, including its bounded adjustments.

Use (39), and fix $0 < u \ll v \ll \min(v_0, \sigma_0)$. Call a boundary active if $D_j \geq S^{1-v}$. At scale $D' = D_j$ take the threshold

$$g(D') = \max(1, \lfloor c_* D' S^{-u} \mathbf{1}_{\text{active}} \rfloor)$$

for small fixed c_* . Since each endpoint scale is $O(P)$ this ensures $h - b - d \asymp P$ at the censored kernel, and pair tests at turns are either vacuous or well below the adjacent far adjustments. Write $F(D') = C_\eta g(D')^{-a+\eta}$ as a pair bound, or 1 at unpaired boundaries with $D' = 1$.

The subdivision has reduced rapid travel to many main pieces of scale S . It remains to extract a power saving from each such piece without spending the same independent end history twice.

We sum each kernel over its *left* boundary height at fixed right height and histories from both ends (left/right here meaning first/last in traversal). Include in its numerical budget also the factor $F(D_L)$ at the first boundary of that piece. We claim a bound S^{-c_1} for this product on main pieces and active-scale inserts, for some $c_1 > 0$ uniformly at all sufficiently small η ; on inactive-scale inserts at most $C_\eta S^{2\eta}$ suffices.

Indeed in the coincident-jump case the summed kernel is at most $C g_L^a g_R^a P^{-a}$, using bounded multiplicity per height. With $F(D_L)$ this leaves $O_\eta(S^\eta (g_R/P)^a)$, giving the claim (on main pieces $g_R/P \leq C \max(S^{-v}, S^{-u})$, and on active inserts $\leq C S^{-u}$). In the distinct case sort $1 + m$ in dyads of scale M . For fixed crossing increments each m can occur boundedly often as the left height varies, with $O(D_L)$ possible heights total. Thus by $B_m \leq C M^{-\vartheta}$ we have kernel bound

$$C \min(D_L M^{-\vartheta}, M^a) = C D_L^a \min((D_L/M)^\vartheta, (M/D_L)^a).$$

Here $D_L^a F(D_L) \leq C_\eta S^\eta$ times S^{au} if active. When $M \geq S^{1-v_0}$ (39) allows multiplying by $S^{-\sigma_0}$, enough. For smaller M on a main piece or active insert, there must be a giant crossing jump $y_i \geq cP$ on some side G . Accordingly we can multiply the bound, after splitting into the two possible sides, by $C(g_G/P)^a$ by its conditional tail. On a main piece this saves S^{-av} if G inactive, otherwise S^{-au} ; on an active insert (both sides active) also S^{-au} . In the latter savings cases, if left itself is active and pays S^{au} , the factor $(M/D_L)^a \leq S^{-a(v_0-v)}$ supplies the gain. Inactive inserts just use the unconstrained dyad estimate with both boundary scales inactive. For clarity, all these cases have a common strictly positive saving before the small η loss:

$$\Lambda = \min\{au, a(v-u), a(v_0-v), \sigma_0 - au\} > 0.$$

Every dyad on a main piece or active insert costs at most $C_\eta S^{\eta-\Lambda}$. Choose η sufficiently small, then sum $O(\log S)$ dyads. This proves the claimed negative power; inactive inserts cost at most $C_\eta S^{2\eta}$.

To combine, first hold all histories fixed and sum out the fine variables in sequence from first to last using the uniform summed-kernel bounds (the last height fixed). Applying the pair constraints (40) in the product of censored-history probabilities contributes exactly the indicated upper costs allocated to the left boundaries. Taking η small, all labeling costs, inactive inserts, and endpoint weight factors are absorbed: the resulting weak-bridge sum is at most $R^{O(1)} \sum_{k \geq cH/S} S^{-c_2 k}$ for some $c_2 > 0$. This is stretched-exponentially small; (42) at the two exterior pieces preserves this, proving (37). □

8 Uniform endpoint-free laws and moments

The absolute estimates of Section 7 give upper spatial tails under every law considered here. The remaining issue for a uniform endpoint-free walk is a lower bound on its endpoint displacement.

In the half-plane we compare nearby lengths by adjoining a long initial bridge. In the plane we insert that bridge at a minimum and count the possible inverses. Both comparisons yield a lower law on a common set of lengths of natural density one. We first record the moment consequences of the absolute upper bound. For a walk γ started at γ_0 , its maximum radius is $\max_j |\gamma_j - \gamma_0|$; it lies between half the diameter and the diameter.

8.1 Upper tails, moments, and the half-plane comparison

Theorem 8.1 (Moment and half-plane consequences). *For the independent-irreducible infinite walk, and for uniform strict bridges at every sufficiently large even center length n , both endpoint distance and maximum radius have positive-moment powers $3p/4$: for either variable R_n ,*

$$\mathbb{E}R_n^p = n^{3p/4+o(1)} \quad (p > 0 \text{ fixed}).$$

The same holds for the vertex-bridge conventions of Section 4.

For endpoint-free uniform walks in the plane or half-plane, the probability of maximum radius exceeding $n^{3/4+\epsilon}$ is smaller than every fixed inverse power of n , for each fixed $\epsilon > 0$. In the half-plane there is one natural-density-one set of lengths along which endpoint distance and maximum radius have exponent $3/4$ in probability and all the preceding positive-moment exponents.

Under the half-plane law of free center length $n \geq 1$ with weights $\rho^n e^{-tn}$, length is $t^{-1+o(1)}$ and endpoint distance and maximum radius are $t^{-3/4+o(1)}$ in probability as $t \downarrow 0$; their positive moments have the corresponding powers. At fixed bridge height h , the critical bridge length is $h^{4/3+o(1)}$ both in probability and in first mean.

Proof. Absolute upper tails and moment bounds. For the infinite walk, consider the first n whole irreducibles; their concatenation contains the first n centers. Except with probability Cn^{1-A} , every one of these pieces has diameter at most n^A , where A is any sufficiently large fixed constant. The concatenation has exactly its critical bridge weight as probability, by unique factorization. Split it at its n th center. If this prefix reaches radius $n^{1/d_*+\epsilon}$, summing its critical mass over diameter dyads costs stretched-exponentially little by (37), with its slack ξ sufficiently small. The remaining suffix has diameter at most Cn^{A+1} and costs at most $CV_{Cn^{A+1}}$, including bounded endpoint weight corrections. Choose the exponent ν in (42) sufficiently small after A ; this suffix factor is absorbed by the stretched-exponential bound. Since A can be arbitrarily large, the upper tail for the first n centers is smaller than every fixed inverse power of n .

The almost-sure lower exponent of Theorem 2.7, this upper tail, and the deterministic linear distance bound give the stated moments for endpoint distance and maximum radius. For uniform strict or vertex bridges, divide the absolute fast-path bound by their previously proved polynomial lower mass bound and use their typical lower cutoff. The same argument gives the upper tail for uniform endpoint-free walks. For plane walks the total edge-weighted mass is at least one, by the connective constant and submultiplicativity. For half-plane walks it has a polynomial lower bound by inclusion of strict bridges, with one outward center appended when required by parity.

Long prefixes compare nearby half-plane lengths. Let h_n be the total ρ^n -mass of the n -center half-plane paths rooted at the prescribed inward center. Fix a small $\epsilon > 0$ and a dyad $N \leq n < 2N$. Put

$$k = \lfloor N^{\alpha-2\xi} \rfloor, \quad L_0 = \lfloor N^{1-\xi} \rfloor, \quad \alpha = 9/16,$$

with $0 < \xi < 3\epsilon/16$. Call a tuple of the first k independent irreducibles good when its total length l is at most L_0 and its height exceeds $N^{3/4-\epsilon/2}$. Its probability w_N tends to one: the length failure is at most $Ckq_L(1/L_0) = o(1)$, and height damping bounds the other failure by $\exp\{1 - cN^{3\epsilon/8-2\xi}\}$.

Adjoin to each good tuple an arbitrary nonempty half-plane continuation above its final port. These events are disjoint for different tuples, as in Theorem 2.8. Their combined probability in the uniform length- n law is therefore

$$p_n = w_N \mathbb{E}_{\text{good}}[h_{n-l}/h_n] \leq 1.$$

The distribution of l in this expectation depends on N , not on n . The volume upper bound and bridge lower bound give $|\log h_j| \leq C_\nu N^\nu$ throughout $N - L_0 \leq j \leq 2N$, for each fixed $\nu > 0$. Jensen's inequality and $1 - p_n \leq -\log p_n$ yield

$$\sum_{N \leq n < 2N} (1 - p_n) \leq -N \log w_N + \mathbb{E}_{\text{good}} \sum_{N \leq n < 2N} (\log h_n - \log h_{n-l}).$$

For each fixed shift l , the interior terms cancel. The remaining two boundary intervals have total length at most $2L_0$, so their contribution is $O_\nu(N^{1-\xi+\nu}) = o(N)$ when $\nu < \xi$. Thus the average failure probability tends to zero. On a good-tuple event, the endpoint stays above the final tuple height and hence has distance at least $n^{3/4-\epsilon}$ for large N .

Together with the all-length upper tail, this proves the claimed lower and upper cutoffs outside a vanishing fraction of each dyad. Choose exponent slacks and probability-error thresholds tending to zero sufficiently slowly over successive dyads. The retained lengths form one natural-density-one set on which the probability conclusion holds. On this same set the lower cutoff and the superpolynomial upper tail, with the linear deterministic bound, give every fixed positive moment.

Half-plane thermal and fixed-height consequences. For the half-plane Boltzmann law take $N = \lceil 1/t \rceil$ and the same good tuples. The suffix partition is exactly the original half-plane partition, so the good-prefix probability is $w_N \mathbb{E}_{\text{good}} e^{-tl} \rightarrow 1$. The volume bound makes lengths larger than $N^{1+\delta}$ superpolynomially unlikely, even after multiplication by any fixed length power. Below this cutoff, condition on length and apply the uniform upper spatial tail. Lengths below $N^{1/2}$ cannot violate the spatial upper cutoff by the linear bound, and the good-prefix lower cutoff makes them atypical. Taking δ and the speed slack sufficiently small proves the spatial upper bound and all positive spatial moments. The endpoint lower bound combined with the speed bound also rules out every length lower power $N^{1-\delta}$, giving the length statement and its moments.

At a fixed bridge height h , the short-length tail is negligible by (37), since the diameter is at least a constant times h and the denominator is $B_h \asymp h^{-\vartheta}$. To bound the mean, first restrict diameter to $r = Ch \log h$. The finite chord bound (38), applied in a common convex box to order- r translates of the fixed source, bounds the first-length mass by $r^{13/12+o(1)}$. For larger diameters, the exponential strip cutoff and the deterministic area bound make the length-weighted tail negligible when C is large. Division by B_h gives mean length at most $h^{4/3+o(1)}$ and hence the typical upper cutoff; the short tail gives the matching lower bound for the mean.

The same length-versus-scale conclusion holds for arches from a fixed half-plane source with terminal gap in $[h, 2h]$ on one side and diameter at most Ch , for sufficiently large fixed C . The confined pointwise arch lower bound gives denominator at least $ch^{-\vartheta}$, and the same translated first-moment and fast-path estimates apply. $\square$

8.2 Unrestricted walks by insertion at a minimum

A full-plane walk need not begin with a long bridge. We create one by reflecting the part before a minimum and separating it from the part after that minimum. An output can admit many such insertions in reverse. We first bound this multiplicity, then compare masses at nearby lengths.

Theorem 8.2 (Unrestricted spatial laws by refolding). *For full-plane uniform walks rooted at a fixed center, there is a set $\mathcal{N} \subset \mathbb{N}$ of natural density one such that, as $n \rightarrow \infty$ through $\mathcal{N}$, endpoint distance and maximum radius are $n^{3/4+o(1)}$ in probability. For every fixed $p > 0$ their p th moments are $n^{3p/4+o(1)}$ on the same set. Under the normalized full-plane law with edge weights $(\rho e^{-t})^n$, as $t \downarrow 0$ the length is $t^{-1+o(1)}$ and both spatial variables are $t^{-3/4+o(1)}$ in probability, with corresponding positive-moment powers. The thermal assertion holds without exceptions in t .*

Write c_m for the total critical mass $\sum \rho^m$ of m -edge walks from a fixed center. Lattice symmetry identifies the two starting center types. We have $c_m \geq 1$ and, by (42),

$$0 \leq \log c_m \leq C_\nu(1+m)^\nu \quad (\nu > 0).$$

Throughout this argument $d_* = 4/3$, $a = 3/4$, and $\vartheta = 1 - a$.

The insertion and its inverse. Fix one lattice normal. Walks whose endpoints realize its two global height extrema, in either order, have total critical mass $o(1)$ at length $m \rightarrow \infty$. Indeed their bounded endpoint extensions are strict port bridges. Their exact-length masses u_l tend to zero: each fixed piece-count term tends to zero, and the remaining terms are dominated by the summable envelope $k^{-1/\alpha+o(1)}$ from (4). Call the walks not omitted eligible.

Write a_0, b_0 for the input endpoints. Choose an internal global height extremum of each eligible walk by a fixed deterministic rule. Choose the sign of the normal coordinate y so it is a minimum v ; the two signs will be counted separately. The two used edges at v are the rising slants. Let P be the wall just below v , with its port adjacent to v . All input vertices lie strictly above P . Reflect the incoming arm, including v , across P , insert a strict upward bridge from that port to a port on a wall Q , and translate the outgoing arm, including v , by the bridge's port increment. The three parts lie in disjoint open slabs and form a self-avoiding walk. Translate it once more to a fixed output center of the type produced by reflection; its mass c_n agrees with the other starting type by lattice symmetry.

Use bridges of center length l satisfying

$$s \leq Q - P \leq 2s, \quad l \leq C_0 s^{d_*}.$$

The resulting edge length is $n = m + l + 1$: the minimum center has been duplicated. Its weight is the input weight times ρ^{l+1} . Let A_l be the total critical mass of the allowed inserted bridges of length l , and put $A = \sum_l A_l$. These masses do not depend on the input. Proposition 4.1 gives confined bridge mass $\asymp s^{-\vartheta}$ and first-length mass at most $C s^{13/12}$ at each height in the bin. Thus, for sufficiently large fixed C_0 ,

$$A \asymp s^a.$$

Figure 5 shows the construction before the common translation that fixes the output root.

Both P and Q are global up-cuts of the output: each is crossed exactly once, with all earlier vertices below and all later vertices above. For fixed sign and cut levels, the inverse is unique. Cut at the two ports, reflect the lower exterior back, translate the upper exterior to align the ports, merge the two copies of v , and anchor the original start. A pair is called *valid* if this refolded walk is simple. It need not satisfy the deterministic rule used to choose the input minimum; counting all valid pairs only enlarges the inverse multiplicity. If the input has absolute endpoint-height increment at most r , then

$$|P + Q - y(\text{start}) - y(\text{end})| \leq r. \quad (43)$$

Indeed before reanchoring the two output endpoint heights are $2P - y(a_0)$ and $y(b_0) + Q - P$, and the expression is translation invariant.

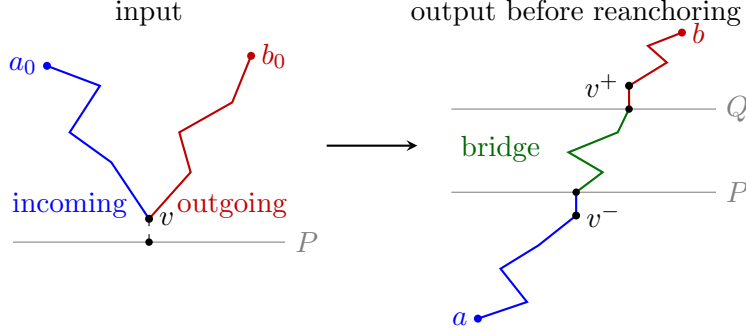


Figure 5: Insertion at an internal minimum, shown schematically rather than as a lattice embedding. Reflect the blue incoming arm across P ; translate the red outgoing arm by the port increment of the green bridge. The two copies v^-, v^+ of the minimum lie outside the inserted slab. The walls P, Q are global up-cuts of the output. Cutting there and reversing these operations recovers the input. The final common translation to fix the output root does not change $P + Q - y(a) - y(b)$.

Two bounds on the inverse labels. For ordered disjoint level intervals I, J , let $M_{I,J}(\omega)$ be the number of valid pairs (P, Q) of an output ω with $P \in I, Q \in J$. The first bound below controls this count in one pair of intervals. The second also bounds how many opposite interval pairs can contain any cuts when (43) holds.

Lemma 8.3 (Recoverable insertion labels). *Fix $\eta > 0$ and consider outputs of edge length at most N^2 . Each assertion below holds outside a family of total critical mass at most $\exp(-N^c)$, for some $c > 0$.*

- If I lies below J and both have size at most CH , where $1 \leq H \leq CN^2$, then $M_{I,J} \leq H^a N^\eta$.
- Among valid pairs satisfying (43) and $s \leq Q - P \leq 2s$, with $1 \leq r \ll s \leq CN^2$ and $s/r \rightarrow \infty$, there are at most $Cs^a (r/s)^\eta N^{2\eta}$ pairs.

The exceptional-mass conclusion is preserved by polynomially many choices of intervals, coordinates, and tests, and by any polynomial multiplicity.

Proof. Valid pairs in fixed bins: the occurrence moment. Set $k = \lfloor N^\gamma \rfloor$, with $\gamma > 0$ to be chosen small. We bound the unnormalized occurrence moment

$$\sum_{\omega: |\omega| \leq N^2} \rho^{|\omega|} M_{I,J}(\omega)^k,$$

where $|\omega|$ is edge length and the output start is fixed. Expanding the power selects an ordered tuple of k valid pairs. Suppose it uses t distinct lower levels and u distinct upper levels. Their incidence pattern costs $\exp(O(k \log k))$. The lowest and highest levels have polynomially many choices. The two exterior pieces cost at most $CV_{CN^2}^2$: sum the first from the anchored source and the last freely from its joining port. Every intervening segment is a strict bridge with free transverse increment. The bridge between the last lower and first upper cut has mass at most $\sup_h B_h \leq C$.

The incidence graph has $t + u$ vertices and k edges, hence at least $t + u - k$ connected components. One edge from each component gives a matching. Remove edges incident to the lowest lower or highest upper level, leaving at least $t + u - k - 2$ matches. For each retained match select the gap immediately below its lower cut and the gap immediately above its upper cut. All selected gaps are distinct. Bin the $t + u - 2$ within-interval gap heights dyadically, at cost at most $(C \log N)^{2k}$. An untested gap of scale D costs $\sum_{h \lesssim D} B_h \leq CD^a$.

Changing the measure on a selected gap. For matched gap scales D_1, D_2 , choose $g = \max(1, \lfloor \min(D_1, D_2)/4 \rfloor)$ using the dyadic lower ends. Read the lower gap backward from its upper cut and the upper gap forward from its lower cut, reflecting to a common upward orientation. Retain only complete irreducibles ending below g . For a relative history ξ of height $b < g$, write $\mu(\xi) = \rho^{L(\xi)}$. Its censored-law probability is

$$P_g(\xi) = \mu(\xi)T(g-b), \quad T(v) = p(H \geq v).$$

Relative to this probability law, the summed weight of a gap of scale D has remaining kernel

$$K_D(\xi) = \mathbb{E} \left[\sum_{h \text{ in the gap bin}} B_{h-b-H} \middle| H \geq g-b \right] \leq CD^a,$$

with negative residual heights contributing zero. The crossing jump is the first omitted irreducible. Renewal factorization proves the kernel identity, and the bound is uniform in the retained history: for each crossing jump the residual heights range over an interval of size $O(D)$, whose bridge masses sum to at most CD^a .

Validity forces the two aligned histories to satisfy (40), for one of the two assignments of roles. They emanate from the same port and center and then follow disjoint arms of the refolded walk. For a lower tested left cut d_1 and a higher right cut $d_2 < g$, continue the left arm to d_2 , still inside its selected gap. This is a strip crosscut with a unique crossing at d_1 , so every right-prefix crossing there lies to its right.

We now sum fine gap heights at fixed relative histories. The central bridge was bounded uniformly, so these heights can be summed freely. There is no transverse matching factor: translation to the next port determines each placement, and the pair constraints use aligned relative histories. Thus the occurrence sum is bounded by an integral under independent censored reference laws, with kernels bounded by CD_i^a . It does not require independence under the law of an output conditioned on its labels. Distinct matches use disjoint gaps, so the survival costs multiply. Each match contributes at most

$$C_\theta D_1^a D_2^a g^{-a+\theta} \leq C'_\theta H^a N^{3\theta}.$$

There are $t + u - 2$ gaps and at least $t + u - k - 2$ such savings. Including untested gaps and all classifications therefore gives

$$\sum_{\omega: |\omega| \leq N^2} \rho^{|\omega|} M_{I,J}(\omega)^k \leq N^{O(1)} V_{CN^2}^2 H^{ak} N^{3\theta k} \exp\{O_\theta(k \log(k \log N))\}.$$

Choose θ , then γ , sufficiently small relative to η , and use (42) with $2\nu < \gamma$. Markov's inequality at $H^a N^\eta$ proves the first exceptional-mass bound.

How many opposite bins contain cuts? Fix the output terminal height at polynomial cost and put $T = y(\text{start}) + y(\text{end})$. Under the displacement constraint, P ranges over an interval of size $O(s)$ below $T/2$ by at least $(s-r)/2$. Partition this range into intervals I_i of width r . The possible partners lie in $J_i = T - I_i + [-r, r]$, above all the lower bins for large s/r . There are $L' \leq Cs/r$ bins. The first part of the lemma bounds valid pairs in each (I_i, J_i) by $Cr^a N^\eta$. It remains to count occupied bins, meaning bins with some global up-cut in each interval, without requiring that the pair be valid.

Separate the indices into a fixed number of residue classes, say modulo ten. Test k increasing occupied indices in one class. The upper intervals occur in reverse order, and a gap of d indices

corresponds to height separation comparable to dr on both sides. Fix the outermost two cut levels at polynomial cost. From each end of the bridge between them, stop successively at the first renewal hit in the next required interval. A hit in an interval of width $O(r)$ at distance comparable to dr has probability at most $Cd^{-\vartheta}$. Indeed, after its first hit the expected number of renewals in the next r levels is $\sum_{j \leq r} B_j \gtrsim r^a$, while the unconditional expected number in the enlarged interval is at most $Cr(dr)^{-\vartheta}$.

These costs multiply over successive first hits. The two prefixes stop at separated levels; their remaining joining bridge costs at most C . The reverse prefix at the upper end is placed by translation, with its outer transverse port free. Hence the two sides together cost $Cd^{-2\vartheta}$ per index gap. Summing the gaps gives

$$\sum_{d \leq L'} Cd^{-2\vartheta} \leq C(s/r)^{1-2\vartheta}.$$

If Y counts occupied bins in this residue class, its weighted binomial moment is consequently at most

$$\sum_{\omega: |\omega| \leq N^2} \rho^{|\omega|} \binom{Y^{(\omega)}}{k} \leq N^{O(1)} V_{CN^2}^2 (C(s/r)^{1-2\vartheta})^k.$$

Take $k = \lfloor N^\gamma \rfloor$ with γ sufficiently small relative to η . The binomial Markov estimate, including its factor k^k , bounds Y by $C(s/r)^{1-2\vartheta} N^\eta$ outside a stretched-exponential exceptional mass. Multiply this by the first bound per bin. Since $a = 1 - \vartheta$, the result is

$$Cr^a(s/r)^{1-2\vartheta} N^{2\eta} = Cs^a(r/s)^\vartheta N^{2\eta},$$

which proves the second bound. All coordinates and labels have polynomial ranges for outputs of length at most N^2 , so the stated unions and multiplicities preserve both exception estimates. $\square$

Proof of Theorem 8.2. Comparing short-displacement and total masses. Fix a small $\epsilon > 0$ and a dyad $[N, 2N)$. Choose fixed positive ζ, η sufficiently small relative to ϵ , and put

$$r = N^{3/4-\epsilon/2}, \quad s = N^{3/4-\zeta}, \quad R = N^{3/4+\zeta}.$$

Let b_j be the critical mass of eligible j -edge inputs whose absolute endpoint-height increment is at most r . Choose l with probability A_l/A , independently of n , and set $j = n - l - 1$. The insertion outputs at length n , counted with multiplicity, have mass $\rho \sum_l A_l b_j$. The second label bound and $A \asymp s^a$ give

$$\mathbb{E}_l[b_j/c_n] \leq C(r/s)^\vartheta N^{2\eta}.$$

The exceptional outputs contribute negligibly: their absolute mass is stretched-exponentially small, their multiplicity is polynomial, and $c_n \geq 1$.

For comparison, restrict all eligible inputs to diameter at most R . They retain $(1 - o(1))c_j$ mass uniformly in the relevant lengths, by the fast-path bound and the $o(1)$ omitted mass. Their lower cut P lies within CR of the output starting height. Cover this range by $O(1 + R/s)$ intervals of size at most $s/10$; each has an associated higher interval of size $O(s)$ for Q . The first label bound gives multiplicity at most $C(1 + R/s)s^a N^\eta$. After dividing by A ,

$$\mathbb{E}_l[c_j/c_n] \leq CN^{2\zeta+\eta}.$$

The logarithmic comparison and its boundary error. Put $\kappa = \vartheta/8$. Jensen's inequality applied to the sum of the two mass ratios gives

$$\begin{aligned} & \mathbb{E}_l \log(1 + N^{\kappa\epsilon} b_j / c_j) + \mathbb{E}_l \log(c_j / c_n) \\ & \leq \log \mathbb{E}_l[(c_j + N^{\kappa\epsilon} b_j) / c_n] \leq (2\zeta + \eta + o(1)) \log N. \end{aligned}$$

For the last inequality it suffices, for example, to choose $\vartheta\zeta + 2\eta < 3\vartheta\epsilon/8$; the scaled short-displacement term then even tends to zero. We have obtained

$$\mathbb{E}_l \log(1 + N^{\kappa\epsilon} b_j / c_j) \leq (2\zeta + \eta + o(1)) \log N + \log c_n - \mathbb{E}_l \log c_j.$$

The maximal shift is

$$D_N := \max(l + 1) = O(s^{d_*}) = O(N^{1-4\zeta/3}) = o(N).$$

For each fixed shift, summing the last difference over $N \leq n < 2N$ cancels all interior terms. The volume bound leaves boundary error

$$O_\nu(D_N N^\nu) = o(N), \quad 0 < \nu < 4\zeta/3.$$

The left summand is between zero and $O(\log N)$ because $b_j \leq c_j$. Reindexing it from $j = n - l - 1$ to the exact dyad therefore costs at most $O(D_N \log N) = o(N)$. For each fixed $\lambda > 0$, it follows that

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \#\{N \leq n < 2N : b_n / c_n > \lambda\} \leq \frac{2\zeta + \eta}{\kappa\epsilon}.$$

Here b_n depends on N, ϵ , but not on ζ, η . Keep all parameters fixed while taking this limit, and then let ζ, η tend to zero. The upper fraction is zero.

One set of lengths for probability and moments. Adding back the omitted walks changes the probability by $o(1)$, since $c_n \geq 1$. Thus the lower cutoff $n^{3/4-\epsilon}$ holds for absolute normal displacement outside a vanishing fraction of each dyad. As in the half-plane argument, let the slacks and probability-error thresholds decrease sufficiently slowly over dyads. The retained lengths have natural density one. Intersecting with the half-plane set retains natural density one and gives a common set for both laws. Endpoint distance dominates normal displacement, and the all-length upper tail already controls maximum radius. This proves both probability exponents. The same lower cutoff and the superpolynomial upper tail, with the deterministic linear bound, give the moments for every fixed $p > 0$ on that one set.

An alternative full-plane thermal proof. The same insertion gives the thermal conclusion by the calibrated and censored estimates, independently of the reserved-height comparison in Section 6. Take $N = \lceil 1/t \rceil$ and the same r, s . The partition sum $Z(t) = \sum_m c_m e^{-tm}$ is at least cN . The volume bound makes lengths $m > N^{1+\delta}$ superpolynomially unlikely, also with any fixed length power, for each fixed small $\delta > 0$. Insert into eligible short-displacement inputs below this cutoff. Outputs have length at most N^2 , and $e^{-t(l+1)} = 1 - o(1)$ uniformly since $l + 1 = o(N)$. The second label bound, now summed with thermal weights, gives

$$\sum_{m \leq N^{1+\delta}} b_m e^{-tm} \leq C(r/s)^\vartheta N^{2\eta} Z(t).$$

The omitted walks have $o(1)$ mass at each large exact length, so their thermal mass is $o(N)$. This proves the endpoint lower cutoff at every large discount scale. The uniform speed bound, conditioned on lengths between $N^{1/2}$ and $N^{1+\delta}$, gives the spatial upper cutoff and its moments. Smaller lengths are atypical by the endpoint lower cutoff. Finally the same lower cutoff and speed bound give the typical length lower power; the volume bound gives its upper tail and all positive length moments. $\square$

8.3 An independent count of favorable exact lengths

The density-one theorem is stronger than a power-cardinality conclusion, but uses censored survival and refolding. We record a separate consequence of the thermal route under its weaker length precision: the deficit $1 - \mathbb{E}e^{-L/N} = N^{-\alpha+o(1)}$, the absolute mass and upper spatial estimates, and the turning-chain bound with a reserved independent group. No insertion-label bound is needed. The additional ingredient is a length atom estimate on that reserved group.

Proposition 8.4 (Power-cardinality of favorable lengths). *Fix $0 < \epsilon < 1$. For unrestricted uniform honeycomb walks rooted at a fixed center, the number of integers $m \in [R^{1-\epsilon}, R]$ for which*

$$\Pr_m\{m^{3/4-\epsilon} \leq |\omega_m - \omega_0| \leq \text{diam}(\omega) \leq m^{3/4+\epsilon}\} \geq 1 - \epsilon$$

is $R^{1+o(1)}$ as $R \rightarrow \infty$. In particular, the limsup of each positive moment exponent of endpoint distance or diameter is $3/4$ times its moment order. If the corresponding spatial exponent in probability exists at all exact lengths, it equals $3/4$.

Proof. Write $\alpha = 9/16$. We first record the length atom estimate under the exponent-precision hypotheses of the turning-chain route. If the length L of one irreducible satisfies $1 - \mathbb{E}e^{-L/r} = r^{-\alpha+o(1)}$, then independent sums satisfy

$$\sup_m p(S_L(j) = m) \leq j^{-1/\alpha+o(1)}. \quad (44)$$

Here is a direct verification, without logarithmic precision. Fix a small $\zeta > 0$ and put $z = r^{1-\zeta/2}$. The Laplace estimate implies $p(L > s) \leq s^{-\alpha+o(1)}$ and $\mathbb{E}[L; L \leq s] \leq s^{1-\alpha+o(1)}$. Hence the contributions of $L < r^{1-\zeta}$ and $L > r/10$ to $\mathbb{E}(1 - e^{-L/z})$ are negligible compared with $z^{-\alpha+o(1)}$. The intermediate interval therefore has probability at least $r^{-(1-\zeta/2)\alpha+o(1)}$. Subtracting an independent bounded-length copy in the characteristic-function symmetrization gives phases between $cr^{-\zeta}$ and a fixed constant below π at frequency $1/r$. Thus the squared characteristic-function modulus is at most $1 - cr^{-\alpha-O(\zeta)+o(1)}$. On the actual length lattice all modulus-one characters form a finite set and translate the same bound. Fourier inversion, followed by $\zeta \downarrow 0$, proves (44).

Now take the full-plane thermal law with edge weights $(\rho e^{-1/N})^m$ and its deterministic minimum-split decomposition. Use the notation of the two-end estimate in Corollary 5.11: for a height scale H_i and count scale k_i , let $K_i = H_i^{3/4}$ and $\kappa_i = \min(k_i, K_i)$. Put $M = \max_i \kappa_i$ on the nonempty bridges, and $M = 1$ if there are none. The endpoint-minimum case is included using its one-end decomposition. Restrict to length at most N^2 , whose complement has negligible thermal probability by the absolute critical mass bound.

The turning-chain estimate without smoothing bounds the critical mass of each dyadic family, including count choices, by $MN^{o(1)}$. The two no-renewal ends cost at most a constant times U_N^2 . Lemma 6.2 bounds the plane normalization below by $(q_N^{-1} - 1)U_N^2$, where $q_N = N^{-\alpha+o(1)}$. Consequently all families with $M < N^{\alpha-\eta}$ have vanishing normalized mass, for every fixed $\eta > 0$.

In the remaining families fix the end shapes, all witness labels, and all pieces except the unused batch belonging to an index attaining M . Proposition 5.10 reserves this batch independently of the avoidance tests and the large-height and short-length tests. Exact total length is therefore one scalar equality in that batch. Applying (44) costs an additional $M^{-1/\alpha+o(1)}$, uniformly in the fixed outside data. The bounded endpoint length corrections merely change the target integer. Damping may be dropped on these bridges before using the critical estimate. After normalization, for every integer m the mass of length m in the retained families is at most

$$N^{-\alpha+o(1)} M^{1-1/\alpha} \leq N^{-1+(1/\alpha-1)\eta+o(1)}.$$

No independence between the length and height of an individual irreducible is used: its entire batch remains unexposed until this scalar test.

Theorem 6.4 gives probability tending to one to $N^{1-s} \leq m \leq N^{1+s}$ and to the corresponding spatial exponent window with fixed slack $s > 0$. Conditional on m , the thermal law is the uniform law. Taking s sufficiently small relative to ϵ and using Markov's inequality on the conditional failure probability, the lengths satisfying the proposition carry probability tending to one. Their intersection with the retained families still does so. The uniform atom bound forces their number to be at least $N^{1-O(\eta)+o(1)}$.

Set $N = R^{1/(1+2s)}$. For small enough s the resulting length interval lies in $[R^{1-\epsilon}, R]$. Since s and η are arbitrarily small, the count is at least $R^{1-o(1)}$, and the trivial bound by R proves the claim. The limsup assertions combine these favorable lengths with the all-length upper spatial tail. This argument gives power-cardinality; the separate refolding argument establishes the density-one conclusion. $\square$

9 Correlation lengths, pulling, and prescribed spatial endpoints

There are two further ways to change the thermal law. One may reward displacement by an exponential factor, or prescribe the terminal vertex. The first change tests convergence of a generating function. The second asks for one coefficient in two spatial coordinates. We first determine the exponential threshold and the associated force response. We then prove the spatial local estimate needed to normalize a prescribed endpoint.

In this section lengths ℓ count edges of ordinary walks from a fixed vertex o . Put $z_t = \rho e^{-t}$, $t > 0$, and define

$$G_t(b) = \sum_{\gamma: o \rightarrow b} z_t^{\ell(\gamma)}, \quad \chi(t) = \sum_b G_t(b).$$

The empty walk is included. For a unit vector e , let

$$m_e(t) = \sup \left\{ s \geq 0 : \sum_b e^{s e \cdot (b-o)} G_t(b) < \infty \right\}, \quad (45)$$

$$m_{\text{rad}}(t) = \sup \left\{ s \geq 0 : \sum_b e^{s|b-o|} G_t(b) < \infty \right\}. \quad (46)$$

Both convergence sets are intervals containing zero. For the directional sum this follows by separating nonnegative and negative projections; the latter contribution is bounded by $\chi(t)$. All constants below allow the fixed conversion between Euclidean units and row spacing.

Directional decay and endpoint tilts have long been studied through renewal methods. Ioffe proved all-direction Ornstein–Zernike asymptotics for self-avoiding walk on $\mathbb{Z}^d$ in the strictly

subcritical fugacity regime [8]. Borgs, Chayes, King and Madras studied directional masses for anisotropic walks and linear displacement of the most likely endpoint under asymmetric positive step weights [2]. Ioffe and Velenik proved ballistic local limit results for self-interacting walks under fixed forces in the ballistic regime [9]. Our estimates concern the approach to zero force and critical fugacity on the honeycomb lattice. We derive the required uniform powers from the quantitative bridge estimates above.

Theorem 9.1 (Exponential thresholds and force). *Uniformly over unit vectors e , as $t \downarrow 0$,*

$$m_e(t) = t^{3/4+o(1)}, \quad m_{\text{rad}}(t) = t^{3/4+o(1)}.$$

More precisely, for every $\varepsilon > 0$ and all sufficiently small t ,

$$t^{3/4+\varepsilon} \leq m_{\text{rad}}(t) \leq m_e(t) \leq Ct^{3/4},$$

where C is independent of e . For each unit vector e and $s \geq 0$, the limit

$$f_e(s) = \lim_{j \rightarrow \infty} \frac{1}{j} \log \sum_{\ell(\gamma)=j} \rho^j e^{se \cdot (\gamma_j - o)}$$

exists. It satisfies $f_e(0) = 0$ and

$$f_e(s) = s^{4/3+o(1)} \quad (s \downarrow 0),$$

uniformly in e .

The proof uses two consequences of the absolute count and fast-path bounds in Propositions 5.12 and 5.13. For every fixed sufficiently small $\xi > 0$,

$$\sum_{\substack{R \leq D(\gamma) \leq 2R \\ L(\gamma) \leq R^{4/3-\xi}}} \rho^{L(\gamma)} \leq C_\xi e^{-R^\xi}, \quad U_R := \sum_{D(\gamma) \leq R} \rho^{L(\gamma)} \leq e^{C_\nu(1+R)^\nu} \quad (\nu > 0). \quad (47)$$

These sums start at a specified vertex and count vertices in L . For the first bound, paths of diameter at least R have length at least cR . Choose $\eta > 0$ so small that $(4/3 - \xi)(3/4 + \eta) < 1$. At each remaining length $n \leq R^{4/3-\xi}$, the refined fast-path estimate applies and gives mass at most

$$\exp(-c_\eta R/n^{3/4}) \leq \exp(-c_\eta R^{3\xi/4}).$$

Summing the polynomial number of lengths gives the first bound. For the second, a self-avoiding trace of diameter at most R visits at most $C(1+R)^2$ lattice vertices. Apply the absolute count bound with exponent smaller than $\nu/2$ and sum over these lengths. Changing either sum to edge weights costs one constant factor.

For the divergence estimate we need a positive mass of bridges whose thermal discount is bounded below. Theorem 2.1 gives $B_h \geq ch^{-1/4}$ and $M_h \leq Ch^{13/12}$. Hence

$$B_h \{L > C_0 h^{4/3}\} \leq \frac{M_h}{C_0 h^{4/3}} \leq \frac{C}{C_0} h^{-1/4}.$$

Choosing the fixed C_0 large enough retains a positive fraction of B_h with $L \leq C_0 h^{4/3}$. This uses the finite first-length mass before any exact-length conditioning.

Proof of Theorem 9.1. Fix $\varepsilon > 0$ and put $R = t^{-3/4-\varepsilon}$. Stop a walk on its first visit to distance at least R from its starting vertex. Its endpoint lies at distance at most $R + a_{\text{lat}}$, where a_{lat} is the lattice edge length, and its diameter lies between R and $2R + a_{\text{lat}}$. Let $W(t, R)$ be the total z_t -weight of these stopped prefixes. The prefixes with at most $R^{4/3-\xi}$ vertices have mass tending to zero by the first estimate in (47), with harmless fixed adjustments of the radii. The remaining prefixes have total mass at most

$$CU_{3R} \exp(-tR^{4/3-\xi}).$$

Choose $\xi > 0$ so small that

$$4/3 - \xi - \frac{1}{3/4 + \varepsilon} > 0,$$

and then choose ν smaller than this difference in the bound for U_{3R} . It follows that $W(t, R) = o(1)$.

Now split any complete walk greedily into such stopped prefixes, restarting the rule at the endpoint of each prefix, followed by one remainder whose displacement is less than R . Edge weights multiply. Dropping avoidance between different pieces gives, at $s = 1/R$,

$$\sum_{\gamma} z_t^{\ell(\gamma)} e^{sA(\gamma)} \leq e\chi(t) \sum_{k \geq 0} (e^{1+a_{\text{lat}}/R} \max_{\text{vertex types}} W(t, R))^k < \infty$$

for all sufficiently small t . The estimates hold at both vertex types by lattice symmetry. Thus $m_{\text{rad}}(t) \geq t^{3/4+\varepsilon}$, and $m_e(t) \geq m_{\text{rad}}(t)$.

To prove divergence, take an integer height $H \asymp t^{-3/4}$. Choose one of the three inward port normals at o whose scalar product with e is at least $1/2$. Consider bridges in that direction with height $h \in [H, 2H]$, length at most $C_1 H^{4/3}$, and horizontal displacement of the sign favorable to e . Reflection in the normal preserves length and critical weight, so the total mass at every such height is at least $cH^{-1/4}$. Every retained displacement has projection on e at least $c_0 H$, where $c_0 > 0$ is independent of e .

From each retained bridge extract the prefix ending at its first renewal cut whose relative height lies in $[H, 2H]$ and whose prefix has the stated sign and length properties. Such a cut exists because the terminal cut qualifies. A fixed extracted prefix may have suffixes to several heights in $[H, 2H]$; their total critical mass is at most

$$\sum_{0 \leq j \leq H} B_j \leq CH^{3/4}.$$

Since the original bridges summed over heights have mass at least $cH^{3/4}$, the distinct extracted prefixes have total critical mass at least a fixed $c_1 > 0$. Their lengths are at most $C_1 H^{4/3}$, so their e^{-tL} -discounted mass is also bounded below by a fixed positive constant.

Concatenate any number of these prefixes in the chosen direction. The pieces lie in successive disjoint open slabs. Their joins are recoverable by the same first-qualifying-cut rule, relative to the current initial port. Later pieces cannot change an earlier cut or its prefix tests. Hence different ordered lists give different port paths. Trimming the initial and final half-edges gives ordinary plane walks from o ; the length and endpoint-weight corrections are bounded independently of the number of pieces. Consequently the directionally tilted plane sum diverges when

$$c_2 e^{sc_0 H} > 1.$$

This proves $m_e(t) \leq C/H \leq Ct^{3/4}$, uniformly in e .

It remains to pass from convergence thresholds to the fixed-length free energy. Write

$$a_j(s, e) = \sum_{\ell(\gamma)=j} \rho^j e^{se \cdot (\gamma_j - o)}.$$

After an even number of steps the terminal vertex has the same lattice type as o . Splitting there and dropping avoidance therefore gives $a_{2j+2k} \leq a_{2j}a_{2k}$. Fekete's lemma gives a limit for $(2j)^{-1} \log a_{2j}$. Single-edge truncation gives

$$a_{j+1} \leq 3\rho e^{sa_{\text{lat}}} a_j.$$

Applying this twice places each odd-length logarithm between the adjacent even-length logarithms up to a constant. Thus the limit defining $f_e(s)$ exists at all lengths. It is finite. The connective-constant identity gives $f_e(0) = 0$.

The directional susceptibility is $\sum_j e^{-tj} a_j(s, e)$. The root test shows that it converges if $t > f_e(s)$ and diverges if $t < f_e(s)$. In particular, $s < m_e(t)$ implies $f_e(s) \leq t$, and $s > m_e(t)$ implies $f_e(s) \geq t$. Applying the threshold bounds with $t = s^{1/(3/4+\delta)}$ and $t = s^{1/(3/4-\delta)}$, with an arbitrarily small fixed $\delta > 0$ and harmless constant adjustments, proves $f_e(s) = s^{4/3+o(1)}$. All threshold constants used here are uniform in e . $\square$

Corollary 9.2 (One-sided force response). *For $s > 0$ the convex function f_e has one-sided derivatives, and*

$$f'_{e,-}(s) = s^{1/3+o(1)}, \quad f'_{e,+}(s) = s^{1/3+o(1)} \quad (s \downarrow 0).$$

At a point where f_e is differentiable, the mean displacement in direction e , divided by length, under the force-weighted j -edge law tends to $f'_e(s)$ as $j \rightarrow \infty$.

Proof. Each $j^{-1} \log a_j(s, e)$ is convex, and its pointwise limit is convex. Since $f_e(0) = 0$, convexity gives

$$\frac{f_e(s)}{s} \leq f'_{e,-}(s) \leq f'_{e,+}(s) \leq \frac{f_e(2s) - f_e(s)}{s} \leq \frac{f_e(2s)}{s}.$$

Here $f_e(s) \geq 0$: rotation by 120 degrees about o makes the mean endpoint of the unforced fixed-length law zero, so Jensen's inequality gives $a_j(s, e) \geq a_j(0, e)$. Theorem 9.1 now proves the derivative exponents. Finally the derivative of $j^{-1} \log a_j$ is the stated normalized mean. Its lower and upper difference quotients converge to those of f_e ; letting their increments tend to zero proves the claim at differentiability points. No differentiability assertion is needed for the one-sided result. $\square$

The exponent $1/3$ agrees with Pincus's force-extension scaling relation $1/\nu - 1$ at $\nu = 3/4$ [25, Eq. (I.4)]; here it follows from the threshold bounds and convexity. This small-force relation is distinct from fixed-force ballistic asymptotics; see also [10, Section 4.2.3].

For prescribed endpoints we will use the same geometric estimates before summing over lengths. In edge-count notation, the absolute critical count bound in Proposition 5.12 is $c_j \rho^j \leq \exp(C_\nu j^\nu)$ for every $\nu > 0$, where c_j counts rooted j -edge walks. The refined fast-path bound in Proposition 5.13 is

$$\sum_{\substack{\ell(\gamma)=j \\ D(\gamma) \geq d}} \rho^j \leq \exp(-c_\eta d/j^{3/4}), \quad d \geq j^{3/4+\eta}, \quad (48)$$

for fixed $\eta > 0$ and sufficiently large j . The conversion from vertex count to edge count changes length by one and weight by a fixed factor; decreasing the slack and the exponential constant absorbs both changes. Both bounds concern unnormalized masses, so they remain available before a prescribed endpoint has been normalized.

Remark 9.3 (A second route to the radial threshold). *The turning-chain estimates give a useful alternative which requires no first-length estimate at a single chosen height. Write p for the critical*

irreducible law, T for its height and L for its vertex length. By Lemma 5.1 and the length deficit in Proposition 2.4,

$$p(e^{-L/N}; T > R) \geq cR^{-3/4} - q_N, \quad q_N := p(1 - e^{-L/N}) = N^{-9/16+o(1)}.$$

At $R = N^{3/4-\delta}$ the right side is at least $c'R^{-3/4}$. A pull $s = N^{-3/4+2\delta}$ in a fixed positive multiple of the height therefore makes the total tilted irreducible weight exceed one. Free concatenation then forces divergence of the radially tilted plane sum, after the bounded endpoint conversion.

For convergence put $s = N^{-3/4-\delta}$ and choose an even block length $j \asymp N^{1+\eta}$ with η sufficiently small after δ . On $D \leq j^{3/4+\eta}$ the pull is bounded, while $e^{-j/N}$ beats the critical count bound. On successive larger diameter ranges, (48) beats the pull because $s \ll j^{-3/4}$. The tilted mass of this block is less than one at both vertex types. Splitting paths into such blocks, with one bounded-length remainder, proves convergence. This recovers $m_{\text{rad}}(1/N) = N^{-3/4+o(1)}$ by an argument directly adapted to the absolute turning-chain bounds.

9.1 A spatial local limit with lattice compatibility

The exponential threshold alone gives no lower bound for $G_t(b)$ at a specified vertex. We now prove that a sum of order $H^{3/4}$ irreducible bridges can reach every admissible target of height H at the expected two-dimensional lattice scale. The proof does not assume convergence to a unique stable law.

Let $Y = (T, X)$ be the displacement of one p -distributed irreducible, with T measured in row spacings and X in units of the spacing between ports on one cut. The ports on row h are shifted horizontally by $h/2$ modulo this spacing. Thus their relative-displacement lattice is

$$\Gamma = \{(h, m + h/2) : h, m \in \mathbb{Z}\}.$$

The two elementary two-vertex bridges have displacements $(1, 1/2)$ and $(1, -1/2)$. They are irreducible and generate Γ , so Γ is also the lattice generated by all possible increments. Let Δ be the Euclidean diameter of an irreducible and L its vertex length. Put $\beta = 3/4$ and $K = H^\beta$.

For Fourier inversion the relevant subgroup is Γ_1 , generated by differences of possible increments. Lemma 5.2 proves that Γ_1 has rank two and hence finite index in Γ . All increments have one common class $a_* + \Gamma_1$; that class generates Γ/Γ_1 . We call k compatible with $y \in \Gamma$ when $ka_* - y \in \Gamma_1$.

Proposition 9.4 (Prescribed displacement with controlled length and diameter). *For each $C < \infty$ and $\eta > 0$ there are $c > 0$, $M_0 < \infty$ and $H_0 < \infty$ such that, for every integer $H \geq H_0$ and relative-port target $y = (H, x) \in \Gamma$ with $|x| \leq CH$,*

$$\sum_{\lceil K \rceil \leq k \leq \lfloor 2K \rfloor} p^{\otimes k} \left\{ \sum_{i=1}^k Y_i = y, \quad \sum_{i=1}^k L_i \leq H^{4/3+\eta}, \quad \sum_{i=1}^k \Delta_i \leq M_0 H \right\} \geq cKH^{-2} = cH^{-5/4}. \quad (49)$$

Incompatible counts contribute zero. In particular the paths counted here remain within distance $M_0 H$ of their initial port.

Proof. We first establish the estimate without the length and diameter restrictions. Although subsequential laws of the rescaled sums may differ, each has full forward support and a strictly positive density in the open forward half-plane. Fourier inversion will then give a uniform lattice lower bound. Finally we mark individual increments to impose the two restrictions.

Subsequential jump limits. Lemma 5.1 gives

$$p(|Y| > r) \leq Cr^{-\beta}, \quad p(|Y| \leq r) \leq Cr^{1-\beta}. \quad (50)$$

Consider any sequence $H \rightarrow \infty$. On a subsequence the measures

$$\mu_H = H^\beta p(Y/H \in \cdot)$$

converge vaguely on $\mathbb{R}^2 \setminus \{0\}$ to a measure μ . This follows by compactness on each closed annulus and a diagonal extraction. The bounds

$$\mu_H(|y| > r) \leq Cr^{-\beta}, \quad \int_{|y| \leq r} |y| d\mu_H(y) \leq Cr^{1-\beta}$$

give the corresponding bounds for μ , by approximation away from zero. In particular $\int (1 \wedge |y|) d\mu(y) < \infty$. The measure is supported in $\{T \geq 0\}$.

If $k/H^\beta \rightarrow u > 0$, expansion of one characteristic function gives

$$\mathbb{E} \exp \left(is \cdot H^{-1} \sum_{i=1}^k Y_i \right) \rightarrow F_u(s) := \exp \left(u \int (e^{is \cdot y} - 1) d\mu(y) \right). \quad (51)$$

For completeness, truncate the integral to $\delta \leq |y| \leq M$. Vague convergence gives the limit there at continuity radii. The small-jump error is at most $C|s|\delta^{1-\beta}$; the large-jump error is at most $CM^{-\beta}$. Send first H to infinity, then δ to zero and M to infinity. The one-factor error is $O(H^{-\beta})$, so its squared term disappears after multiplication by k .

The right side is a characteristic function. Construct independent Poisson point collections on disjoint annuli, with intensity $u\mu$, and sum their points. There are finitely many points outside the unit disk, and the sum of the norms inside it has finite expectation. Thus the small-jump series converges absolutely almost surely. This is the finite-variation form of the Poisson construction of a pure-jump infinitely divisible law; see Schilling [26, Chapter 7] for the general construction. Tightness of the rescaled sums also follows directly from (50): remove jumps above M , and use the truncated first moment for the others. Fourier uniqueness therefore proves weak convergence to the constructed law. We prove the support and uniform lattice lower bound needed here next.

Full forward support. For any $u > 0$ the support of the limiting law is the closed additive semigroup generated by the support of μ , including zero, and is independent of u . One inclusion follows by truncating the Poisson sum. Conversely, a specified finite list of jumps can be approximated with positive probability, excluding every other jump above a small cutoff. The remaining small jumps have arbitrarily small total norm with positive probability, by their first-moment bound. Hence each finite sum of support points belongs to the support.

This semigroup contains the entire open forward half-plane. To prove it, fix an open ball B whose closure lies in $\{T > 0\}$ and then a smaller closed ball B' in B . The free-terminal guide estimate in Section 5.1, applied over a fixed positive interval of target heights, gives

$$\sum_{k \geq 0} p^{*k}(HB') \geq c_B H^\beta. \quad (52)$$

Indeed a narrow straight guide places the terminal port in HB' , uniformly over order H heights, and contributes $cH^{-1/4}$ at each height. Counts $k \leq \xi H^\beta$ have total contribution at most $C_B \xi^2 H^\beta$: to reach HB' their height sum must exceed $c_B H$, and the truncated-sum bound from (50) gives probability at most $C_B k H^{-\beta}$. Counts $k \geq SH^\beta$ contribute at most $C_B H^\beta e^{-c_B S}$, because

$$p \left(\sum_{i=1}^k T_i \leq C_B H \right) \leq e^{C_B} (p(e^{-T/H}))^k \leq C_B e^{-c_B k/H^\beta}.$$

Choose ξ small and S large. The remaining mass in (52) is at least $c'_B H^\beta$, spread over at most $SH^\beta + 1$ counts. Thus for some $k_H \in [\xi H^\beta, SH^\beta]$, the probability of $H^{-1} \sum_{i \leq k_H} Y_i \in B'$ is bounded below independently of H . On a further subsequence $k_H/H^\beta \rightarrow u \in [\xi, S]$. Weak convergence and the closedness of B' show that the limiting law gives positive mass to B' . Because all positive u have the same support, every such ball meets the support of every limiting law.

Fourier inversion on the displacement lattice. We have obtained full support in the open forward half-plane. To turn this into a positive lattice lower bound, we need a density and Fourier domination. Write $\varphi(\theta) = p(e^{i\theta \cdot Y})$ for the increment characteristic function. Lemma 5.2 gives, near each of its modulus-one characters, the uniform estimate

$$|\varphi(\theta)| \leq 1 - c \operatorname{dist}(\theta, \mathcal{P})^\beta, \quad (53)$$

where $\mathcal{P}$ is the finite character group annihilating Γ_1 on the dual torus of Γ . Away from fixed neighborhoods of $\mathcal{P}$ the modulus is at most a constant strictly smaller than one. Passing to the limit near zero in (53) gives $|F_u(s)| \leq e^{-cu|s|^\beta}$. Thus the limiting law has a continuous density g_u .

In fact $g_u(v) > 0$ whenever the first coordinate of v is positive. The convolution identity $g_u = g_{u/2} * g_{u/2}$ follows from (51). Choose a small ball of heights strictly between zero and the first coordinate of v . Full support and continuity provide a point in that ball where $g_{u/2} > 0$, and a smaller ball on which it is bounded below. The reflected difference ball about v remains in the forward half-plane and has positive mass under the other half-time law. The convolution is therefore positive at v .

Take compatible k and y with $k/H^\beta \rightarrow u \in [1, 2]$ and $y/H \rightarrow v$. Fourier inversion on Γ yields

$$H^2 p^{*k}(y) \longrightarrow c_\Gamma g_u(v), \quad c_\Gamma > 0. \quad (54)$$

Here c_Γ is the covolume of Γ_1 in the chosen coordinates. To check the periodicity factor, for $\theta_0 \in \mathcal{P}$ all increments have the same phase $e^{i\theta_0 \cdot a_*}$, so $\varphi(\theta_0 + \theta) = e^{i\theta_0 \cdot a_*} \varphi(\theta)$. Compatibility makes the phase in the inversion integrand equal to one. Thus all the finitely many neighborhoods contribute identically. In each neighborhood rescale θ by H^{-1} and dominate by $e^{-c'|s|^\beta}$ using (53); the complement is exponentially small in k . This proves (54) with its stated constant.

The same subsequence argument proves a uniform positive lower bound when $|x|/H \leq C$ and $k/H^\beta \in [1, 2]$ is compatible. If it failed, choose offending targets and counts, extract $x/H \rightarrow v_2$, $k/H^\beta \rightarrow u$, and then a vague limit μ as above. Equation (54) would contradict $g_u(1, v_2) > 0$. A fixed positive fraction of the integers in $[K, 2K]$ are compatible with each $y \in \Gamma$, because Γ/Γ_1 is finite cyclic. Consequently

$$\sum_{K \leq k \leq 2K} p^{*k}(y) \geq c_C K H^{-2}. \quad (55)$$

Length and diameter restrictions. We now remove paths whose length or total increment diameter is too large. If $\sum_i L_i > M$, then

$$\sum_i (1 - e^{-L_i/M}) \geq 1 - e^{-\sum_i L_i/M} \geq 1 - e^{-1}.$$

Mark one increment by $1 - e^{-L/M}$. Its total marked mass is $q_M := p(1 - e^{-L/M})$. The other $k - 1$ increments match the remaining displacement with probability at most $C(k - 1)^{-8/3} \leq CH^{-2}$ by Lemma 5.2, independently of the value of the marked increment. Summing the mark and then the count gives the excluded mass bound

$$CK^2 H^{-2} q_M.$$

For $M = H^{4/3+\eta}$, the weak deficit bound $q_M = M^{-9/16+o(1)}$ makes this $o(KH^{-2})$.

The same argument controls spatial excursions without a new independence assumption. If $\sum_i \Delta_i > BH$, then $\sum_i \min(1, \Delta_i/(BH)) \geq 1$. The diameter tail implies

$$p(\min(1, \Delta/(BH))) \leq C(BH)^{-\beta}.$$

Marking by this function excludes mass at most $CB^{-\beta}KH^{-2}$. Choose a fixed $B = M_0$ large enough to leave at least half the lower bound in (55), and then take H large enough for the length exclusion. The entire concatenated path is within its sum of increment diameters of its first port. This proves (49). $\square$

9.2 Thermal walks ending at a specified vertex

We now use the local estimate to construct actual paths in a corridor between two prescribed vertices. This supplies a lower denominator before any conditional tail estimate is used.

Theorem 9.5 (Prescribed spatial offsets). *At each fixed $N > 0$ put $t = 1/N$. Define*

$$\underline{m}(N) = \liminf_{|b-o| \rightarrow \infty} \frac{-\log G_{1/N}(b)}{|b-o|}, \quad \overline{m}(N) = \limsup_{|b-o| \rightarrow \infty} \frac{-\log G_{1/N}(b)}{|b-o|},$$

where both limits range over all lattice vertices and directions. Then

$$\underline{m}(N) = N^{-3/4+o(1)}, \quad \overline{m}(N) = N^{-3/4+o(1)} \quad (N \rightarrow \infty).$$

If a sequence of terminal vertices satisfies $|b_N - o| = N^{3/4+o(1)}$, the probability law

$$\mathbb{P}_{N,b_N}(\gamma) = \frac{(\rho e^{-1/N})^{\ell(\gamma)}}{G_{1/N}(b_N)}, \quad \gamma : o \rightarrow b_N,$$

satisfies $\ell = N^{1+o(1)}$ and $D = N^{3/4+o(1)}$ in probability. For each fixed $r > 0$ it also satisfies

$$\mathbb{E}_{N,b_N} \ell^r = N^{r+o(1)}, \quad \mathbb{E}_{N,b_N} D^r = N^{3r/4+o(1)}.$$

Proof. Fix a small $\delta > 0$, and put $H = \lfloor N^{3/4-\delta} \rfloor$. Choose one of the three inward port normals at o whose projection of $b - o$ is at least $c|b - o|$. Call the vertex type of o type P , and the other type R . Start the port path on the cut immediately below o . If b has type R , end at its incident port on the cut immediately above b ; delete the initial and terminal half-edges to obtain the ordinary walk. If b has type P , end at its incident port immediately below b ; delete the initial half-edge and append the final half-edge from that port to b . The appended vertex lies beyond the final open slab and is therefore unvisited. If the port path has L internal vertices, the resulting edge count is $L - 1$ in the first case and L in the second. These operations are recoverable from the fixed orientation and terminal vertex. The associated port displacement differs from $b - o$ by a bounded amount, has positive height comparable to $|b - o|$, and has horizontal component at most a constant times that height.

When $|b - o| \gg H$, divide this displacement along its straight guide into $q = O(|b - o|/H)$ exact port displacements, each with integer height between H and $2H$ and bounded horizontal-to-height ratio. Round intermediate positions to ports on the chosen cuts; this changes each horizontal displacement by a bounded amount and leaves the final port exact. Proposition 9.4, with a fixed length slack $\eta > 0$ sufficiently small after δ , gives critical mass at least $cH^{-5/4}$ for every segment, using only lengths at most $(2H)^{4/3+\eta}$ and diameter at most CH . Since

$$H^{4/3+\eta} = o(N),$$

the thermal discount on each segment is bounded below by a fixed positive constant.

Concatenate the segments in their consecutive slabs. They cannot share vertices across slabs. The predetermined intermediate cuts are each crossed once, so the pieces are recoverable from their concatenation. They remain in a corridor of width CH about the straight guide. After the bounded endpoint conversion, we obtain

$$G_{1/N}(b) \geq \exp \left[-C_\delta \left(1 + \frac{|b-o|}{H} \right) \log N \right] \quad (|b-o| \gg H). \quad (56)$$

For fixed N and $s < m_{\text{rad}}(1/N)$,

$$G_{1/N}(b) \leq \left(\sum_v e^{s|v-o|} G_{1/N}(v) \right) e^{-s|b-o|}.$$

The parenthesized quantity is finite and independent of b . Hence the lower spatial rate is at least $m_{\text{rad}}(1/N)$, which is at least $N^{-3/4-\varepsilon}$ for every fixed $\varepsilon > 0$ and sufficiently large N . On the other hand (56) gives the upper spatial rate at most $C_\delta H^{-1} \log N$. Letting the fixed exponent slacks tend to zero proves both asserted rate scales. No equality or existence of a single directional spatial limit is required.

Now suppose $|b_N - o| = N^{3/4+o(1)}$. For every fixed $\zeta > 0$, choose $\delta < \zeta/3$ in (56). For all sufficiently large N this gives

$$G_{1/N}(b_N) \geq e^{-N^\zeta}. \quad (57)$$

The choice of ζ will be made after the desired tail slack, and stays fixed as N tends to infinity.

Let $c_j \rho^j$ be the total critical mass at edge length j . Proposition 5.12 gives $c_j \rho^j \leq \exp(C_\nu j^\nu)$ for every $\nu > 0$. Thus, for every $\varepsilon > 0$, the absolute thermal mass on $j > N^{1+\varepsilon}$, even multiplied by j^r for any fixed r , is at most $\exp(-cN^\varepsilon)$ after decreasing ν . Dividing by (57) proves the length upper tail and its moment version.

For the spatial upper tail choose a small $\varepsilon' > 0$ relative to ε . On lengths $j \leq N^{1+\varepsilon'}$, the event $D > N^{3/4+\varepsilon}$ is covered by (48), with a sufficiently small fixed η there. Its absolute mass is at most $\exp(-N^c)$ for some $c > 0$, after summing the polynomial number of lengths. The lengths above this cutoff have just been treated. Choose $\zeta < c$ in (57). This proves the spatial upper tail. The same proof, using $D \leq a_{\text{lat}} j$ and the weighted length tail, gives its positive-moment version.

Finally every path to b_N has $D \geq |b_N - o|$. If $j < N^{1-\varepsilon}$, this required displacement is at least $N^{3/4-o(1)}$, so (48) again bounds the absolute mass by $\exp(-N^{c'})$ for some $c' > 0$; lengths smaller than the graph distance contribute nothing. Division by (57) proves the length lower tail. The diameter lower bound is deterministic. Integrating the proved weighted upper tails and using these lower bounds gives both moment exponents. $\square$

Corollary 9.6 (Length responses at a prescribed endpoint). *For each fixed vertex b and $t > 0$, $G_t(b)$ can be differentiated term by term any fixed number of times. If $|b_N - o| = N^{3/4+o(1)}$, then for each positive integer j ,*

$$\left. \frac{(-1)^j \partial_t^j G_t(b_N)}{G_t(b_N)} \right|_{t=1/N} = N^{j+o(1)}, \quad -\partial_t \log G_t(b_N)|_{t=1/N} = N^{1+o(1)}.$$

The terminal vertex is held fixed while taking each derivative.

Proof. On a compact subinterval of $t > 0$, every differentiated series is dominated by a constant times the untilted susceptibility at a smaller positive value of t . Hence $(-1)^j \partial_t^j G_t(b)/G_t(b)$ is

exactly the j th edge-length moment under the normalized endpoint law. Apply Theorem 9.5. The logarithmic derivative is the first moment. Higher logarithmic derivatives are cumulants and are not identified by this argument. $\square$

10 Confined renewal rewards

The length tail of one irreducible does not specify where its expected length is accumulated. We now retain that length in two kinds of spatial region. The polynomial vacuum estimates give an irreducible reward in a logarithmically enlarged box and an expected number of completed vertices before exit. The disk representation gives the expected completed-prefix mass inside a ball whose radius is a fixed multiple of the height scale. These conclusions use the common renewal identities and their own finite inputs, stated below; the weaker hypotheses in the first argument do not require the sharp length deficit of Section 2.

10.1 Length rewards with logarithmic confinement

We use the honeycomb lattice dual to a triangular tiling of side one. Successive horizontal cuts are at Euclidean distance $\sqrt{3}/2$; their integer separation is the height. As before, a port has no weight, and a path visiting L vertices has weight ρ^L . Write b_h for the critical mass of bridges from a fixed bottom port to all top ports at height h , $b_0 = 1$, and m_h for the same mass with an insertion of L , $m_0 = 0$. The polynomial vacuum representation gives the following finite inputs [23, Theorem 1.1, Proposition 1.3 and Lemma 8.2]:

$$b_h = h^{-1/4+o(1)}, \quad m_h = h^{13/12+o(1)}. \quad (58)$$

For a fixed-height bridge let S be its largest absolute horizontal displacement from its source. There are absolute positive c, C such that

$$\sum_{\gamma: H(\gamma)=h, S(\gamma)>t} \rho^{L(\gamma)} \leq C e^{-ct/(h+1)}. \quad (59)$$

The same bound holds on the subclass of irreducibles. The mixed exit law in the strip has total first length mass at most $h^{13/12+o(1)}$; the mass of half-plane arches not contained in that strip is $O(b_h)$. For the latter assertion, let A_h be the strip return-arch mass and $c_0 = \cos(3\pi/8)$. The strip flux identity in [23, Section 2] is $b_h + c_0 A_h = 1$. Since $b_h \rightarrow 0$, monotone exhaustion gives $c_0 A_\infty = 1$ and $A_\infty - A_h = b_h/c_0$. All these are fixed-source, unnormalized masses. The logarithmic confinement in (59) will be used explicitly below.

Let p be the irreducible probability from Proposition 2.2. Its normalization proof only uses the concatenation identity and divergence of $\sum_h b_h$, so (58) suffices for this application. Under p , write H, L, D for height, vertex length and diameter, and S for largest horizontal displacement. Put $\beta = 3/4$ and $d = 4/3$. The following reward estimate is also useful when the terminal port is subsequently prescribed.

Proposition 10.1 (Truncated and discounted irreducible rewards). *For this irreducible law,*

$$p(H > r) = r^{-\beta+o(1)}, \quad \mathbb{E}_p[L; H \leq r] = r^{d-\beta+o(1)}, \quad (60)$$

$$\mathbb{E}_p[Le^{-H/r}] = r^{d-\beta+o(1)}. \quad (61)$$

The lower bound in (61) is retained after imposing $H \leq r \log^2 r$ and $S \leq r \log^4 r$. Furthermore,

$$p(L > n) \leq n^{-9/16+o(1)}, \quad p(D > r) \leq r^{-3/4+o(1)}, \quad 1 - \mathbb{E}_p e^{-L/n} \geq n^{-17/24+o(1)}. \quad (62)$$

Proof. For $s > 0$, the renewal and reward identities give

$$B(s) := \sum_{h \geq 0} b_h e^{-sh} = \frac{1}{1 - \mathbb{E}_p e^{-sH}}, \quad \sum_{h \geq 0} m_h e^{-sh} = B(s)^2 \mathbb{E}_p [Le^{-sH}].$$

Positive coefficient summation of (58) yields $B(1/r) = r^{\beta+o(1)}$ and $\sum m_h e^{-h/r} = r^{d+\beta+o(1)}$. This proves (61). The height deficit is $r^{-\beta+o(1)}$, so $p(H > r) \leq r^{-\beta+o(1)}$ and $\mathbb{E}[H; H \leq r] \leq r^{1-\beta+o(1)}$. To obtain the reverse height-tail bound, fix $\delta > 0$ and use the deficit at scale $r^{1+\delta}$. Heights at most r contribute at most $r^{-\beta-\delta+o(1)}$, whereas that deficit is $r^{-\beta(1+\delta)+o(1)}$. The former is negligible because $\beta < 1$. Thus $p(H > r) \geq r^{-\beta(1+\delta)-o(1)}$; let δ decrease to zero.

Write $j_h = \mathbb{E}_p[L; H = h]$. Coefficientwise the reward identity is $m = b * j * b$. At height $3r$, the convolution multiplier of each j_h with $h \leq r$ is

$$(b * b)_{3r-h} \geq r^{2\beta-1-o(1)},$$

because order r terms have both indices comparable to r . Since $m_{3r} \leq r^{d+\beta-1+o(1)}$, this proves the upper bound in (60). Splitting $p(L > n)$ according as $H > n^{1/d}$ or not gives the first bound in (62).

We verify that the discounted reward is spatially retained. The elementary packing bound for a self-avoiding path in a strip is $L \leq Ch(1+S)$ at height $h \geq 1$. Integrating (59) therefore gives, after decreasing c ,

$$\mathbb{E}_p[L; H = h, S > t] \leq Ch(1+t+h)e^{-ct/(h+1)}.$$

For $h > r \log^2 r$, use $j_h \leq m_h$ and the factor $e^{-h/r}$; the resulting tail is smaller than every inverse power of r . For $h \leq r \log^2 r$ and $S > r \log^4 r$, the last display has the same property after summing over h . Thus removing these two tails retains $r^{d-\beta+o(1)}$. Evaluating this conclusion at $r = h^{1-\delta}$ gives $\sum_{j \leq h} j_j \geq h^{d-\beta-o(1)}$ after letting $\delta \downarrow 0$. This proves the other half of (60).

On the retained set $L = O(r^2 \log^6 r)$. Take $r = n^{1/2-\delta}$, so eventually $L \leq n$ on that set. Since $1 - e^{-L/n} \geq (1 - e^{-1})L/n$ for $L \leq n$, its retained reward gives

$$1 - \mathbb{E}_p e^{-L/n} \geq n^{-1} r^{7/12-o(1)} = n^{-17/24-(7/12)\delta-o(1)}.$$

Let $\delta \downarrow 0$. Finally, split the event $D > r$ at $H = r/\log^2 r$. The height tail costs $r^{-\beta+o(1)}$; on the complement, diameter greater than r forces $S \geq c_0 r$. Summing (59) over the remaining heights is superpolynomially small. This proves the diameter bound. $\square$

To distinguish a completed prefix from the part retained before an exit, write η_k for the k th irreducible, $H_k = H(\eta_k)$, $L_k = L(\eta_k)$, $S_H(k) = \sum_{i \leq k} H_i$, and $\Gamma_k = \eta_1 \circ \dots \circ \eta_k$. For $h, w > 0$ put

$$V_h = \sum_{k \geq 1} L_k \mathbf{1}_{\{S_H(k) \leq h\}}, \quad V_h^{(w)} = \sum_{k \geq 1} L_k \mathbf{1}_{\{S_H(k) \leq h, S(\Gamma_k) \leq w\}}.$$

Thus $V_h^{(w)}$ counts a block only when its entire past, through that block's endpoint, remains in the lateral interval $[-w, w]$. Later blocks do not change whether an earlier block is counted.

Corollary 10.2 (Consequences under the weaker finite inputs). *Let γ_0 be the source port, γ_n the n th visited vertex of the infinite concatenation, and Y_n its forward height from the source. Almost surely,*

$$Y_n \geq n^{3/4-o(1)}, \quad |\gamma_n - \gamma_0| \leq n^{17/18+o(1)}.$$

For the normalized finite bridge law summed over heights with weight $\rho^L e^{-L/n}$, its height is at least $n^{3/4-\varepsilon}$ with probability tending to one, for every $\varepsilon > 0$. The unnormalized mass of half-plane arches

of length greater than n is at most $n^{-3/16+o(1)}$. The completed-vertex rewards just defined satisfy, for some fixed $C > 0$,

$$\mathbb{E}V_h = h^{4/3+o(1)}, \quad \mathbb{E}V_h^{(Ch \log^2 h)} = h^{4/3+o(1)}. \quad (63)$$

Consequently the expected number of vertices before first exit from a disk of radius R is at least $R^{4/3-o(1)}$.

Proof. We give the estimates that distinguish these conclusions from the sharp spatial law. At a dyadic vertex time n , let $m = \lfloor n^{9/16-\varepsilon} \rfloor$. The length tail in (62), a union bound above n , and Markov's inequality for lengths truncated at n imply

$$p\left(\sum_{i \leq m} L_i > n\right) \leq n^{-\varepsilon+o(1)}.$$

The probability that all of those m heights are below $n^{3/4-2\varepsilon/\beta}$ is stretched exponentially small, by the matching height tail. Once these pieces have been completed, every later vertex is above their slabs. Borel–Cantelli on dyadics, followed by monotonicity between dyadics, proves the asserted height lower bound.

Conversely take $m = \lceil n^{17/24+\varepsilon} \rceil$. The last estimate in (62) and exponential Markov give

$$p\left(\sum_{i \leq m} L_i \leq n\right) \leq e(\mathbb{E}_p e^{-L/n})^m \leq \exp(-n^{\varepsilon/2})$$

for large n . The diameter-tail bound, a union bound and truncated first moments show that $\sum_{i \leq m} D_i \leq m^{1/\beta+\varepsilon}$ except with a power-decaying probability. These bounds are summable on dyadics, and the union of the first m blocks contains every vertex up to time n . Sending ε to zero gives $(17/24)/\beta = 17/18$.

Including the empty bridge, the height-summed tilted law has normalizer

$$Z_n = (1 - \mathbb{E}_p e^{-L/n})^{-1} \geq n^{9/16-o(1)}.$$

The bound uses the upper length tail. Excluding the empty bridge replaces Z_n by $Z_n - 1$ and does not change the conclusion. Its part of height at most h is bounded by $\sum_{j \leq h} b_j = h^{\beta+o(1)}$. Take $h = n^{3/4-\varepsilon}$. For arches, split at height $h = n^{3/4}$: the part leaving that strip has mass $O(b_h)$; the remaining long arches have mass at most $n^{-1}h^{13/12+o(1)}$. Both contributions have exponent $-3/16$.

For the completed prefix, put $j_h = \mathbb{E}_p[L; H = h]$ and mark the start height l of each counted block. The expected count is exactly

$$\mathbb{E}V_h = \sum_{l+j \leq h} b_l j_j.$$

The cumulative bounds in Proposition 10.1 prove the upper bound. For the lower restrict $l \leq h/2$ and apply the retained reward at scale $r = h^{1-\delta}$, so eventually $j \leq h/2$. The prefix masses with $l \leq h/2$ total $h^{\beta+o(1)}$. Their contribution with lateral excursion greater than $h \log^2 h$ is superpolynomially small by (59); the retained last block has excursion at most $r \log^4 r = o(h)$. For each marked block, its preceding prefix stays within $h \log^2 h$ of the source and the marked block stays within $o(h)$ of its own start. Hence the whole prefix through that block is confined to $[-Ch \log^2 h, Ch \log^2 h]$ for one fixed C . These are precisely contributions to $V_h^{(Ch \log^2 h)}$, of expected total mass at least $h^{\beta+o(1)} r^{d-\beta-o(1)}$. Let $\delta \downarrow 0$ and use $V_h^{(w)} \leq V_h$ to obtain both parts of (63). Taking h a sufficiently small multiple of $R/\log^3 R$ puts every retained prefix inside the disk. All its counted vertices occur before first exit, which proves the last assertion. $\square$

10.2 Completed-prefix mass in a fixed-proportion ball

The disk transfer representation supplies a stronger confinement input [21, Theorem 1.1]. In its unit-edge embedding a layer has height $3/2$, and its height- h bridge masses satisfy

$$b_h \asymp (1+h)^{-1/4}, \quad m_h = h^{13/12+o(1)}. \quad (64)$$

For every sufficiently large fixed C , the lower bound for m_h remains valid when the entire bridge has lateral wandering at most Ch . Multiplying the preceding triangular-side-one coordinates by $\sqrt{3}$ gives this unit-edge embedding: the layer distance becomes $3/2$. Vertex counts and critical weights are unchanged. We keep layer height integral in each embedding.

Proposition 10.3 (A fixed-ball completed-prefix expectation). *Under the irreducible law from (64), let $\mathcal{V}_R$ be the vertices through the last block ending at cumulative layer height at most R , excluding the overshooting block, and let $\mathcal{V}_\infty$ be the full infinite trace. With the starting port as origin,*

$$\mathbb{E}|\mathcal{V}_R| = R^{4/3+o(1)}, \quad \mathbb{E}|\mathcal{V}_R \cap B(0, R)| = R^{4/3+o(1)},$$

and almost surely

$$|\mathcal{V}_\infty \cap B(0, R)| \leq R^{4/3+o(1)}.$$

Changing the ball radius by any fixed positive factor preserves the expectation exponent.

Proof. Again put $j_h = \mathbb{E}_p[L; H = h]$ and $\beta = 3/4$. The renewal and reward identities with (64) give

$$B(1/R) \asymp R^\beta, \quad \sum_h m_h e^{-h/R} = R^{25/12+o(1)}, \quad \sum_h j_h e^{-h/R} = R^{7/12+o(1)}.$$

They imply $\sum_{h \leq R} j_h = R^{7/12+o(1)}$. For the lower bound, evaluate at $R^{1-\varepsilon}$: the tail beyond R is exponentially negligible because $j_h \leq m_h$ has polynomial growth; then let $\varepsilon \downarrow 0$. Consequently $\sum_{l+h \leq R} b_l j_h = R^{4/3+o(1)}$, proving the first expectation and the upper bound for the second.

The fixed-ball lower bound needs the confined bridge estimate, rather than only these transforms. Choose $c_0 > 0$ so small that each height- m bridge with $c_0 R \leq m \leq 2c_0 R$ and lateral wandering at most Cm lies in $B(0, R)$ and has height at most R . The sum of their length masses is

$$\sum_{c_0 R \leq m \leq 2c_0 R} \sum_{\substack{\gamma: H(\gamma)=m \\ \gamma \subset B(0, R)}} L(\gamma) \rho^{L(\gamma)} \geq R^{25/12-o(1)}. \quad (65)$$

Under the iid irreducible law, the left side counts visits in prefixes ending at such renewal heights, with a visit repeated at later permissible prefix endings. For a vertex in a block ending at height $s \leq R$, condition on that block and its past. Dropping every future confinement condition, the expected number of repetitions at later renewal endings is at most $\sum_{h \leq R} b_h = O(R^\beta)$, by independence of the future pieces. Every counted vertex belongs to $\mathcal{V}_R \cap B(0, R)$. Tonelli's theorem therefore bounds the left side of (65) by $CR^\beta \mathbb{E}|\mathcal{V}_R \cap B(0, R)|$. Since $25/12 - \beta = 4/3$, this proves the lower bound. The same choice of a smaller c_0 handles any fixed positive multiple of the ball radius.

For the almost-sure statement, the bounded-factor height deficit gives $p(H > R) \asymp R^{-\beta}$: its upper bound is immediate, and after using that upper bound the contribution of $H \leq \eta R$ to the deficit at R is at most $C\eta^{1-\beta} R^{-\beta}$. Choose a fixed small η ; the remaining contribution gives $p(H > \eta R) \geq cR^{-\beta}$. Replacing R by R/η proves the lower bound at R . Splitting according to height and using the cumulative reward gives $p(L > t) \leq t^{-9/16+o(1)}$. The first $k = \lceil R^{\beta+\varepsilon} \rceil$ blocks reach a height larger than the greatest layer meeting $B(0, R)$ except with probability $\exp(-cR^\varepsilon)$. Their total length is at most $R^{4/3+3\varepsilon(4/3)/\beta}$ except with probability $R^{-2\varepsilon+o(1)}$, by a union bound for a length above that threshold and Markov's inequality for truncated lengths. After those blocks, the path remains above the ball. The estimates are summable on dyadic radii; monotonicity and $\varepsilon \downarrow 0$ complete the proof. $\square$

11 The irreducible length tail from signed cylinder estimates

This section derives the irreducible length tail using only the signed cylinder estimates stated below and the elementary renewal identities. The main difficulty is a lower bound: large mean length for finite chords must force a long irreducible piece, even after the chords are converted into bridges. We control the second length moment by closing chords into polygons, and control the multiplicity of the conversion by folding arches around a fixed irreducible piece.

11.1 Finite inputs and the proof strategy

We use ports on the triangular tiling of side one, with height measured in whole tessellation layers and length equal to visited vertex count. The finite inputs come from the signed cylinder representation [22, Section 2, Propositions 9.1–9.2 and Theorems 1.1–1.2]. We state them here in the form used below. Let $\vartheta = \pi/8$. If $K(a, b)$ is the positive critical chord kernel in a convex tessellation polygon, its signed boundary balance is

$$\sum_b e^{3iT(a,b)/8} K(a, b) = 1,$$

where $T(a, b)$ is the total tangent turn of a chord from the inward edge at a to the outward edge at b . Convexity gives $|T(a, b)| \leq \pi$, so the real coefficients are uniformly positive. The same local identity cancels the two orientations of a self-collision whose incoming stem lies outside its loop. In the upper half-plane put $H_\vartheta(d) = K(0, d)$ for the kernel to the port d lattice units to the right, and let W_R be the mass of arches from one port reaching distance at least R . The finite boundary theorem gives

$$2 \sin \vartheta \sum_{d \geq 1} H_\vartheta(d) = 1, \quad W_R = R^{-1/4+o(1)}. \quad (66)$$

For a finite convex tessellation polygon D and a honeycomb vertex v not adjacent to a boundary port, the raw critical visit mass summed over all ordered first-exit boundary chords of D through v , with both sources and terminals summed, is comparable with absolute positive constants to the sum of the three nesting partitions at its incident face centers. Each nesting partition counts disjoint enclosing polygons in D , with weight $2\rho^{|\lambda|}$ per polygon. Writing $G_M(g)$ here for the companion's $G_{M/2}(g)$, where the full physical period M belongs to $4\mathbb{N}$ and the two marked faces are half a period apart, we have

$$G_M(2) = M^{1/6+o(1)}. \quad (67)$$

The visit and localization theorems give the following bounds in a regular hexagon D_R of side R : the mass of ordered chords visiting the middle disk of radius $R/10$ is at most $R^{3/4+o(1)}$, and their first length mass is at least $R^{25/12-o(1)}$. Indeed, every chord visiting the middle disk belongs to the companion's macroscopic chord ensemble, which gives the mass upper bound. Its uniform raw visit mass at each middle vertex is $R^{1/12+o(1)}$. Summing over the order R^2 vertices in the disk gives the first-length lower bound. Finally the marked-polygon theorem gives

$$\sum_{\substack{\lambda \text{ modulo translations} \\ \text{diam } \lambda \leq Q}} \rho^{|\lambda|} |\lambda|^2 \leq Q^{2/3+o(1)}. \quad (68)$$

Polygons in this display are simple, unoriented and unrooted, with each lattice-translation class counted once. Counting both orientations changes only a factor of two; rooting at every vertex would change the moment being estimated. These finite inputs precede the renewal argument. For an infinite concatenation, let γ_0 denote its starting port.

Theorem 11.1 (Length tail from boundary folding). *Under the irreducible critical bridge law p , the finite estimates above imply*

$$p(L \geq n) = n^{-9/16+o(1)}.$$

They also give $p(H > r) = r^{-3/4+o(1)}$ and $p(D > r) \leq r^{-3/4+o(1)}$. Consequently the iid infinite bridge satisfies $|\gamma_n - \gamma_0| = n^{3/4+o(1)}$ and has $R^{4/3+o(1)}$ vertices in a radius- R disk, almost surely. At every integer strip height h , the critical fixed-source, free-terminal bridge law has length $h^{4/3+o(1)}$ in probability and mean length $h^{4/3+o(1)}$. Under the bridge law summed over heights with weight $\rho^L e^{-L/t}$, length is $t^{1+o(1)}$ and diameter is $t^{3/4+o(1)}$ in probability.

All masses in the proof have the positive weight ρ^L unless a tilt is explicitly specified. Turning phases occur only in the identities used to compare these masses.

The decisive comparison can be stated before the geometric details. Sum over $H \leq R \leq 2H$ and over ordered chords of D_R visiting its middle disk, and let $M_j(H)$ be the resulting mass with insertion L^j . The inputs give

$$M_0(H) \leq H^{7/4+o(1)}, \quad M_1(H) \geq H^{37/12-o(1)}.$$

We shall prove

$$M_2(H) \leq H^{53/12+o(1)}.$$

Removing chords shorter than $H^{4/3-\delta}$ does not remove the first moment, and Cauchy–Schwarz then leaves long-chord mass $H^{7/4-o(1)}$. Depending on their endpoint sides, these chords give long bridge mass $H^{3/4-o(1)}$ summed over a height interval, or long arch mass $H^{-1/4-o(1)}$ from a fixed source. Folding an arch at its highest point makes a bridge. The additional estimate needed for this operation bounds its weighted multiplicity by $H^{1/2+o(1)}$, uniformly after one irreducible piece is prescribed. These powers will give the lower irreducible tail $p(L \geq H^{4/3-\eta}) \geq H^{-3/4-o(1)}$.

The proof develops the estimates in their dependency order. Boundary balances first give strip masses, strip moments and a slit bound. A two-source identity then controls pairs of ordered renewal arms; this proves the uniform folding estimate. Finally we close the finite chords, extract the lower tail, and apply the renewal identities to the three laws in Theorem 11.1.

11.2 Boundary comparisons and strip moments

Levels are horizontal tessellation lines, with heights in units of their spacing. Distances along a line are in side units. Let b_h be the mass of strict bridges from a fixed bottom port to the top of the infinite height- h strip, and put $b_0 = 1$. Let a_j be the fixed-source half-plane arch mass with maximum in layer j , the region between levels $j-1$ and j . Its highest visited triangle is down-pointing and has center at height $j-1/3$: an up-pointing triangle in that layer has only one branch leading down and cannot be an internal maximum. Write

$$A(d) = \sum_{q \geq d} H_{\partial}(q).$$

Lemma 11.2 (Comparison with a half-plane cap). *In any convex tessellation domain, finite or infinite, the mass of chords from a fixed port which reach distance at least S from that port is at most CW_{cS} , for absolute positive constants c, C . Convex boundary balances extend to half-planes, strips and wedges by exhaustion.*

Proof. First exhaust a tangent half-plane by aligned tessellation hexagons. Boundary saturation in (66) and positive real parts show that the artificial exits tend to zero. Artificial exits in any smaller

convex domain are a subset of these exits, so they too vanish. This justifies its limiting boundary balance.

For the quantitative assertion, cut the domain by a tessellation hexagon about a nearby lattice vertex, whose inner and outer radii about the source lie between cS and $S/2$. Subtract the balances before and after the cut. A chord valid in both domains has the same terminal phase and cancels; all remaining old chords are precisely those lost on imposing the cut. Every chord reaching distance S is among these lost chords. The new exits lie on the cap, and convex positivity compares their mass with the lost mass in both directions, with absolute constants. Their paths are also cap exits in the cut tangent half-plane. Applying the same subtraction there bounds their mass by the lost half-plane arch mass, at most W_{cS} . The exhaustion argument above applies if the original domain is infinite. $\square$

Lemma 11.3 (Boundary tails and bridge mass). *One has*

$$A(h) = h^{-1/4+o(1)}, \quad b_h = h^{-1/4+o(1)}. \quad (69)$$

For a wedge of angle 60° or 120° , a source on one ray at apex distance at most CH has mass at least $H^{-1/4-o(1)}$ to the other ray. This lower bound is retained within distance $H^{1+\epsilon}$ of the source, for every fixed $\epsilon > 0$.

Proof. Consider first the 60° wedge between the positive horizontal ray and the ray at angle $\pi/3$, with source at half-integer coordinate $a > 0$ on the horizontal ray. Relative to the full upper half-plane, let $L_<(a)$ and $L_>(a)$ be the lost masses to the left and right of the source, including endpoints beyond the apex. Let $D(a, b)$ be the lost mass when both endpoint coordinates are positive. Reversal gives $D(a, b) = D(b, a)$.

The phases of a left base exit, a right base exit and an exit on the other ray are respectively $e^{3i\vartheta}$, $e^{-3i\vartheta}$ and $e^{i\vartheta}$. If $K_w(a)$ is the mass to the other ray, subtraction of the balances gives

$$e^{3i\vartheta} L_<(a) + e^{-3i\vartheta} L_>(a) = e^{i\vartheta} K_w(a).$$

Multiplication by $e^{-i\vartheta}$ and imaginary parts yield $L_<(a) = \sqrt{2} L_>(a)$. Summing this relation over $a \leq R$ and using symmetry of D gives

$$(\sqrt{2} - 1) \sum_{0 < a < b \leq R} D(a, b) + \sqrt{2} \sum_{0 < a \leq R < b} D(a, b) = \sum_{0 < a \leq R} A(\lceil a \rceil).$$

The total loss is at most CW_{ca} by Lemma 11.2. For a lower bound, take also the reflected wedge with its apex on the other side of the source. Their intersection is a triangle of diameter $O(a)$. An arch escaping this triangle is lost in at least one wedge; reflection makes their loss masses equal. Thus the total loss in either wedge is at least cW_{Ca} , and has exponent $-1/4$.

Since $L_>(a) = \sum_{b > a} D(a, b)$, the preceding identity is bounded above and below by fixed multiples of $\sum_{a \leq R} L_>(a)$. Consequently

$$\sum_{a \leq R} A(\lceil a \rceil) = R^{3/4+o(1)}.$$

Monotonicity gives the upper bound for $A(R)$. For the lower bound, subtract the cumulative sum at R from that at $R^{1+\delta}$ and bound the difference above by $R^{1+\delta} A(R)$. Letting $\delta \downarrow 0$ proves the first assertion in (69).

Half-plane-strip subtraction gives the exact relation

$$b_h = \sin \vartheta \sum_{j>h} a_j.$$

An arch in this sum reaches distance comparable to h , so the cap bound gives $b_h \leq h^{-1/4+o(1)}$. To obtain the reverse bound, let $K_w(a, v)$ be the other-ray wedge kernel. Reversal and reflection make it symmetric in a, v , and the balance above shows $\sum_v K_w(a, v) = a^{-1/4+o(1)}$. Cut the wedge at height h . All exits with $v > Ch$ are lost, and the new exits have mass at most b_h per source. By symmetry,

$$\sum_{a \leq R} \sum_{v > Ch} K_w(a, v) \geq R^{3/4-o(1)} - h^{3/4+o(1)}.$$

The subtracted term bounds the sum with $v \leq Ch$ by summing first over v and then over all a . Therefore

$$Rb_h \geq c(R^{3/4-o(1)} - h^{3/4+o(1)}).$$

Set $R = h^{1+\epsilon}$ and let $\epsilon \downarrow 0$.

For either stated wedge angle, the missing same-side endpoints beyond the apex have mass at least $A(CH) = H^{-1/4+o(1)}$. Its boundary balance compares the total same-side loss with the other-ray mass, with fixed positive constants. Lemma 11.2 removes at most $H^{-(1+\epsilon)/4+o(1)}$ outside the prescribed radius, which is negligible. This proves the confined assertion. $\square$

Lemma 11.4 (Strip moments and fixed-width summability). *Let $H_h(d)$ be the strip same-side kernel at separation d . Then*

$$m_B(h) := \sum_{\text{bridges of height } h} \mathsf{L}\rho^{\mathsf{L}} \leq h^{13/12+o(1)},$$

and the same upper bound holds for the first-length mass of strip arches. Moreover,

$$\sum_{d>0} dH_h(d) \leq h^{3/4+o(1)}. \quad (70)$$

At each fixed strip height, the total critical mass and first-length mass of all walks from a fixed vertex staying in that strip are finite.

Proof. The interior visit identity bounds the sum of positive chord visits to any non-boundary-adjacent vertex by a constant times its three incident $+2$ nesting partitions. Each such one-face partition in the full strip has square at most $G_M(2)$, for a physical period $M \asymp Ch$ divisible by four in a transverse 60° direction. To see the square, translate one strip and its marked face to each of the two cylinder marks. For large fixed C , the vertical separation of the half-period translates exceeds the strip width. The two strips are disjoint, their cycles embed without intersections, and every cycle separates the marks. Their independent union is therefore counted by $G_M(2)$.

It follows from (67) that the raw visit mass at each such vertex is at most $h^{1/12+o(1)}$. This bound applies first in finite convex strip exhaustions and then, by positivity, to genuine strip chords. For a fixed source, sum visits along a horizontal row. Horizontal translation identifies this sum with the sum over the corresponding boundary sources at one representative vertex of that row; it is bounded by the raw all-source visit estimate. There are $O(h)$ rows. Visits in a port-adjacent row can be charged, with bounded multiplicity, to slant neighbors on the path that are not adjacent to the boundary, when $h \geq 2$. This proves the first-length bounds. Smaller heights have finite masses and moments as well: append a fixed bridge to reach a larger height; for arches use monotonicity.

For (70), cut the strip on a transverse 60° tessellation line. Compare balances in the strip and in one remaining side of this cut, then sum over bottom sources on that side. Retain only same-side chords whose endpoints are on opposite sides of the cut; all are lost. A chord with endpoint separation d has exactly d such source placements, up to a fixed lattice factor. The total lost mass thus bounds the left side of (70). It is controlled by new cut exits. Reversing these paths gives, at each cut port of height y , a mass of bottom exits at most $O(1)$, and at most CW_{cy} when y is bounded away from zero, by Lemma 11.2. Summing over $0 \leq y \leq h$ gives $h^{3/4+o(1)}$.

We finally record why arbitrary fixed-width walks, needed in the slit exhaustion below, are summable. Split such a walk at a global highest vertex. On each resulting chain, cut at the last occurrence of its lowest level, then the last highest level in the remainder, and continue. Vertex heights in thirds of a layer are integral and change at every step. The positive spans of these extremal pieces decrease strictly, except possibly for equality of the first two. Thus the number of pieces is bounded in terms of the strip height. Each piece can be extended at its extreme ends to horizontal ports by adding at most one new vertex per end, giving a bridge with bounded multiplicity and bounded weight change. Drop avoidance between different pieces and compensate for shared endpoints. There are finitely many possible lists of spans. For each list, the mass and its first-length insertion are bounded by products of finite bridge masses and moments. There are only finitely many possible rows of the split vertex; after fixing a representative in that row, the original initial vertex determines at most one horizontal translation. This proves both asserted finiteness statements. $\square$

11.3 Slit estimates and ordered pairs of renewal arms

The next estimates will control how many arches can fold into a bridge containing a prescribed irreducible. We first bound paths that go around the tip of a slit. We then use the same bound for pairs of bridge prefixes whose endpoints are far apart at the shorter height.

Lemma 11.5 (A slit bound uniform in the lower depth). *In the strip between levels $-s_0$ and L , where s_0, L are positive integers, cut the bonds at ports on a horizontal ray from a tessellation vertex to the right at level zero. The cut has upper and lower shores. For an upper-shore source at distance $a = 1/2, 3/2, \dots$ from the tip, entering upward, let $V(a)$ be the mass of exits on the bottom wall. Then*

$$\sum_a V(a) \leq L^{3/4+o(1)}, \quad (71)$$

uniformly in s_0 .

Proof. The slit is attached to the exterior. A stem from its boundary still approaches a first self-collision from outside the new loop, so the same local cancellation proves the signed balance. Figure 6 shows the domain. The following table gives the boundary-determined turns from an upward-entering source; a thin opening of the slit makes the concave tip turn $-\pi$ explicit. Put $\theta_0 = 3\pi/8$.

exit	upper left	upper right	top	bottom	lower shore
T	π	$-\pi$	0	π	2π
$e^{3iT/8}$	$e^{i\theta_0}$	$e^{-i\theta_0}$	1	$e^{i\theta_0}$	$e^{2i\theta_0}$

Compare with the isolated upper strip of height L . Let $E_<(a)$ and $E_>(a)$ be the nonnegative gains to other upper-shore ports on the left and right of the source, and let $E_T(a)$ be the gain to the top. Let $A_0(a)$ be the reference mass to same-side endpoints left of the tip, which are missing from the

slit shore. Project the balance difference by

$$\mathcal{P}(w) = -\frac{\Im(e^{-2i\theta_0}w)}{\sin \theta_0}.$$

It kills every lower-shore term, including a return to the same cut location. The other coefficients give

$$E_{<} + V + 2 \cos \theta_0 E_T = A_0 + \frac{\sin(\pi/8)}{\sin \theta_0} E_{>}.$$

These identities may first be written in remote finite truncations. Lemma 11.4 makes their exit errors vanish and allows absolute summation over sources. For the latter assertion, every gained or bottom-exiting path goes around the tip. Translate to a fixed source and ignore the slit: its possible offsets relative to the tip are at most a constant times its length. The fixed-width first-length sum therefore bounds the summed gains and bottom exits.

Reversal of an upper-shore gained chord gives $\sum_a E_{<}(a) = \sum_a E_{>}(a)$. Since $\sin(\pi/8)/\sin \theta_0 = \sqrt{2} - 1 < 1$, summation and positivity imply

$$\sum_a V(a) \leq \sum_a A_0(a) = \sum_{d>0} dH_L(d).$$

Lemma 11.4 proves the desired bound. Its right side depends only on the upper height L , which explains the uniformity in s_0 . $\square$

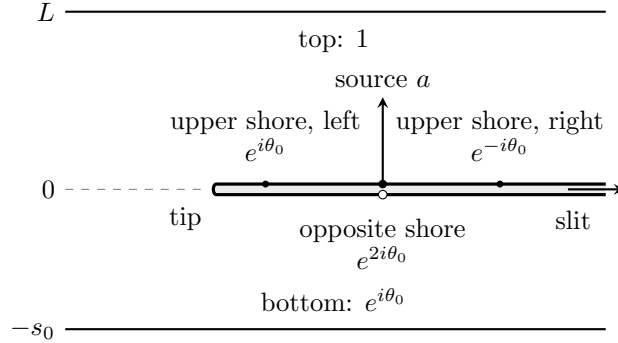


Figure 6: The slit domain and the exit phases for an upward-entering source at distance a from the tip, where $\theta_0 = 3\pi/8$. The shores are separated in the drawing to distinguish the two ports at each cut bond. A chord to the opposite shore has total tangent turn 2π ; its phase is therefore removed by projection perpendicular to $e^{2i\theta_0}$.

Lemma 11.6 (Two sources and positive boundary kernels). *Let K be the positive chord kernel in a convex tessellation polygon. For adjacent ports a, b in counterclockwise order on a straight boundary segment and another port e , with cyclic order $a < b < e$, let $K_2(ac, bd)$ denote the mass of mutually vertex-disjoint chord pairs. Put $r = e^{2i\theta}$ and $s = 3/8$. Then*

$$K(a, e) - rK(b, e) = \sum_{b < d < e} e^{isT(b, d)} K_2(ae, bd) - r \sum_{e < c < a} e^{isT(a, c)} K_2(ac, be). \quad (72)$$

Consequently same-side kernels decrease as the terminal port moves away from a fixed source. For every fixed small $\epsilon > 0$, the half-plane kernel confined within distance $H^{1+5\epsilon}$ of its source satisfies

$$K_{\text{conf}}(c, b) \geq H^{-5/4 - C\epsilon - o(1)} \quad (1 \leq |b - c| \leq H^{1+3\epsilon}). \quad (73)$$

Proof. Because a, b lie on the same straight side, $T(a, e) = T(b, e)$; give both trunks the common phase $e^{isT(a, e)}$ during the following exploration. Fix a trunk from a to e and grow a turning-weighted stem from b , stopping at its first trunk contact as well as at boundary exits. Compare this with trunk b to e and stem from a . Self-collisions cancel by the local rule. A contact term is an embedded Y tree with outward arms in cyclic order a, b, e . At its junction, the turn from the a arm onto the e arm is $+\pi/3$, whereas from the b arm it is $-\pi/3$. The other turns and all vertex weights agree, so the ratio of the two contact phases is $e^{is(2\pi/3)} = r$. This also covers immediate contact. After multiplying the second exploration by r , all contact terms cancel. The remaining disjoint boundary exits have exactly the two cyclic ranges in (72); division by the common trunk phase gives that identity. In strip exhaustions, remote disjoint exits vanish by product domination and the cap bound.

If e lies beyond b on the same base segment, multiply the identity by $e^{3i\vartheta}$ and take imaginary parts. The first sum has zero imaginary part. In the subtracted sum the sine phases are nonnegative, since $5\vartheta + sT \in [2\vartheta, 8\vartheta]$. As $\sin 3\vartheta = \sin 5\vartheta > 0$, this gives $K(a, e) \leq K(b, e)$. Reversal makes it outward decrease of the terminal from a fixed source; a tessellation reflection gives the other direction.

To prove the confined lower bound, use a tangent half-hexagonal cap whose radius is a small constant times $H^{1+5\epsilon}$. Along its base, the sum of kernels to ports beyond distance $H^{1+4\epsilon}$ is at least the half-plane tail $A(H^{1+4\epsilon})$ minus the lost mass $CW_{cH^{1+5\epsilon}}$. By Lemmas 11.2 and 11.3, the second term is negligible and the first has exponent $-(1+4\epsilon)/4$. There are $O(H^{1+5\epsilon})$ ports on the cap base. The outward monotonicity just proved bounds the kernel to every nearer port below by this sum divided by that number. The cap lies inside the required confinement radius, giving (73). $\square$

Lemma 11.7 (Top endpoints of two bridges). *Take adjacent bottom ports a, b , ordered left to right. Let $b_h(e)$ be the height- h top kernel from a , and put $D_h(c, d) = K_2(ac, bd)$ for top ports $c < d$, in left-to-right order. Then*

$$b_h(e) - b_h(e-1) = \sum_{d>e} D_h(e, d) - \sum_{c<e} D_h(c, e), \quad b_h(e) = \sum_{c \leq e < d} D_h(c, d),$$

and

$$\sum_{h \geq H} \sum_{c < d} D_h(c, d) \leq H^{-1/4+o(1)}. \quad (74)$$

Proof. Apply the imaginary-part projection in the preceding proof with e on the top wall. Bottom exits to the right of b and to the left of a vanish. The top coefficients agree because $\sin 3\vartheta = \sin 5\vartheta$, giving the difference identity. Summing it from the left end of the strip gives the second identity; the kernel tends to zero at infinity by summability.

For the last bound, join the adjacent bottom ports below the strip by a three-center arch. Together with the two bridges this gives a downward half-plane arch from c to d . Translate its first port to a fixed source. The connector has a unique deepest vertex, so the output recovers the connector, the translation and the original height. Thus the map is injective over all h , with constant weight factor ρ^3 . Its image reaches distance at least a constant times H , and (66) proves the bound. $\square$

We now pass to irreducible pieces, still without any lower length-tail estimate. Unique decomposition at once-crossed levels gives

$$B(z) = \sum_{h \geq 0} b_h z^h = \frac{1}{1 - I(z)}, \quad I(z) = \sum_{\gamma \in \mathcal{I}} \rho^{L(\gamma)} z^{H(\gamma)}.$$

Lemma 11.3 gives $B(e^{-1/H}) = H^{3/4+o(1)}$; letting $z \uparrow 1$ proves $I(1) = 1$. Thus the critical irreducible weights define the same law p as in Proposition 2.2. Write H, X, L for a block's height, signed horizontal displacement and length, and let S be its horizontal coordinate range. Its diameter is at most $C(H + S)$.

It will be convenient to name the normalized law of all finite bridges with weight $\rho^L z^H$, including the empty bridge: call it $\mathbb{Q}_z$. Under $\mathbb{Q}_z$, the number of pieces is geometric with probabilities $(1 - I(z))I(z)^k$, $k \geq 0$, and, given this number, their shapes are independent with law

$$p_z(d\gamma) = I(z)^{-1} z^{H(\gamma)} p(d\gamma).$$

This statement concerns different pieces; no independence of the coordinates within one piece is asserted.

Lemma 11.8 (Increment estimates before the lower length bound). *Under p ,*

$$\mathbb{P}_p(H > t) = t^{-3/4+o(1)}, \quad \mathbb{P}_p(S > t) \leq t^{-3/4+o(1)}, \quad \mathbb{P}_p(L > n) \leq n^{-9/16+o(1)}. \quad (75)$$

In addition, $\mathbb{P}_p(|X| > t) = t^{-3/4+o(1)}$.

Proof. The height deficit is $\mathbb{E}_p(1 - e^{-H/t}) = t^{-3/4+o(1)}$, giving the height upper tail and $\mathbb{E}[H; H \leq t] \leq t^{1/4+o(1)}$. Evaluate the deficit at $t^{1+\epsilon}$. Heights at most t contribute only $t^{-3/4-\epsilon+o(1)}$, which is smaller than the deficit $t^{-3(1+\epsilon)/4+o(1)}$. This proves the lower tail as $\epsilon \downarrow 0$.

Call a block wide if $S > t$, and put $u = \mathbb{E}_p[z^H; S > t]$. A bridge containing such a block is itself wide, so at each height its mass is at most CW_{ct} by the cap bound. Under $\mathbb{Q}_z$, the probability of at least one wide block is exactly

$$\frac{B(z)u}{1 + B(z)u}.$$

Take $z = e^{-1/H}$ with $H = t^{1-\epsilon}$. This probability is at most $CHW_{ct}/B(z) = o(1)$, and hence $u \leq H^{-1/2+o(1)}t^{-1/4+o(1)}$. On $H \leq H$ the tilt is bounded below. Add the height-tail bound on the complement and let $\epsilon \downarrow 0$ to obtain the span bound.

The marked-reward identity gives

$$\sum_h m_B(h) z^h = B(z)^2 \mathbb{E}_p[L z^H], \quad \mathbb{E}_p[Le^{-H/H}] \leq H^{7/12+o(1)}.$$

Splitting according to $H > n^{3/4}$ and applying Markov to the remaining length gives the upper length tail in (75).

It remains to prove that the signed displacement has a matching lower tail. By Lemma 11.7, bridge weights to any endpoint windows of size G per row, summed over $R \leq h \leq 2R$, are at most $C(G+1)R^{-1/4+o(1)}$. For $z = e^{-1/R}$, the whole height bin has $\mathbb{Q}_z$ -probability $R^{-o(1)}$, whereas the part with $|X_{\text{total}}| \leq R^{1-\epsilon}$ has probability at most $R^{-\epsilon+o(1)}$. Thus displacement larger than this threshold has at least subpower probability.

If every piece had displacement at most $t_* = R^{1-\delta}$, with $\epsilon \ll \delta$, such a total displacement would be superpolynomially unlikely. Here are the estimates. Reflection preserves height and reverses displacement, so the truncated increments $X_{\text{trunc}} = X \mathbf{1}_{\{|X| \leq t_*\}}$ are centered under p_z . Their variance is at most $t_*^{5/4+o(1)}$ by the span tail. The geometric number of pieces exceeds $R^{3/4+c\delta}$ with superpolynomially small probability for small fixed $c > 0$. For $|v| \leq 1/t_*$,

$$\log \mathbb{E}_{p_z} e^{v X_{\text{trunc}}} \leq C v^2 \mathbb{E}_{p_z} X_{\text{trunc}}^2.$$

Both the displacement threshold divided by t_* and its square divided by the resulting total variance grow as positive powers of R . Exponential Markov therefore gives the claimed negligible probability. A union bound, using the geometric mean $R^{3/4+o(1)}$, now forces

$$\mathbb{E}_p[z^H; |X| > R^{1-\delta}] \geq R^{-3/4-o(1)}.$$

Let $\delta \downarrow 0$ and combine with the span upper tail. $\square$

Lemma 11.9 (Two ordered renewal arms). *Start independent infinite p -sequences at adjacent bottom ports, designated left and right. Let P_m be the following event:*

- *the paths formed by the first m pieces of the two arms are disjoint;*
- *for each pair of positive renewal indices at most m , if their heights are $n \leq p$, every crossing of line n by the height- p prefix lies strictly on its designated side of the other arm's endpoint at n . When $n = p$, this means that the two endpoints are ordered.*

Then

$$p_m := \mathbb{P}(P_m) \leq m^{-1+o(1)}. \quad (76)$$

Proof. Fix a small $\delta > 0$, and set

$$L = \lceil m^{4/3+\delta} \rceil, \quad G = m^{4/3-\delta}.$$

We first show that, outside a superpolynomially small event, on each arm at most $m/4$ renewal endpoints lie within horizontal distance G of the pertinent extreme crossing of the other full infinite path at that height. The pertinent crossing is the leftmost on the right arm or the rightmost on the left arm; it exists because the path reaches every height.

Condition on the other path. Divide the renewal indices into residue classes modulo $m_0 = \lfloor m^{1-c\delta} \rfloor$, with a small fixed $c > 0$. Successive sampled endpoints in one class are separated by a fresh chunk of m_0 independent pieces. By the displacement lower tail in Lemma 11.8, such a chunk contains a positive power of m increments with magnitude greater than $2G$, except with probability tending to zero. Conditional on all chunk heights and absolute displacements, their nonzero signs are independent fair signs. The target crossing position depends on the heights and the other path, but not these signs.

If there are k increments of magnitude greater than $2G$, their signed sum lands in an interval of width $2G$ with probability at most $C/\sqrt{k}$. Indeed, after fixing the other signs, the subsets of positive large signs that yield a sum in that interval form an antichain: adding one such sign changes the sum by more than $4G$. Counting maximal chains bounds its size by the middle binomial coefficient. Thus the conditional window probability at each sampled endpoint tends uniformly to zero. Successive conditioning makes a positive fraction of hits in any residue class exponentially unlikely in m/m_0 . Union over the classes proves the assertion about at most $m/4$ close endpoints; the initial incomplete chunk costs at most $m_0 = o(m)$ indices.

Suppose now that P_m holds, both total heights are at most L , and the exception above does not occur. There are at least cm^2 pairs of renewal indices for which the two prefixes are disjoint and their gap at the shorter height exceeds G . Crossings there by the longer prefix equal those by its full path: later pieces lie strictly above its endpoint. We claim that the expected number of all such height-bounded pairs is

$$O(1) + O(L^{7/4+o(1)}/G). \quad (77)$$

Summing over renewal prefixes gives ordinary bridge weights. Equal-height pairs have bounded total mass by (74). For unequal heights, fix their difference $s_0 = p - n > 0$ and which arm is shorter. Join their adjacent bottom starts by the three-center connector used in Lemma 11.7.

At row n , choose a tessellation vertex between the shorter endpoint and the extreme crossing of the longer prefix. The gap gives at least cG choices. From that tip cut the horizontal ray pointing away from the longer path's crossings, through the shorter endpoint. The joined path avoids the slit: the shorter bridge is strict below its endpoint, and all crossings of the longer bridge at this row are on the other side. It starts at the slit shore, goes toward the connector, and terminates on the line of height p .

The map is illustrated in Figure 7. Send the tip (t, n) to the origin. If the shorter endpoint is on the right, use $(x', y') = (x - t, n - y)$; if it is on the left, use $(x', y') = (t - x, n - y)$. These tessellation symmetries make the slit rightward and the initial shore upper. The endpoint is on $y' = -s_0$, while the unique highest vertex, on the connector, has $y' = n + 2/3$. The path is therefore counted by the slit estimate with upper wall $L + 1$. Conversely, its unique highest vertex recovers n , the connector and the original translation, and hence both bridges and the chosen tip. For the fixed assignment this map is injective, with a fixed weight factor. Summing (71) over the slit sources and dividing by the cG tip choices bounds the pair mass, already summed over $n \leq L$, by $CL^{3/4+o(1)}/G$. Sum over the at most L height differences to obtain (77).

Division by cm^2 now bounds p_m in the restricted height range by $m^{-1+C\delta+o(1)}$. To remove that restriction, set $t = m^{4/3+\delta/2}$. Exponential Markov at $1/t$, together with $\mathbb{E}[\mathbf{H}; \mathbf{H} \leq t] \leq t^{1/4+o(1)}$, shows that a sum of m heights exceeding L with no increment above t has superpolynomially small probability. A height above t at index j on either arm is independent of P_{j-1} . Thus

$$p_m \leq Cm^{-1+C'\delta+o(1)} + Cm^{-1-c'\delta} \sum_{j < m} p_j, \quad p_0 = 1,$$

for some $c' > 0$. Induction gives $p_m = O_\eta(m^{-1+\eta})$ whenever $\eta > C'\delta$: under this bound the second term is $O(m^{-1+\eta-c'\delta})$. Choose δ arbitrarily small to obtain (76). $\square$

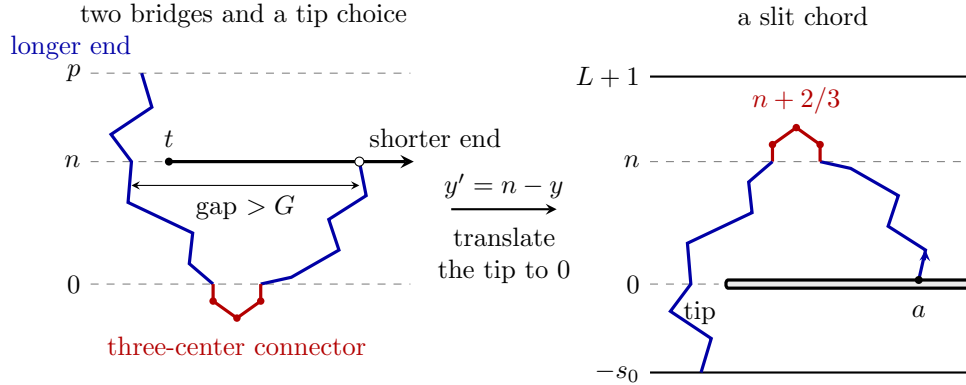


Figure 7: The pair-to-slit map when the shorter arm is on the right. Choose a tessellation vertex t in the gap at height n , and direct the slit through the shorter endpoint. After reflection across that level and translation of the tip, the (shorter) endpoint is an upper-shore source and the longer endpoint is on the bottom line $-s_0 = -(p - n)$. The unique maximum, at height $n + 2/3$, identifies the connector and recovers n . If the shorter arm is on the left, a half-turn about t gives the same orientation. Heights and paths are schematic, and the two slit shores are separated only for display.

11.4 Folding uniformly around a prescribed irreducible

Fold a half-plane arch of exact maximum layer k at its leftmost highest down triangle. Split there, ending the first arm and starting the second at the port above that triangle on line k , and reflect the second arm across that line. The result is a height- $2k$ bridge, with a renewal at k . The highest triangle was counted twice, so the folded weight is ρ times the arch weight. Unfolding at that middle renewal recovers the arch, making this map injective.

Call a segment of an ordinary bridge *valid* if its endpoints and middle are renewal positions, at heights $c - k, c, c + k$, and it is the image of this fold up to translation. Endpoints of the whole bridge are allowed. Fix positive constants c_1, C_1, C_2 and a small $\epsilon > 0$. Let K count valid segments whose half-height satisfies

$$c_1 H \leq k \leq H^{1+C_1\epsilon}.$$

The essential bound is uniform in the shape and length of a single fixed irreducible; this is what will permit marking a long block later.

Lemma 11.10 (Folding with a fixed irreducible). *Fix an irreducible ξ of height d . Place independent $\mathbb{Q}_z$ bridges below and above it, translating ξ to the joining port, where $z = e^{-1/T}$ and $T = H^{1+C_2\epsilon}$. Write $\mathbb{E}_\xi^*$ for this law. Uniformly in ξ ,*

$$\mathbb{E}_\xi^* K \leq H^{1/2+O(\epsilon)+o(1)}. \quad (78)$$

Proof. First count valid segments wholly outside ξ . Marking such a segment in either exterior bridge gives two ordinary bridge factors and the folded arch factor. One bridge normalization cancels, so this contribution is at most

$$2\rho B(z) \sum_{k \text{ in range}} a_k z^{2k} \leq 2\rho B(z) \sum_{k \geq c_1 H} a_k \leq H^{1/2+O(\epsilon)+o(1)}.$$

The last inequality uses $B(z) = T^{3/4+o(1)}$ and the arch radius tail.

Every other valid segment contains the whole defect. Its middle renewal cannot lie strictly inside an irreducible. Suppose first that this renewal is $l \geq 0$ levels below the start of ξ ; the other case follows by reversal and height reflection, which preserve the leftmost rule and replace ξ by a fixed transformed defect if necessary.

Here is the normalization cancellation for these segments. Cut at their two endpoint renewals. The ordinary pieces outside the segment are free and contribute $B(z)^2$, canceling the two exterior normalizations. Thus for this orientation their contribution to $\mathbb{E}_\xi^* K$ is the sum, over valid internal segments containing the fixed defect, of

$$\rho^{L(\text{segment})-L(\xi)} z^{2k-d}.$$

There is no weight for the prescribed defect in this expression and no sum over an independently chosen joining port. Unfolding the segment at its middle gives two arms of height k : one ordinary bridge, and one ordinary bridge of height l , followed by ξ , followed by an ordinary bridge of height $k - l - d$. The pieces have total ordinary height $2k - d$, as in the displayed weight. Reversed or reflected ordinary pieces can be rooted at their first port by translation. Figure 8 shows these heights.

If $l = 0$, dropping avoidance already bounds the aligned arm sum by $T^{1/2+o(1)}$, using the squared-sum estimate below. Suppose $l > 0$. The unfolded arms initially meet only at their common port and first down triangle, and take its distinct slant branches. They must be turned into disjoint arms from adjacent ports before Lemma 11.9 can be used.

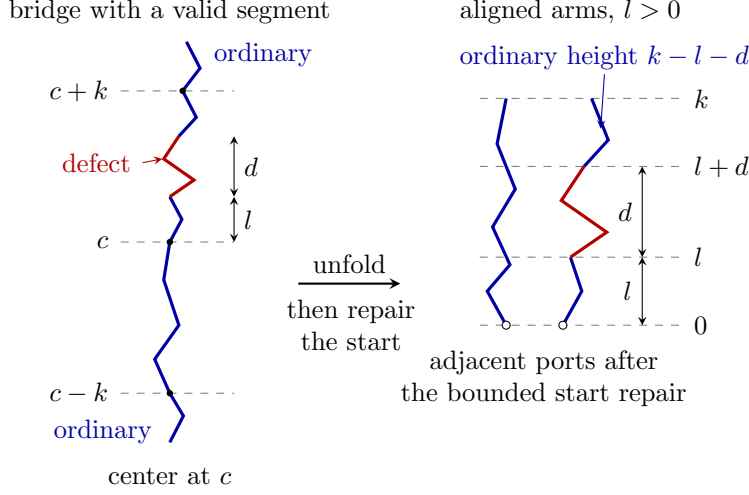


Figure 8: Unfolding a valid segment containing a fixed irreducible defect, in the case $l > 0$ and $l + d \leq k$. The ordinary pieces outside the segment are omitted on the right. The red defect is unchanged; the bounded repair acts at the common start and produces the adjacent disjoint starts shown on the right. Heights are schematic and individual lattice edges are suppressed. The two exterior factors $B(z)$ cancel the normalizations by the cutting argument in the text, independently of this drawing.

Orient the unfolding downward and put its common port at $P = (0, 0)$. Write $h_0 = \sqrt{3}/2$. The first down center is $D = (0, -h_0/3)$; the left branch next visits $U = (-1/2, -2h_0/3)$. Replace the initial portion P, D, U of this branch by

$$P_- = (-1, 0), \quad D_- = (-1, -h_0/3), \quad U.$$

These are actual neighboring ports and centers, as shown in Figure 9. The center D_- is unused by both arms, because D was chosen as the leftmost highest down triangle of the arch. From U the next edge cannot return to D or visit D_- , so it continues downward. The replacement therefore preserves self-avoidance and makes the two arms disjoint. It substitutes one vertex for one vertex, preserving each length and weight. All changed edges lie at depths less than one; every positive integral-depth crossing and its order is unchanged. Since $l \geq 1$, the fixed defect is untouched.

The inverse restores the left initial portion using the unchanged right port and first center. Keep the finitely many choices of arm and symmetry as separate assignments. After reanchoring the adjacent starts at fixed ports, the unchanged far endpoint and the original anchor determine the translation back. This gives bounded inverse multiplicity with no height-dependent factor.

A simple strip crosscut meeting a level just once separates its two horizontal rays. Consequently, whenever either repaired full bridge renews, all crossings of the other at that level lie on the prescribed side of its endpoint; the endpoints on the two exterior lines are ordered as well. This supplies the ordering condition used next.

Let the ordinary prefix on the defect arm contain m blocks and end at height l . Expose the other arm through its first renewal at height $p \geq l$. Sum the remaining ordinary portions, discarding their avoidance constraints. Cauchy-Schwarz applied to the shifted sequence $(b_j z^j)_{j \geq 0}$ bounds the alignment cost uniformly in l, p, d by

$$\sum_{k \geq \max(p, l+d)} b_{k-p} b_{k-l-d} z^{(k-p)+(k-l-d)} \leq \sum_{j \geq 0} b_j^2 z^{2j} \leq T^{1/2+o(1)}.$$

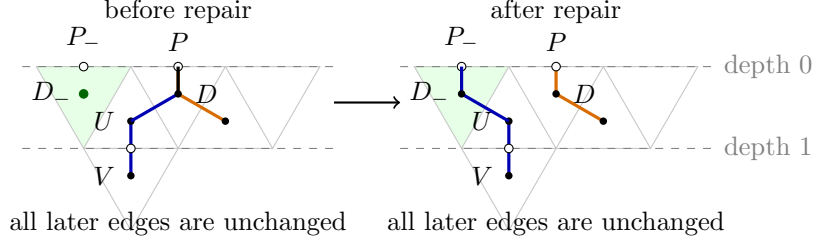


Figure 9: The exact local start repair in the downward unfolding orientation. Put $h_0 = \sqrt{3}/2$. The relevant centers are $D = (0, -h_0/3)$, $D_- = (-1, -h_0/3)$, $U = (-1/2, -2h_0/3)$ and $V = (-1/2, -4h_0/3)$; the ports are $P = (0, 0)$ and $P_- = (-1, 0)$. The two arms originally share P and D ; the green triangle contains D_- , which is unused before the repair. Replace P, D, U on the left branch by P_-, D_-, U . The picture takes $k \geq 2$, as holds in the defect case $l > 0$; for a height-one arm the displayed continuation stops at the port $(-1/2, -h_0)$ before V . Only the displayed local path is specified; no continuation choice is asserted beyond the displayed endpoints.

After this summation the exposed portions can be counted as independent p -sequences: the first m blocks on the defect arm and the other arm stopped at its first renewal at or above l . Their necessary disjointness and ordering conditions imply P_m . If at least m blocks of the other arm have been exposed, restrict to its first m . If fewer have been exposed, extend that arm by ordinary blocks: they stay beyond $p \geq l$ and neither meet the defect arm's ordinary prefix nor alter crossings at its renewal levels. Thus every such extension has the required conditions for the first m blocks of both arms. This explains why a probability bound for two equally indexed prefixes applies to the unequal stopping rule.

Retain the tilt z^l on the m ordinary blocks of the defect arm, and drop the other exposed tilt factors. For $m \leq T^{3/4+\delta'}$, Lemma 11.9 bounds the sum of exposed-prefix weights by

$$\sum_{m \leq T^{3/4+\delta'}} p_m = T^{o(1)}.$$

For larger m , discard ordering instead and use $I(z)^m$; since $1 - I(z) = T^{-3/4+o(1)}$, that tail is negligible for each fixed $\delta' > 0$. Multiplying by the alignment bound proves (78) for segments containing the defect, and hence for all valid segments. $\square$

11.5 Recoverable cuts and the second moment of finite chords

We have obtained the folding estimate needed to transfer a long finite path to a long irreducible. We now prove that there are enough long finite paths. The remaining moment bound comes from closing a chord into an unrooted polygon. To keep its inverse multiplicity small, we must first control how many levels can serve as recoverable cuts of that polygon.

Lemma 11.11 (The number of recoverable cuts). *For every fixed $\epsilon > 0$, polygons of diameter at most Q , counted once per translation class, have the following properties outside a set whose total critical weight is smaller than every inverse power of Q :*

- in each tessellation direction, at most $Q^{1/2+\epsilon}$ levels are crossed exactly twice;
- for each such two-crossing level and either polygon arc that it cuts off, at most $Q^{3/4+\epsilon}$ levels in either other direction are crossed exactly once by that arc.

The second bound also holds for the single crossings of a fixed-source half-plane arch of diameter at most Q in a direction nonparallel to its boundary, with the same exceptional-mass conclusion. These conclusions remain true after multiplication of the exceptional weights by any fixed polynomial in the lengths.

Proof. If a polygon has many exact-two levels in one direction, cut at their two extremes. There is an outer arch on each side, and between the extremes there are two through-bridges: the other pairing would disconnect the single cycle. Every intermediate exact-two level is a common renewal of these bridges. Fixing the extreme levels and ports modulo translation costs a polynomial in Q . The outer arches have bounded mass. Drop avoidance between the through-bridges and count their prefixes as two independent p -sequences.

Let N_Q count their common renewal heights up to CQ . For every fixed positive integer r , the renewal property gives

$$\mathbb{E} \binom{N_Q}{r} \leq \left(\sum_{j \leq CQ} b_j^2 \right)^r \leq Q^{r/2+o(1)}.$$

One obtains the first inequality by listing the r common hits in order and summing the independent height increments between consecutive hits. Therefore

$$\mathbb{P}\{N_Q \geq Q^{1/2+\epsilon}\} \leq \frac{Q^{r/2+o(1)}}{\binom{\lfloor Q^{1/2+\epsilon} \rfloor}{r}} = Q^{-r\epsilon+o(1)}.$$

Taking r as large as needed absorbs the polynomial specification cost and any fixed polynomial length factor.

For the second assertion, fix the original cut level and one arc. Its endpoints are ports on that line, so neither lies on a tessellation line in a different direction. Take the first and last, in level order, among the alleged many exact-one crossings in the second direction. The portion between them is a strict through-bridge and all intermediate exact-one levels are its renewals. For a single sequence the same argument uses

$$\sum_{j \leq CQ} b_j \leq Q^{3/4+o(1)}$$

in place of the squared sum, and the threshold is $Q^{3/4+\epsilon}$. The end portions lie between the cutting-line constraint and the corresponding extreme-level constraint, so they are boundary chords in convex wedges and have bounded mass. The other polygon arc is a bounded-mass half-plane arch. Fixing their cut ports and levels costs only a polynomial. For a fixed-source arch use its base line in place of the polygon's first cutting line; the same decomposition applies. Finally every path of diameter at most Q has $O(Q^2)$ vertices, so fixed polynomial length factors are harmless. Crossings throughout are at bond midpoints. $\square$

Lemma 11.12 (Chord second moment). *For the middle-visiting chord ensemble summed over $H \leq R \leq 2H$,*

$$\sum_{R, \text{ chords}} \rho^{\mathbf{L}} \mathbf{L}^2 \leq H^{53/12+o(1)}. \quad (79)$$

Proof. Use the regular hexagons D_R in the finite input, summed over integers $H \leq R \leq 2H$. Consider ordered chords that visit the middle region (some center within $R/10$ of the middle). Their mass is at most $H^{7/4+o(1)}$ by the boundary estimate. Their length sum is at least $H^{37/12-o(1)}$ by the imported bulk-visit lower bound (at least $R^{2+1/12-o(1)}$ per hexagon). Close chords with additional paths, using a small $\epsilon > 0$ and polygons of diameter $\leq Q = H^{1+6\epsilon}$. All pieces joined at ports

below have disjoint center sets and weights just multiply. Fix endpoint side choices, finitely many possibilities.

- If both endpoints are on the same side, join them by an independent arch in its exterior half-plane. By (73) the confinement toll (total available factor) is at least $H^{-5/4-C\epsilon-o(1)}$. That side level has exactly two crossings in the polygon. For a given nonexceptional polygon up to translation and fixed R , there are at most $Q^{1/2+\epsilon}$ choices for the cut level, and $O(H)$ more for translation along it (place a port on the side); the pieces can then be read from the polygon up to bounded choices.
- If the sides are nonparallel, denote their lines by $v = 0$ (source side) and $u = 0$ (destination side), signed oblique level coordinates chosen so the interior lies at $u < 0, v > 0$. These tessellation lines meet at a lattice vertex at distances $O(H)$ from the two endpoints a, b . Join a into the exterior wedge $u < 0, v < 0$ to an exit port c on $u = 0$. Both possible wedge angles are covered by the confined other-ray bound after (69); use radii $\leq H^{1+\epsilon}$. Then join c to b in the half-plane $u > 0$. By (73) the joint toll is at least $H^{-3/2-C\epsilon-o(1)}$. Now $u = 0$ has exactly two crossings, and in the arc in $u < 0$ the level $v = 0$ has exactly one. Thus the translation choices at each R cost at most $Q^{1/2+\epsilon}Q^{3/4+\epsilon}$ for nonexceptional polygons; the two level placements determine the translation, and the three pieces are recovered from the cuts.
- For opposite parallel sides orient the strip vertically in height, the chord a bridge from a to b of height $h = 2R$. Use an independent second height- h bridge, started to the right of a at an integer gap in $[H^{1+2\epsilon}, 2H^{1+2\epsilon}]$, confined within radius $H^{1+\epsilon}$ of its start. This has mass $h^{-1/4+o(1)}$ by (69) and the cap comparison, and is separate from the chord. Join the pairs of ends by arches below and above the strip using (73). Including the gap choices, the toll is at least $H^{-7/4-C\epsilon-o(1)}$. Across all R , the two cut levels both have exact-two crossings, and determine h , hence R ; thus multiplicity for a nonexceptional polygon is at most $CQ^{1+2\epsilon}H$, including translation along the source side. The original pieces and the gap are then read off up to bounded choices.

In each case the number of inverse specifications even in the exceptional set is only polynomial (port assignments and translations in bounded ranges). Thus by (68) and the observations above, counting polygon lengths squared (which majorize the chord lengths squared) costs $Q^{2/3+o(1)}$ times the stated multiplicities, plus negligible errors. To display the losses, suppress the already specified $O(\epsilon) + o(1)$ corrections. The leading exponents from the inverse toll and the inverse count (including the R sum) are

$$\begin{aligned} \text{same side:} & \quad \frac{5}{4} + \frac{5}{2} = \frac{15}{4}, \\ \text{nonparallel sides:} & \quad \frac{3}{2} + \frac{9}{4} = \frac{15}{4}, \\ \text{opposite sides:} & \quad \frac{7}{4} + 2 = \frac{15}{4}. \end{aligned}$$

In the first two cases one sums over R ; in the opposite-side case the two recovered cut levels already determine R . Including the polygon second-moment exponent gives $15/4 + 2/3 = 53/12$. Let $\epsilon \downarrow 0$ to obtain (79).

□

Corollary 11.13 (Mass of long middle-visiting chords). *For every small fixed $\delta > 0$, the middle-visiting chords with $\mathsf{L} \geq H^{4/3-\delta}$, summed over $H \leq R \leq 2H$, have critical mass at least $H^{7/4-o(1)}$.*

Proof. The shorter chords carry first-length mass at most $H^{4/3-\delta}M_0(H) \leq H^{37/12-\delta+o(1)}$, negligible relative to $M_1(H) \geq H^{37/12-o(1)}$. Thus the long chords retain first-length mass $H^{37/12-o(1)}$. Cauchy-Schwarz and Lemma 11.12 give their mass at least

$$\frac{H^{2(37/12)-o(1)}}{H^{53/12+o(1)}} = H^{7/4-o(1)}.$$

□

11.6 The lower length tail and its changes of law

We can now combine the finite moment estimate with the uniform folding bound. The bridge law $\mathbb{Q}_z$ introduced above is useful here because marking one large irreducible leaves two independent ordinary bridges.

Proof of Theorem 11.1. Fix a small $\eta > 0$, choose $0 < \epsilon \ll \delta \ll \eta$, and put

$$n = H^{4/3-\eta}, \quad T = H^{1+3\epsilon}, \quad z = e^{-1/T}, \quad u = \mathbb{E}_p[z^{\mathbf{H}}; \mathbf{L} \geq n].$$

Call the chords supplied by Corollary 11.13 *long*; their lengths are at least $H^{4/3-\delta}$.

From long chords to bridges or arches. There are finitely many assignments of endpoint sides. At least one assignment carries mass $H^{7/4-o(1)}$, though the assignment may vary with H . Its conversion is as follows.

- For opposite sides, the chord is already a bridge of height $h = 2R$. There are $O(H)$ possible source placements at each R , and h determines R . Translating to a fixed source gives long bridge mass, summed over $2H \leq h \leq 4H$, at least $H^{3/4-o(1)}$.
- For the same side, regard the chord as an arch in its inward half-plane. Visiting the middle forces its maximum layer between c_0H and CH . There are $O(H^2)$ choices of R and source placement, so after translation its long arch mass is at least $H^{-1/4-o(1)}$.
- For nonparallel sides, use the oblique coordinates from the closure proof: the source side is $v = 0$, the destination side is $u = 0$, and the hexagon lies in $u < 0, v > 0$. Append at the destination b a chord in the wedge $u > 0, v > 0$ ending on $v = 0$. By Lemma 11.3, it has mass at least $H^{-1/4-o(1)}$ within radius $H^{1+\epsilon}$. The result is a long arch above $v = 0$, with maximum layer in $[c_0H, H^{1+2\epsilon}]$ and diameter at most $Q = H^{1+6\epsilon}$. Its crossing of $u = 0$ is unique.

Anchor the arch's initial port. At each R , different source placements would give different single-crossing cut levels $u = 0$. The arch version of Lemma 11.11 thus bounds inverse multiplicity, summed over R , by $CHQ^{3/4+\epsilon}$ outside a negligible set. Consequently the long arch mass in this height range is at least $H^{-1/4-C\epsilon-o(1)}$.

The exceptional sets remain negligible after these conversions because all inverse specifications cost only fixed powers of Q .

A long tilted bridge must contain a large block. Under $\mathbb{Q}_z$, the geometric number of blocks exceeds $H^{3/4+4\epsilon}$ with superpolynomially small probability, because $1 - I(z) = T^{-3/4+o(1)}$. Set larger lengths to zero and truncate at n . Their mean under p_z is at most $n^{7/16+o(1)}$ by (75). Exponential Markov at $1/n$ therefore bounds the probability of length at least $H^{4/3-\delta}$ with no block of length at least n by an exponential whose exponent is at most

$$-H^{4/3-\delta}/n + CH^{3/4+4\epsilon}n^{-9/16+o(1)}.$$

The negative term is $-H^{\eta-\delta}$ and the positive term is $H^{9\eta/16+4\epsilon+o(1)}$. Choose $\delta + 4\epsilon < 7\eta/16$; the bound is then superpolynomially small. The same remains true with any fixed polynomial weight in the block count, by the geometric tail.

If the bridge alternative above holds, its tilt costs a constant factor in the relevant height bin. Marking a large block bounds its long mass above by $B(z)^2 u$ plus a negligible error. Thus it yields $u \geq H^{-3/4-O(\epsilon)-o(1)}$.

In either arch alternative, fold the long arches. Their tilted mass is their original mass multiplied by ρz^{2k} , where $c_1 H \leq k \leq H^{1+2\epsilon}$. These extra factors are bounded below by a positive constant. Insert such a folded segment between two arbitrary ordinary bridges. Cutting at its endpoint renews identifies the inserted segment, so the resulting weighted count is its folded mass times $B(z)^2$. It is bounded by the full tilted bridge mass with insertion K and with total length at least $H^{4/3-\delta}$.

Terms with no block of length at least n are negligible by the preceding estimate, since K is at most the cube of the block count plus one. In every remaining term, mark a large block ξ . The weight sum of the two ordinary exterior bridges is $B(z)^2$, and their conditional expected insertion is bounded by Lemma 11.10, uniformly in ξ . Dividing by $B(z)^2$ therefore bounds the folded mass by

$$u H^{1/2+O(\epsilon)+o(1)} + \text{negligible}.$$

Its lower bound is $H^{-1/4-O(\epsilon)-o(1)}$. Thus every alternative gives $u \geq H^{-3/4-O(\epsilon)-o(1)}$. Let ϵ decrease to zero at fixed η , and then take η arbitrarily small. Since u is bounded above by the untilted tail, the upper bound in (75) proves

$$\mathbb{P}_p(\mathbf{L} \geq n) = n^{-9/16+o(1)}. \quad (80)$$

The height and diameter assertions follow from Lemma 11.8 and $D \leq C(\mathbf{H} + S)$.

Infinite concatenation. The matching length and height tails and the span upper tail now give, on one probability-one event,

$$\sum_{i \leq m} \mathbf{L}_i = m^{16/9+o(1)}, \quad \sum_{i \leq m} \mathbf{H}_i = m^{4/3+o(1)}, \quad \sum_{i \leq m} S_i \leq m^{4/3+o(1)}.$$

For upper bounds, moments just below $9/16$ or $3/4$, subadditivity of these powers, Markov's inequality and dyadic Borel–Cantelli suffice. Independent maxima and the matching lower tails give the lower bounds. Equivalently one may apply Lemma 2.6 to the resulting Laplace deficits. Within the next block, every vertex lies above the last completed height and within the sum of the spans and heights. Consecutive length sums have the same exponent, so if vertex j is in block $m+1$, then $m = j^{9/16+o(1)}$. Hence

$$|\gamma_j - \gamma_0| = j^{3/4+o(1)}, \quad \#\{j \geq 1 : |\gamma_j - \gamma_0| \leq R\} = R^{4/3+o(1)}$$

almost surely. The second assertion follows by inverting the two eventual power bounds in the first.

This infinite law is also the port-to-vertex fugacity prefix limit: Lemma 6.1 includes the final half-plane remainder, whose mass cancels from a prescribed-prefix probability. For this application, finiteness at each subcritical fugacity follows already from the connective-constant root bound of [4]; no stronger counting estimate or thermal spatial conclusion is required.

Conditioning on an exact strip height. Critical bridges of height h are iid prefixes conditioned on hitting h , an event of probability $b_h = h^{-1/4+o(1)}$. We first show that this conditioning leaves at least $h^{3/4-\epsilon}$ blocks with high probability. The arch identity makes b_h decreasing, including $b_0 = 1$. Define the series with nonnegative coefficients

$$C(z) = \sum_{h \geq 1} (b_{h-1} - b_h) z^h, \quad 1 - C(z) = (1 - z)B(z).$$

Since $B = (1 - I)^{-1}$, one has $(1 - I)(1 - C) = 1 - z$. Taking negative logarithms as formal series gives

$$[z^h]\{-\log(1 - I(z))\} \leq [z^h]\{-\log(1 - z)\} = \frac{1}{h}.$$

For any integer $m \geq 1$, coefficient positivity consequently yields

$$[z^h] \sum_{r=1}^m I(z)^r \leq m[z^h]\{-\log(1 - I(z))\} \leq \frac{m}{h}.$$

Divide by b_h and take $m = \lfloor h^{3/4-\varepsilon} \rfloor$.

Given $\delta > 0$, choose $\varepsilon > 0$ sufficiently smaller than δ . By (80), the probability that the first m iid blocks all have length below $h^{4/3-\delta}$ is superpolynomially small; division by b_h leaves it negligible. This proves the typical length lower bound. The strip first moment gives

$$\mathbb{E}_h \mathbf{L} = \frac{m_B(h)}{b_h} \leq h^{4/3+o(1)},$$

and Markov gives the typical upper bound. The typical lower bound also gives the matching lower mean. The bridge diameter is at least a constant times h , and the cap comparison, divided by b_h , makes diameter above $h^{1+\delta}$ negligible. In particular the length and mean assertions hold at every sufficiently large integer height, not merely along a subsequence.

Summing heights with a length tilt. Under the bridge law with weight $\rho^{\mathbf{L}} e^{-\mathbf{L}/t}$, the number of blocks is geometric with termination parameter

$$q_t = \mathbb{E}_p(1 - e^{-\mathbf{L}/t}) = t^{-9/16+o(1)}.$$

The upper bound follows by integrating the length tail, or comparing $1 - e^{-x}$ with $\min(1, x)$; the lower bound follows from $p(\mathbf{L} \geq t)$. Conditional on the count, block shapes are independent with law proportional to $e^{-\mathbf{L}/t} p$. For each fixed small $\delta > 0$, the count lies between $t^{9/16-\delta}$ and $t^{9/16+\delta}$ with probability tending to one. Removing the empty bridge changes none of these assertions.

For any small fixed $\eta > 0$, each of the two events

$$t^{1-\eta} \leq \mathbf{L} \leq t, \quad \mathbf{H} \geq t^{3/4-\eta}, \quad \mathbf{L} \leq t$$

has enough tilted mass that it occurs among $t^{9/16-\delta}$ independent blocks with high probability, if δ is sufficiently small. For the first event, subtract $p(\mathbf{L} > t)$ from the matching length tail at $t^{1-\eta}$. For the second, subtract that same upper tail from the height lower tail at $t^{3/4-\eta}$. In each subtraction the first term has a strictly larger power, and on $\mathbf{L} \leq t$ the tilt costs at most a constant factor. These events give the lower length and diameter bounds.

The tilted block laws have uniformly bounded length moments at every power below $9/16$, and height and span moments at every power below $3/4$. Subadditivity at those powers and Markov's inequality, applied to at most $t^{9/16+\delta}$ samples, bound their summed length by $t^{1+o(1)}$ and their summed height and span by $t^{3/4+o(1)}$ in probability. These sums bound the bridge diameter. Taking the slacks arbitrarily small proves the claimed thermal bridge laws and completes the theorem. $\square$

12 Geometric transfers to infinite and thermal paths

Finite bridge moments and planar avoidance give another route to the infinite-path and thermal exponents. For the infinite path, a future renewal supplies a useful averaging device. For the

thermal path, inserting a tall bridge separates the endpoints; the main issue is to control how many insertions can produce the same walk. We give the two transfers together because they use the same strip geometry, but their probability normalizations remain separate.

We use the port convention of Proposition 2.2. For a strict bridge η , $L(\eta)$ counts visited vertices and its weight is $\rho^{L(\eta)}$. Set $B_0 = 1$ and let B_h be the total weight of bridges of height $h \geq 1$ from a fixed bottom port, with the terminal port free. Write $\beta = 3/4$. The finite inputs are

$$B_h \asymp (1+h)^{\beta-1}, \quad (81)$$

$$F_h := \sum_{\eta: H(\eta)=h} \rho^{L(\eta)} L(\eta) \leq h^{13/12+o(1)}, \quad (82)$$

$$\sum_{\substack{\eta: H(\eta)=h \\ \text{lat}(\eta) > M}} \rho^{L(\eta)} \leq ChM^{-5/4} \quad (M \geq C_1 h). \quad (83)$$

Here lat is the maximal horizontal distance from the source, up to an immaterial fixed change of lattice coordinates. The first estimate is [24, Theorem 1.1]; the second is [20, Proposition 11.5]. The third is [19, Lemma 4.1]. The first-length mass in (82) is unnormalized: no division by B_h has been made.

We also use three geometric results from [19]. Lemma 5.1 there says that a critical renewal path, stopped at its first renewal above height h , contains at least $h^{4/3-\xi}$ vertices with probability $h^{-o(1)}$, for each fixed $\xi > 0$. Proposition 3.1 there gives constants C_0, C_1 such that the critical mass of all vertex walks confined to a box of side R , from any specified vertex and with arbitrary endpoint and length, is at most $C_0(1+R)^{C_1}$. Finally, let $S_2(r)$ be the probability that two independent renewal paths from adjacent ports are disjoint through their respective first visits to height r . Corollary 6.6 there gives

$$S_2(r) \leq C_\varepsilon r^{-\beta+\varepsilon} \quad (84)$$

for every $\varepsilon > 0$. These statements use critical weights and the renewal probability law; no thermal or uniform-length law is involved.

12.1 Averaging over future renewals

Let $S_0 = 0$ and $S_k = H_1 + \dots + H_k$ be the renewal heights under the iid irreducible law p , and put

$$\tau_h = \min\{k : S_k > h\}, \quad \Gamma_h = \eta_1 \circ \dots \circ \eta_{\tau_h}.$$

The last block can overshoot h . In particular, estimates for bridges conditioned to end exactly at h cannot be substituted directly for estimates of Γ_h .

Proposition 12.1 (Stopped renewal paths). *For every $\varepsilon > 0$ there is $c_\varepsilon > 0$ such that*

$$\mathbb{P}\{h^{4/3-\varepsilon} \leq L(\Gamma_h) \leq h^{4/3+\varepsilon}, \quad \text{diam}(\Gamma_h) \leq Ch^{1+\varepsilon}\} \geq 1 - O_\varepsilon(h^{-c_\varepsilon}).$$

Consequently, if ω_n is the n -th vertex of the infinite concatenation and its initial port is o , then almost surely

$$|\omega_n - o| = n^{3/4+o(1)}, \quad \#\{n : |\omega_n - o| \leq R\} = R^{4/3+o(1)}.$$

Proof. The factorization in Proposition 2.2 gives $\mathbb{P}\{m \text{ is a renewal height}\} = B_m$. If I_j is the probability that one irreducible has height j , the same factorization and (81) give

$$\sum_{j \geq x} I_j \leq Cx^{-\beta}.$$

Indeed, $\sum_m B_m e^{-m/x} \asymp x^\beta$, its reciprocal is $1 - \sum_j I_j e^{-j/x}$, and this deficit bounds $(1 - e^{-1}) \sum_{j \geq x} I_j$ from below. Decomposing according to the last renewal before h , for $x \geq 2h$ we obtain

$$\mathbb{P}\{S_{\tau_h} > x\} = \sum_{i=0}^{\lfloor h \rfloor} B_i \sum_{j > x-i} I_j \leq C(h/x)^\beta. \quad (85)$$

For the length lower bound choose small $\theta, \xi > 0$ and set $a = \lfloor h^{1-\theta} \rfloor$, $b = \lfloor h^{1-\theta/2} \rfloor$, $k = \lfloor h^{\theta/16} \rfloor$. Run k successive trials of height a , restarting at each terminal renewal. They are independent by the iid law after stopping renewal indices. If their height advances are $\Delta_1, \dots, \Delta_k$, then

$$\mathbb{P}\left\{\sum_{i=1}^k \Delta_i > b\right\} \leq Ck(ak/b)^\beta \leq Ch^{-17\theta/64}.$$

The renewal-trial lemma gives success probability at least $h^{-\theta/32}$ for each trial when h is large. The probability that all fail is at most $\exp(-ch^{\theta/32})$. Outside these two exceptional events a trial of length at least $a^{4/3-\xi}$ is contained in Γ_b , hence in Γ_h . Choose θ, ξ so that $(1-\theta)(4/3-\xi) > 4/3-\varepsilon$. This proves the required polynomial lower-tail estimate.

For the upper bounds put $s = \lfloor h^{1+\delta} \rfloor$, where $\delta > 0$ will be small, and let $u = S_{\tau_h}$. The exceptional probability $\mathbb{P}\{u > s\}$ is at most $Ch^{-\beta\delta}$. Starting from u , the expected number of continuation renewals in the next s bands, including the renewal at u , is

$$D_s = \sum_{j=0}^s B_j \asymp s^\beta.$$

Conditionally on Γ_h , this future renewal count has expectation D_s . On $u \leq s$ every such continuation ends at a height at most $2s$ and contains Γ_h . Therefore

$$\begin{aligned} D_s \mathbb{E}[L(\Gamma_h) \mathbf{1}_{\{u \leq s\}}] &\leq \sum_{m \leq 2s} \sum_{\eta: H(\eta)=m} \rho^{L(\eta)} L(\eta) \\ &\leq s^{25/12+o(1)}. \end{aligned}$$

The first inequality counts a subset of all renewal prefixes ending at heights at most $2s$. Their distribution is the unnormalized bridge measure, by the pathwise factorization; it does not require independence between the length and the height of one irreducible. Markov's inequality now gives

$$\mathbb{P}\{L(\Gamma_h) > h^{4/3+\varepsilon}, u \leq s\} \leq h^{4\delta/3-\varepsilon+o(1)}.$$

The same device controls horizontal excursions. If Γ_h reaches horizontal distance $M = h^{1+\varepsilon}$, all its continuation prefixes do so. For $\delta < \varepsilon$, summing (83) over their possible heights gives

$$D_s \mathbb{P}\{\text{lat}(\Gamma_h) > M, u \leq s\} \leq C \sum_{m \leq 2s} mM^{-5/4} \leq Cs^2 M^{-5/4}.$$

After division by D_s , this is at most $Ch^{-5(\varepsilon-\delta)/4}$. On $u \leq s$ the vertical span is at most a fixed multiple of s . Taking $\delta = \varepsilon/4$, with a smaller slack first if necessary, proves the assertion.

Apply Borel–Cantelli on dyadic values of h , simultaneously for a countable sequence of positive slacks tending to zero. For each fixed small $\eta > 0$, eventually

$$h^{4/3-\eta} \leq L(\Gamma_h) \leq h^{4/3+\eta}, \quad \text{diam}(\Gamma_h) \leq Ch^{1+\eta}$$

at every dyadic scale. A dyadic h a fixed factor below $n^{1/(4/3+\eta)}$ has $L(\Gamma_h) < n$; every subsequent vertex lies above its terminal renewal line, whose height exceeds h . A dyadic h a fixed factor above $n^{1/(4/3-\eta)}$ has $L(\Gamma_h) > n$. Hence, eventually,

$$c_\eta n^{1/(4/3+\eta)} \leq |\omega_n - o| \leq C_\eta n^{(1+\eta)/(4/3-\eta)}.$$

Let $\eta \downarrow 0$. The resulting pointwise bounds also show that all sufficiently early indices, up to $R^{4/3-o(1)}$, are in the ball, and all sufficiently late indices, beyond $R^{4/3+o(1)}$, are outside. This proves the visit-count assertion. $\square$

The fugacity interpretation of this same infinite law is supplied by Lemma 6.1. For $0 < x < \rho$, let $Z_{\text{hp}}(x)$ be the normalizer of nonempty port-to-vertex half-plane walks with weight x per visited vertex. To check the conventions here, a specified tuple of j irreducibles of total vertex length ℓ followed by a nonempty path above its last seam has unnormalized mass $x^\ell Z_{\text{hp}}(x)$. Its probability is exactly x^ℓ . The tuple events are disjoint and their probabilities sum to $J(x)^j$, where $J(x) = \sum_\eta x^{L(\eta)} \uparrow 1$ as $x \uparrow \rho$. Thus the complement has probability tending to zero and the tuple weights tend to their critical values. Taking $j \geq m$ determines the first m visited vertices and proves local convergence, including vanishing probability of termination before the m -th vertex. No uniform finite-length local limit is used in this argument.

12.2 Tall-bridge insertions and their multiplicities

We now work with ordinary walks rooted at a fixed vertex o , and write $\ell(\gamma)$ for their number of edges. An insertion separates the two arms of a walk by a bridge of height in $[H, 2H]$ and diameter at most C_*H , where C_* is fixed. Its added mass will be of order H^β . To use that mass, we must bound the number of inputs producing one output.

Split the input at its first lowest vertex. At an upward-pointing minimum, extend both arms to the horizontal port immediately below it. A downward-pointing minimum must be an endpoint, because an internal vertex of that type cannot have both incident path edges above it. In that case extend through the adjacent lower triangle and then to its bottom port. Every added vertex lies below the original minimum and is therefore unvisited. Reflect the backward arm below this common starting line, insert the bridge, and place the forward arm above its upper line. The three portions occupy disjoint open height layers, as in Figure 10.

The reflection interchanges the two honeycomb vertex types. Fix an output root o^* of the reflected type and translate the output so that the reflected original root is o^* . Walk partition sums at o and o^* are equal by a lattice reflection and translation. That symmetry preserves edge length, diameter, and absolute endpoint-height difference.

Cutting the output at the two joining levels recovers the input and inserted bridge up to a bounded local record. The port cuts subdivide edges into half-edges; ports carry no vertex weight. Thus the output edge weight differs from the product of the input edge weight and bridge vertex weight by a bounded factor, uniformly for activities in $[\rho/e, \rho]$. For a class $\mathcal{A}$ of input walks, let $M_{\mathcal{A}}(\omega)$ count the insertion representations of an output ω with input in $\mathcal{A}$. Each pair of joining levels determines at most a fixed number of representations, independently of h, H .

Lemma 12.2 (Insertion multiplicities). *Suppose $h \rightarrow \infty$ and $h/H \rightarrow 0$. Let $\mathcal{A}_D$ be the walks rooted at o with diameter at most h . Every fixed integer $k \geq 1$ satisfies*

$$\sum_{\omega} \rho^{\ell(\omega)} M_{\mathcal{A}_D}(\omega)^k \leq C_k H^{C+o_k(1)} h^{k\beta}. \quad (86)$$

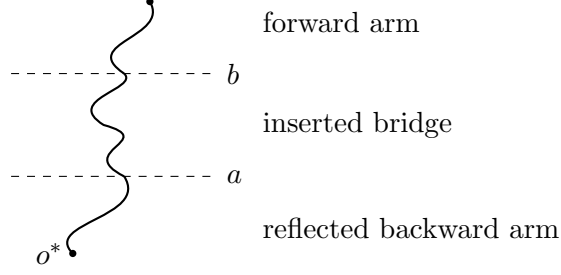


Figure 10: One insertion, with joining levels $a < b$. Cutting at these levels and undoing the lower reflection recovers the input up to a bounded local record. The drawing suppresses the modifications at the ports.

For fixed $K > 0$, let $\mathcal{A}_E$ be the walks rooted at o of length at most H^K and absolute vertical endpoint difference at most h . Then

$$\sum_{\omega} \rho^{\ell(\omega)} M_{\mathcal{A}_E}(\omega)^k \leq C_{k,K} H^{C_K + o_k(1)} \{h^\beta (H/h)^{2\beta-1}\}^k. \quad (87)$$

The sums run over outputs rooted at o^* . The exponents C, C_K are independent of k .

Proof. Write $M = M_{\mathcal{A}}$ for either input class and expand M^k into ordered k -tuples of representations of the same output. We first locate their seams. Let z_0, z_f be the vertical coordinates of the output endpoints. Undoing the lower reflection sends the root to height $2a - z_0$, while removing the inserted height sends the other endpoint to $z_f - (b - a)$, up to bounded local changes. The original endpoint difference is therefore $z_0 + z_f - a - b + O(1)$. Both input classes satisfy

$$|a + b - z_0 - z_f| \leq C_2 h, \quad H \leq b - a \leq 2H. \quad (88)$$

All heights here are in row units, with fixed conversion factors absorbed in C_2 .

Put $S = z_0 + z_f$ and $E = C_2 h$. The possible seams lie in the intervals

$$a \in \left[\frac{S - E - 2H}{2}, \frac{S + E - H}{2} \right], \quad b \in \left[\frac{S - E + H}{2}, \frac{S + E + 2H}{2} \right].$$

Every upper seam thus exceeds every lower seam by at least $H - E > 0$ for large H . Sort the lower seams increasingly and the upper seams decreasingly:

$$a_1 \leq \dots \leq a_k < b_k \leq \dots \leq b_1.$$

Record ties and the original representation pairing by a permutation π : the representation with lower seam a_i has upper seam $b_{\pi(i)}$. These records cost a factor depending only on k .

Every positive gap between consecutive seams in either group is a strict bridge. The two exterior portions and the portion between a_k and b_k are the only three portions left over. In the diameter case the whole output lies in a box of side $O(H)$; in the endpoint case it lies in a box of side $O(H + H^K)$. Fixing the output endpoint and the four extreme seam ports costs a polynomial in H whose exponent is independent of k . The full-box susceptibility bound then sums the three leftover portions at cost H^C or H^{C_K} . Internal joining ports introduce no additional position sums: once the relative gap shapes are specified, each has a unique translation to its joining port.

For the remaining gaps, use heights

$$s_1 = t_1 = 0, \quad s_i = a_i - a_{i-1}, \quad t_i = b_{i-1} - b_i \quad (2 \leq i \leq k).$$

The zero slots account for the two extreme seams. Repeated seams give further zero gaps, with bounded weight. The lower gap of height s_i ends at a_i , and the upper gap of height t_j starts at b_j . Consequently the original representation i pairs the slots $(s_i, t_{\pi(i)})$. Every positive gap appears in exactly one such pair. Figure 11 distinguishes this pairing from the rank pairing used later to sum endpoint constraints.

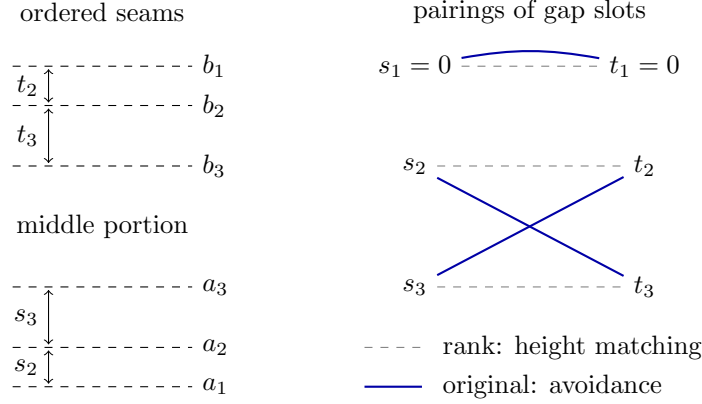


Figure 11: Three representations of one output. The left panel shows the seam order and the positive gaps; the two extreme slots have height zero. The right panel illustrates the original permutation $\pi(1) = 1$, $\pi(2) = 3$, $\pi(3) = 2$. Rank pairs constrain the two gap heights; original pairs inherit avoidance from an input walk. Connecting lines in the right panel match variables and do not represent walk segments. The diagram is not to scale.

The diameter class. All its gap heights are $O(h)$: each lower seam is within $O(h)$ of the root height, and each upper seam is within $O(h)$ of the final height. Put the two gap heights in an original pair into dyadic bins of scales $\sigma, \tau \leq Ch$, with scale one for absent or bounded gaps. The total bridge mass in a bin of scale σ is at most $C\sigma^\beta$, by (81).

For two large gaps in an original pair, reverse and reflect the lower one from its upper endpoint. The two resulting pieces are initial portions of the input arms at its minimum and avoid one another after their common first vertex. Move one start to the adjacent port and pass through the neighboring upward triangle to the same first upper neighbor. If that triangle was already visited, strictness at the lower wall forces the visit immediately after the first upper neighbor; start there and delete the short initial segment. The other arm cannot use that triangle because it would also use the same upper neighbor. This recoding has bounded inverse ambiguity and preserves avoidance through first visits to height $q = c \min(\sigma, \tau)$, for a fixed small $c > 0$. Thus it gives the adjacent-port event in (84).

Expose each gap from its assigned end through its first renewal at or above q . If the exposed height is u , the remaining critical mass at terminal height v is B_{v-u} , with $B_j = 0$ for $j < 0$. Uniformly in the exposed shape,

$$\sum_{v \leq C\sigma} B_{v-u} \leq C\sigma^\beta.$$

The exposed strings have the independent renewal probability law. After integrating the completions, the paired integral is therefore at most

$$C\sigma^\beta \tau^\beta \min(\sigma, \tau)^{-\beta} H^{o(1)} = C \max(\sigma, \tau)^\beta H^{o(1)} \leq Ch^\beta H^{o(1)}.$$

If one gap is absent or bounded, the ordinary mass bound for the other proves the same estimate. Normalize every gap at its assigned test end and drop all other intersection restrictions. The pairs $(s_i, t_{\pi(i)})$ use disjoint gap variables, so their bounds multiply by Tonelli's theorem. The $(1 + \log H)^{O(k)}$ dyadic choices contribute $H^{o_k(1)}$. Together with the three leftover portions, this proves (86).

The endpoint class. Its gaps can have height of order H , but their rank pairing retains a constraint. The original pairs satisfy $|a_i + b_{\pi(i)} - S| \leq E$. Two ordered multisets admitting a matching within E also admit the monotone matching within E . Applied to a_i and $S - b_j$, this gives

$$|a_i + b_i - S| \leq E, \quad |s_i - t_i| \leq 2E \quad (2 \leq i \leq k).$$

We use these rank pairs to integrate the completion heights, while keeping π for the subsequent avoidance tests.

For a gap in a dyadic bin of scale σ , put $w_\sigma = \min(h, \sigma)$ and expose its renewal string through height cw_σ , using threshold zero for a bounded gap. Condition on all exposed strings. A rank pair with one height $O(h)$ has both heights $O(h)$ and completion mass at most $Cw_\sigma^\beta w_\tau^\beta$. For a rank pair at comparable scales $r, r' \gg h$, uniformly in the exposed heights u_0, v_0 , the completion mass is at most

$$\begin{aligned} & \sum_{\substack{u \succ r, v \succ r' \\ |u-v| \leq 2E, u \geq u_0, v \geq v_0}} B_{u-u_0} B_{v-v_0} \\ & \leq Chr^{2\beta-1} \leq Cw_r^\beta w_{r'}^\beta (H/h)^{2\beta-1}. \end{aligned}$$

For the first inequality fix $u - v$, apply Cauchy-Schwarz to the two translated sequences, and use $\sum_{j \leq Cr} B_j^2 \leq Cr^{2\beta-1}$; there are $O(h)$ allowed differences. Here $2\beta - 1 = 1/2 > 0$. An exposed string that overshoots its possible endpoint contributes zero, so the bound is uniform without an extra overshoot condition.

Integrating all completion variables by rank pairs leaves one factor w_σ^β per gap and at most k factors $(H/h)^{2\beta-1}$. The remaining measure is the product of the independent stopped-string laws. Regroup them now by their original pairs $(s_i, t_{\pi(i)})$. Each avoidance test costs $\min(w_\sigma, w_\tau)^{-\beta} H^{o(1)}$, and hence each original pair contributes at most

$$w_\sigma^\beta w_\tau^\beta \min(w_\sigma, w_\tau)^{-\beta} H^{o(1)} \leq h^\beta H^{o(1)}.$$

Every string is used once. Multiplying these bounds, summing the dyadic bins, and restoring the polynomial cost of the three leftover portions proves (87). $\square$

12.3 The thermal deduction

Let c_m count unrestricted m -edge walks from a fixed vertex, including the empty walk, and for $T > 1$ set

$$x_T = \rho e^{-1/T}, \quad Z_T = \sum_{m \geq 0} c_m x_T^m, \quad \mathbb{P}_T^{\text{th}}(\gamma) = Z_T^{-1} x_T^{\ell(\gamma)}.$$

We also use the diameter clause of [19, Theorem 1.1]: for every $\eta > 0$, every finite $A > 0$ and every integer $m \geq 1$, the uniform law on m -edge walks satisfies

$$\mathbb{P}_m\{\text{diam}(\gamma) > m^{3/4+\eta}\} \leq C_{\eta,A} m^{-A}. \quad (89)$$

This is an all-length upper bound; the thermal deduction does not require an all-length uniform endpoint lower bound.

Proposition 12.3 (Thermal stretching by insertion). *Under $\mathbb{P}_T^{\text{th}}$, as $T \rightarrow \infty$,*

$$\begin{aligned}\ell(\gamma) &= T^{1+o(1)}, & \text{diam}(\gamma) &= T^{3/4+o(1)}, \\ |\gamma_{\ell(\gamma)} - \gamma_0| &= T^{3/4+o(1)}\end{aligned}$$

in probability. Each statement means convergence within every fixed positive exponent slack.

Proof. Every m -edge walk lies in a box of side $O(m)$, so full-box susceptibility gives $c_m \rho^m \leq C(1+m)^C$. In particular Z_T is finite, $Z_T \geq 1$, and for every $\xi > 0$,

$$\mathbb{P}_T^{\text{th}}\{\ell(\gamma) > T^{1+\xi}\} \leq \sum_{m > T^{1+\xi}} C(1+m)^C e^{-m/T} = o(1). \quad (90)$$

We first prove the endpoint lower bound on lengths at most T^2 . This restriction loses $o(1)$ probability and permits use of the second multiplicity estimate.

Fix a small $\zeta > 0$ and set $h = T^{3/4-\zeta}$, $H = T^{3/4-\zeta/2}$, rounding heights to integers. The critical mass of bridges of height in $[H, 2H]$ is comparable to H^β . By (83), choosing the fixed confinement constant C_* sufficiently large retains a fixed fraction with diameter at most C_*H . The mass of bridges in this height window with $L(\eta) > T$ is at most $T^{-1}H^{25/12+o(1)}$, by (82). Its ratio to H^β is $H^{4/3+o(1)}/T = o(1)$. On the remainder, $x_T^{L(\eta)} \geq e^{-1}\rho^{L(\eta)}$. Thus every input admits insertions of total x_T -mass at least cH^β .

Let $\mathcal{A}$ consist of inputs of length at most T^2 whose absolute vertical endpoint difference is at most h , and let V be their x_T -mass. Since $T^2 \leq H^K$ for a fixed sufficiently large K , Lemma 12.2 applies to $M_{\mathcal{A}}$. Put

$$A_h = h^\beta (H/h)^{2\beta-1} \leq H^\beta, \quad Q = A_h H^\delta,$$

where $\delta > 0$ will be chosen small after ζ . For every fixed integer $k \geq 1$, the lemma gives

$$\begin{aligned}\sum_{M_{\mathcal{A}}(\omega) > Q} x_T^{\ell(\omega)} M_{\mathcal{A}}(\omega) &\leq Q^{1-k} \sum_{\omega} \rho^{\ell(\omega)} M_{\mathcal{A}}(\omega)^k \\ &\leq C_{k,K} A_h H^{C_K + o_k(1) - \delta(k-1)}.\end{aligned}$$

Choose k so that $\delta(k-1) > C_K + \beta + 1$, and then let $T \rightarrow \infty$. The last expression is $o(1)$. This order is possible because C_K is independent of k .

Sum all insertions of inputs in $\mathcal{A}$. For outputs with $M_{\mathcal{A}} \leq Q$, there are at most Q representations per output. Their unrestricted partition sum, rooted at o^* , equals Z_T by the root symmetry established above. The preceding estimate handles the remaining outputs. Recoverability and the bounded weight factors therefore give

$$cH^\beta V \leq C A_h H^\delta Z_T + o(1). \quad (91)$$

Divide by $H^\beta Z_T$ and use

$$\frac{A_h}{H^\beta} = (h/H)^{1-\beta}.$$

For sufficiently small δ , the bound $V/Z_T \leq C(h/H)^{1-\beta} H^\delta + o(1)$ tends to zero. Adding the discarded length tail proves that the absolute vertical endpoint difference exceeds $T^{3/4-\zeta}$ with probability tending to one. The Euclidean endpoint distance has the same lower exponent after absorbing the fixed row-spacing factor into the slack.

It remains to obtain the diameter upper bound and the length lower bound. Conditioned on its length, the thermal law is uniform. Fix $\varepsilon > 0$ and choose $\eta, \xi > 0$ so that

$$(1 + \xi)(3/4 + \eta) < 3/4 + \varepsilon.$$

For $m < T^{1/2}$ the diameter is deterministically at most $CT^{1/2}$. For $T^{1/2} \leq m \leq T^{1+\xi}$, the probability of violating $D \leq T^{3/4+\varepsilon}$ is, by (89), at most $C_{\eta,A}T^{-A/2}$ under each conditional length law. The remaining lengths have probability $o(1)$ by (90). This proves the upper diameter exponent, and hence the upper endpoint exponent.

For the lower length bound it suffices to take $0 < \xi < 1/2$. Choose $\eta, \zeta > 0$ such that

$$(1 - \xi)(3/4 + \eta) < 3/4 - \zeta, \quad \zeta < 1/4.$$

The typical endpoint lower bound excludes lengths $m < T^{1/2}$ deterministically. For $T^{1/2} \leq m < T^{1-\xi}$, it forces $D > m^{3/4+\eta}$, up to a fixed factor absorbed by the strict exponent inequality. Equation (89) bounds the conditional probability of this event uniformly by $C_{\eta,A}T^{-A/2} = o(1)$. Thus $\ell \geq T^{1-\xi}$ with probability tending to one. Together with the length upper tail and $A \leq D$, this proves all three claims. $\square$

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